Supplementary Material

Uncertainty Evaluation to Support Safety of the Intended Functionality Analysis for Determining the Applicability of Performance Targets Across Triggering Conditions in ML-Based LiDAR Object Detection

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Contents

1 How to read this document **1**

2 Condition classes **2**

3 Per-split outcome tables **2**
3.1 nuScenes, distance (m3) 2
3.2 KITTI, distance, labels as run and with the KITTI rule (m6 part A) 7
3.3 Change of dataset and detector, KITTI and nuScenes (m4, one split; m6 part B, five splits) . . . 11
3.4 nuScenes, recorded rain against clear daytime (m6 part C) 16

4 False acceptance proportion per split at the exploratory limit 0.1 **18**

5 KITTI rule: counts **18**

6 Accepted detections per distance bin **18**

7 Division by recorded sequence **20**

8 Division of KITTI by recorded drive **22**

9 Retention, shares of wrong detections, intervals, frames and the common threshold (division by frame) **24**

10 Post-hoc analyses **29**

11 Predictions fixed in advance and their outcomes **31**

12 Frames shown in Fig. 1 **31**

13 Source files **32**

1 How to read this document

This document lists the recorded results behind the paper in full: the outcome of every cell on every split, the intervals and retention behind each outcome, the counts behind the KITTI rule, the division by recorded sequence, the predictions fixed in advance, the selection of the frames shown in Fig. 1 and the post-hoc analyses. Nothing in it is recomputed: `tools/supplementary/make_supplementary.py` reads every number from a file of the repository and lists it with file, line or row, and column in `docs/supplementary/supplementary_sources.csv`; its option `--check` re-reads each listed value and compares it with the print.

One cell. A threshold τ_c is selected in the *condition of selection* c (columns `established` or `source` in the files) on the selection part, at limit α and error probability $\delta = 0.05$, and tested on the test part of the *condition of use* c' (`applied` or `target`). The one-sided Hoeffding–Bentkus bounds L and U of the false acceptance proportion (FAP) $A_{c'}(\tau_c)$, each at $\delta/2$, decide the outcome: **A** applicable ($U \leq \alpha$), **N** not applicable ($L > \alpha$), **?** not determined (otherwise), **-** no threshold selected in c. A code string such as `A??NN` gives the outcome on splits 0 to 4 from left to right. A cell with $c' = c$ tests a threshold in its own condition of selection and checks the selection; a cell with $c' \neq c$ is a *transfer*.

Limits. The target limits 10^{-2}, 10^{-3} and 10^{-4} were fixed before the first run. The limits 0.05 and 0.1 were fixed at the same time as exploratory limits, to show which outcomes the data can decide; they are not safety targets. No threshold could be selected at 10^{-3} or 10^{-4} in any condition.

Terms and files. The repository's `docs/glossary.md` defines every term and maps it to the column names of the files, which keep the names they were written with (for example `applies` for applicable, `eval` for the test part). Run names m1 to m8 identify the scripts in `experiments/` and the folders in `results/`. Distance conditions are assigned per detection (boundary: median distance of the car detections in the grid part); weather and dataset are assigned per frame. KITTI results apply the KITTI rule unless stated otherwise.

2 Condition classes

Each condition is represented by a class of frames or detections in which one process acts on the point cloud. Neither dataset labels classes C3 and C4, and they are not measured.

Table 1: Condition classes, defined by the process acting on the point cloud.

class	process	effect on detections	measured
C1	airborne scatterers	returns without a surface behind them; fewer distant returns	with C2, as recorded rain
C2	surface water, spray	non-rigid clusters; fewer ground returns	with C1, as recorded rain
C3	vehicle-like structure	consistent geometry where no vehicle is	no
C4	occlusion	partial evidence on a present object	no
C5	distance	fewer returns per object	both datasets

3 Per-split outcome tables

Each line is one cell of an applicability matrix: the threshold selected in one condition (*sel.*, condition of selection) tested in the condition of use (*use*) at limit α. The code string gives the outcome per split, seeds 0 to 4 from left to right: `A` applicable, `N` not applicable, `?` not determined, `-` no threshold selected. The detailed tables give, per split with a threshold, the threshold τ, the false acceptance proportion $\hat{A}$ with its two-sided HB interval ($\delta = 0.05$) and the retention $\hat{R}$, as recorded. Limits 10^{-2}, 10^{-3}, 10^{-4} are the target limits; 0.05 and 0.1 are exploratory limits.

3.1 nuScenes, distance (m3)

Source: `results/division_by_frame/nuscenes_distance/matrix_all_variants.csv`. Variants: overlap setting (IoU 0.5), object set (car; vehicles), cut quantile of car range in the grid part (0.25, 0.5, 0.75). The primary variant is car, cut quantile 0.5.

Table 2: nuScenes distance (m3), all variants: outcome per split.

objects	cut q.	ind.	α	sel.	use	seeds 0–4
car	0.25	u_e	10^{-2}	far	far	`?A?AA`
car	0.25	u_e	10^{-2}	far	near	`?????`
car	0.25	u_e	10^{-2}	near	far	`AAAAA`
car	0.25	u_e	10^{-2}	near	near	`AAAA?`
car	0.25	u_e	10^{-3}	far	far	---
car	0.25	u_e	10^{-3}	far	near	---
car	0.25	u_e	10^{-3}	near	far	---
car	0.25	u_e	10^{-3}	near	near	---
car	0.25	u_e	10^{-4}	far	far	---
car	0.25	u_e	10^{-4}	far	near	---
car	0.25	u_e	10^{-4}	near	far	---
car	0.25	u_e	10^{-4}	near	near	---
car	0.25	u_e	0.05	far	far	`?A??A`
car	0.25	u_e	0.05	far	near	`AAAAA`
car	0.25	u_e	0.05	near	far	`NNNNN`
car	0.25	u_e	0.05	near	near	`AAAAA`
car	0.25	u_e	0.1	far	far	`?A??A`
car	0.25	u_e	0.1	far	near	`AAAAA`
car	0.25	u_e	0.1	near	far	`NNNNN`
car	0.25	u_e	0.1	near	near	`AA?AA`
car	0.25	u_s	10^{-2}	far	far	`AA-AA`
car	0.25	u_s	10^{-2}	far	near	`AA-AA`
car	0.25	u_s	10^{-2}	near	far	`?AAA?`
car	0.25	u_s	10^{-2}	near	near	`?AAA?`
car	0.25	u_s	10^{-3}	far	far	---
car	0.25	u_s	10^{-3}	far	near	---

objects	cut q.	ind.	α	sel.	use	seeds 0–4
car	0.25	u_s	10^{-3}	near	far	---
car	0.25	u_s	10^{-3}	near	near	---
car	0.25	u_s	10^{-4}	far	far	---
car	0.25	u_s	10^{-4}	far	near	---
car	0.25	u_s	10^{-4}	near	far	---
car	0.25	u_s	10^{-4}	near	near	---
car	0.25	u_s	0.05	far	far	?AAAA
car	0.25	u_s	0.05	far	near	AAAAA
car	0.25	u_s	0.05	near	far	NNNNN
car	0.25	u_s	0.05	near	near	?A?A?
car	0.25	u_s	0.1	far	far	????A
car	0.25	u_s	0.1	far	near	AAAAA
car	0.25	u_s	0.1	near	far	NNNNN
car	0.25	u_s	0.1	near	near	AA?A?
car	0.5	u_e	10^{-2}	far	far	?A?AA
car	0.5	u_e	10^{-2}	far	near	N??NN
car	0.5	u_e	10^{-2}	near	far	AAAAA
car	0.5	u_e	10^{-2}	near	near	AAAA?
car	0.5	u_e	10^{-3}	far	far	---
car	0.5	u_e	10^{-3}	far	near	---
car	0.5	u_e	10^{-3}	near	far	---
car	0.5	u_e	10^{-3}	near	near	---
car	0.5	u_e	10^{-4}	far	far	---
car	0.5	u_e	10^{-4}	far	near	---
car	0.5	u_e	10^{-4}	near	far	---
car	0.5	u_e	10^{-4}	near	near	---
car	0.5	u_e	0.05	far	far	?A?AA
car	0.5	u_e	0.05	far	near	?????
car	0.5	u_e	0.05	near	far	AAAAA
car	0.5	u_e	0.05	near	near	?A??A
car	0.5	u_e	0.1	far	far	????A
car	0.5	u_e	0.1	far	near	AAAAA
car	0.5	u_e	0.1	near	far	NNNNN
car	0.5	u_e	0.1	near	near	AAA?A
car	0.5	u_s	10^{-2}	far	far	AA?A?
car	0.5	u_s	10^{-2}	far	near	?????
car	0.5	u_s	10^{-2}	near	far	AAAAA
car	0.5	u_s	10^{-2}	near	near	AAAA?
car	0.5	u_s	10^{-3}	far	far	---
car	0.5	u_s	10^{-3}	far	near	---
car	0.5	u_s	10^{-3}	near	far	---
car	0.5	u_s	10^{-3}	near	near	---
car	0.5	u_s	10^{-4}	far	far	---
car	0.5	u_s	10^{-4}	far	near	---
car	0.5	u_s	10^{-4}	near	far	---
car	0.5	u_s	10^{-4}	near	near	---
car	0.5	u_s	0.05	far	far	?A??A
car	0.5	u_s	0.05	far	near	?A?A?
car	0.5	u_s	0.05	near	far	AA?AA
car	0.5	u_s	0.05	near	near	?AAA?
car	0.5	u_s	0.1	far	far	????A
car	0.5	u_s	0.1	far	near	AAAAA
car	0.5	u_s	0.1	near	far	NNNNN
car	0.5	u_s	0.1	near	near	A??AA
car	0.75	u_e	10^{-2}	far	far	?AAA?
car	0.75	u_e	10^{-2}	far	near	NNNNN
car	0.75	u_e	10^{-2}	near	far	AAAAA
car	0.75	u_e	10^{-2}	near	near	AAAA?
car	0.75	u_e	10^{-3}	far	far	---
car	0.75	u_e	10^{-3}	far	near	---
car	0.75	u_e	10^{-3}	near	far	---
car	0.75	u_e	10^{-3}	near	near	---
car	0.75	u_e	10^{-4}	far	far	---
car	0.75	u_e	10^{-4}	far	near	---
car	0.75	u_e	10^{-4}	near	far	---
car	0.75	u_e	10^{-4}	near	near	---
car	0.75	u_e	0.05	far	far	?AAAA
car	0.75	u_e	0.05	far	near	NNNNN
car	0.75	u_e	0.05	near	far	AAAAA
car	0.75	u_e	0.05	near	near	?A?AA
car	0.75	u_e	0.1	far	far	?A?A?
car	0.75	u_e	0.1	far	near	?A???
car	0.75	u_e	0.1	near	far	AAAAA
car	0.75	u_e	0.1	near	near	?A?AA
car	0.75	u_s	10^{-2}	far	far	AAAA?
car	0.75	u_s	10^{-2}	far	near	NNN?N
car	0.75	u_s	10^{-2}	near	far	AAA-A
car	0.75	u_s	10^{-2}	near	near	AAA-A
car	0.75	u_s	10^{-3}	far	far	---
car	0.75	u_s	10^{-3}	far	near	---
car	0.75	u_s	10^{-3}	near	far	---
car	0.75	u_s	10^{-3}	near	near	---
car	0.75	u_s	10^{-4}	far	far	---
car	0.75	u_s	10^{-4}	far	near	---

objects	cut q.	ind.	α	sel.	use	seeds 0–4
car	0.75	u_s	10^{-4}	near	far	---
car	0.75	u_s	10^{-4}	near	near	---
car	0.75	u_s	0.05	far	far	?????
car	0.75	u_s	0.05	far	near	NNNNN
car	0.75	u_s	0.05	near	far	AAAAA
car	0.75	u_s	0.05	near	near	?AAA?
car	0.75	u_s	0.1	far	far	????A
car	0.75	u_s	0.1	far	near	?????
car	0.75	u_s	0.1	near	far	AAAAA
car	0.75	u_s	0.1	near	near	?AAAA
vehicles	0.25	u_e	10^{-2}	far	far	?AAA?
vehicles	0.25	u_e	10^{-2}	far	near	?????
vehicles	0.25	u_e	10^{-2}	near	far	AAAAA
vehicles	0.25	u_e	10^{-2}	near	near	AAA??
vehicles	0.25	u_e	10^{-3}	far	far	---
vehicles	0.25	u_e	10^{-3}	far	near	---
vehicles	0.25	u_e	10^{-3}	near	far	---
vehicles	0.25	u_e	10^{-3}	near	near	---
vehicles	0.25	u_e	10^{-4}	far	far	---
vehicles	0.25	u_e	10^{-4}	far	near	---
vehicles	0.25	u_e	10^{-4}	near	far	---
vehicles	0.25	u_e	10^{-4}	near	near	---
vehicles	0.25	u_e	0.05	far	far	?A??A
vehicles	0.25	u_e	0.05	far	near	AAAAA
vehicles	0.25	u_e	0.05	near	far	N?NNN
vehicles	0.25	u_e	0.05	near	near	A???A
vehicles	0.25	u_e	0.1	far	far	?A??A
vehicles	0.25	u_e	0.1	far	near	AAAAA
vehicles	0.25	u_e	0.1	near	far	NNNNN
vehicles	0.25	u_e	0.1	near	near	?????
vehicles	0.25	u_s	10^{-2}	far	far	--??
vehicles	0.25	u_s	10^{-2}	far	near	--AA
vehicles	0.25	u_s	10^{-2}	near	far	AA?-?
vehicles	0.25	u_s	10^{-2}	near	near	AAA-A
vehicles	0.25	u_s	10^{-3}	far	far	---
vehicles	0.25	u_s	10^{-3}	far	near	---
vehicles	0.25	u_s	10^{-3}	near	far	---
vehicles	0.25	u_s	10^{-3}	near	near	---
vehicles	0.25	u_s	10^{-4}	far	far	---
vehicles	0.25	u_s	10^{-4}	far	near	---
vehicles	0.25	u_s	10^{-4}	near	far	---
vehicles	0.25	u_s	10^{-4}	near	near	---
vehicles	0.25	u_s	0.05	far	far	?AA?A
vehicles	0.25	u_s	0.05	far	near	AAAAA
vehicles	0.25	u_s	0.05	near	far	NNN?N
vehicles	0.25	u_s	0.05	near	near	?AAAA
vehicles	0.25	u_s	0.1	far	far	?AA?A
vehicles	0.25	u_s	0.1	far	near	AAAAA
vehicles	0.25	u_s	0.1	near	far	NNNNN
vehicles	0.25	u_s	0.1	near	near	A??A?
vehicles	0.5	u_e	10^{-2}	far	far	AAAAA
vehicles	0.5	u_e	10^{-2}	far	near	NN?NN
vehicles	0.5	u_e	10^{-2}	near	far	AAAAA
vehicles	0.5	u_e	10^{-2}	near	near	AAA??
vehicles	0.5	u_e	10^{-3}	far	far	---
vehicles	0.5	u_e	10^{-3}	far	near	---
vehicles	0.5	u_e	10^{-3}	near	far	---
vehicles	0.5	u_e	10^{-3}	near	near	---
vehicles	0.5	u_e	10^{-4}	far	far	---
vehicles	0.5	u_e	10^{-4}	far	near	---
vehicles	0.5	u_e	10^{-4}	near	far	---
vehicles	0.5	u_e	10^{-4}	near	near	---
vehicles	0.5	u_e	0.05	far	far	?A?AA
vehicles	0.5	u_e	0.05	far	near	?????
vehicles	0.5	u_e	0.05	near	far	AAAAA
vehicles	0.5	u_e	0.05	near	near	?A??A
vehicles	0.5	u_e	0.1	far	far	?A?AA
vehicles	0.5	u_e	0.1	far	near	AAAAA
vehicles	0.5	u_e	0.1	near	far	NNNNN
vehicles	0.5	u_e	0.1	near	near	?AA?A
vehicles	0.5	u_s	10^{-2}	far	far	AA?AA
vehicles	0.5	u_s	10^{-2}	far	near	N???N
vehicles	0.5	u_s	10^{-2}	near	far	--A
vehicles	0.5	u_s	10^{-2}	near	near	--?
vehicles	0.5	u_s	10^{-3}	far	far	---
vehicles	0.5	u_s	10^{-3}	far	near	---
vehicles	0.5	u_s	10^{-3}	near	far	---
vehicles	0.5	u_s	10^{-3}	near	near	---
vehicles	0.5	u_s	10^{-4}	far	far	---
vehicles	0.5	u_s	10^{-4}	far	near	---
vehicles	0.5	u_s	10^{-4}	near	far	---
vehicles	0.5	u_s	10^{-4}	near	near	---
vehicles	0.5	u_s	0.05	far	far	?A??A
vehicles	0.5	u_s	0.05	far	near	?????

objects	cut q.	ind.	α	sel.	use	seeds 0–4
vehicles	0.5	u_s	0.05	near	far	AAAAA
vehicles	0.5	u_s	0.05	near	near	?AAA?
vehicles	0.5	u_s	0.1	far	far	????A
vehicles	0.5	u_s	0.1	far	near	AAAAA
vehicles	0.5	u_s	0.1	near	far	????A
vehicles	0.5	u_s	0.1	near	near	AAAAA
vehicles	0.75	u_e	10^{-2}	far	far	?AA?A
vehicles	0.75	u_e	10^{-2}	far	near	NNNNN
vehicles	0.75	u_e	10^{-2}	near	far	AAAAA
vehicles	0.75	u_e	10^{-2}	near	near	AAAA?
vehicles	0.75	u_e	10^{-3}	far	far	---
vehicles	0.75	u_e	10^{-3}	far	near	---
vehicles	0.75	u_e	10^{-3}	near	far	---
vehicles	0.75	u_e	10^{-3}	near	near	---
vehicles	0.75	u_e	10^{-4}	far	far	---
vehicles	0.75	u_e	10^{-4}	far	near	---
vehicles	0.75	u_e	10^{-4}	near	far	---
vehicles	0.75	u_e	10^{-4}	near	near	---
vehicles	0.75	u_e	0.05	far	far	?AA?A
vehicles	0.75	u_e	0.05	far	near	NNNNN
vehicles	0.75	u_e	0.05	near	far	AAAAA
vehicles	0.75	u_e	0.05	near	near	?A??A
vehicles	0.75	u_e	0.1	far	far	AAAAA
vehicles	0.75	u_e	0.1	far	near	?????
vehicles	0.75	u_e	0.1	near	far	AAAAA
vehicles	0.75	u_e	0.1	near	near	?A?AA
vehicles	0.75	u_s	10^{-2}	far	far	??AAA
vehicles	0.75	u_s	10^{-2}	far	near	NNNNN
vehicles	0.75	u_s	10^{-2}	near	far	--A
vehicles	0.75	u_s	10^{-2}	near	near	--?
vehicles	0.75	u_s	10^{-3}	far	far	---
vehicles	0.75	u_s	10^{-3}	far	near	---
vehicles	0.75	u_s	10^{-3}	near	far	---
vehicles	0.75	u_s	10^{-3}	near	near	---
vehicles	0.75	u_s	10^{-4}	far	far	---
vehicles	0.75	u_s	10^{-4}	far	near	---
vehicles	0.75	u_s	10^{-4}	near	far	---
vehicles	0.75	u_s	10^{-4}	near	near	---
vehicles	0.75	u_s	0.05	far	far	??A?A
vehicles	0.75	u_s	0.05	far	near	NNNNN
vehicles	0.75	u_s	0.05	near	far	AAAAA
vehicles	0.75	u_s	0.05	near	near	?AAAA
vehicles	0.75	u_s	0.1	far	far	????A
vehicles	0.75	u_s	0.1	far	near	NNNNN
vehicles	0.75	u_s	0.1	near	far	AAAAA
vehicles	0.75	u_s	0.1	near	near	?AAAA

Table 3: nuScenes distance (m3), primary variant (car, median cut): values per split.

ind.	α	sel.	use	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	out.
u_e	10^{-2}	far	far	0	0.1823	0.0068	[0.0033, 0.0124]	0.038	?
u_e	10^{-2}	far	far	1	0.1708	0.0030	[0.0009, 0.0071]	0.039	A
u_e	10^{-2}	far	far	2	0.1728	0.0052	[0.0023, 0.0100]	0.042	?
u_e	10^{-2}	far	far	3	0.1685	0.0037	[0.0012, 0.0082]	0.037	A
u_e	10^{-2}	far	far	4	0.1733	0.0045	[0.0018, 0.0094]	0.037	A
u_e	10^{-2}	far	near	0	0.1823	0.0191	[0.0128, 0.0276]	0.344	N
u_e	10^{-2}	far	near	1	0.1708	0.0143	[0.0088, 0.0218]	0.325	?
u_e	10^{-2}	far	near	2	0.1728	0.0156	[0.0097, 0.0231]	0.338	?
u_e	10^{-2}	far	near	3	0.1685	0.0157	[0.0101, 0.0237]	0.320	N
u_e	10^{-2}	far	near	4	0.1733	0.0161	[0.0103, 0.0239]	0.328	N
u_e	10^{-2}	near	far	0	0.0925	0.0000	[0.0000, 0.0016]	0.003	A
u_e	10^{-2}	near	far	1	0.0773	0.0000	[0.0000, 0.0016]	0.002	A
u_e	10^{-2}	near	far	2	0.0872	0.0000	[0.0000, 0.0016]	0.004	A
u_e	10^{-2}	near	far	3	0.1002	0.0000	[0.0000, 0.0016]	0.005	A
u_e	10^{-2}	near	far	4	0.1141	0.0002	[0.0000, 0.0024]	0.011	A
u_e	10^{-2}	near	near	0	0.0925	0.0043	[0.0016, 0.0092]	0.101	A
u_e	10^{-2}	near	near	1	0.0773	0.0018	[0.0004, 0.0053]	0.070	A
u_e	10^{-2}	near	near	2	0.0872	0.0032	[0.0009, 0.0074]	0.097	A
u_e	10^{-2}	near	near	3	0.1002	0.0051	[0.0022, 0.0099]	0.130	A
u_e	10^{-2}	near	near	4	0.1141	0.0074	[0.0037, 0.0133]	0.169	?
u_e	0.05	far	far	0	0.3133	0.0470	[0.0368, 0.0585]	0.177	?
u_e	0.05	far	far	1	0.2944	0.0304	[0.0225, 0.0405]	0.154	A
u_e	0.05	far	far	2	0.3121	0.0440	[0.0342, 0.0553]	0.192	?
u_e	0.05	far	far	3	0.3001	0.0387	[0.0298, 0.0496]	0.168	A
u_e	0.05	far	far	4	0.3070	0.0354	[0.0268, 0.0460]	0.178	A
u_e	0.05	far	near	0	0.3133	0.0489	[0.0387, 0.0613]	0.650	?
u_e	0.05	far	near	1	0.2944	0.0440	[0.0340, 0.0556]	0.622	?
u_e	0.05	far	near	2	0.3121	0.0465	[0.0364, 0.0587]	0.662	?
u_e	0.05	far	near	3	0.3001	0.0464	[0.0361, 0.0583]	0.633	?
u_e	0.05	far	near	4	0.3070	0.0455	[0.0355, 0.0574]	0.639	?

ind.	α	sel.	use	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	out.
u_e	0.05	near	far	0	0.2693	0.0288	[0.0210, 0.0384]	0.108	A
u_e	0.05	near	far	1	0.2513	0.0175	[0.0114, 0.0253]	0.101	A
u_e	0.05	near	far	2	0.2630	0.0238	[0.0167, 0.0327]	0.118	A
u_e	0.05	near	far	3	0.2820	0.0320	[0.0237, 0.0419]	0.141	A
u_e	0.05	near	far	4	0.2579	0.0191	[0.0129, 0.0274]	0.110	A
u_e	0.05	near	near	0	0.2693	0.0413	[0.0319, 0.0528]	0.564	?
u_e	0.05	near	near	1	0.2513	0.0302	[0.0221, 0.0403]	0.525	A
u_e	0.05	near	near	2	0.2630	0.0405	[0.0311, 0.0521]	0.565	?
u_e	0.05	near	near	3	0.2820	0.0447	[0.0348, 0.0567]	0.595	?
u_e	0.05	near	near	4	0.2579	0.0372	[0.0283, 0.0483]	0.535	A
u_e	0.1	far	far	0	0.4305	0.0990	[0.0846, 0.1150]	0.395	?
u_e	0.1	far	far	1	0.4149	0.0863	[0.0726, 0.1012]	0.365	?
u_e	0.1	far	far	2	0.4136	0.0863	[0.0728, 0.1013]	0.377	?
u_e	0.1	far	far	3	0.4095	0.0880	[0.0743, 0.1030]	0.366	?
u_e	0.1	far	far	4	0.4102	0.0811	[0.0678, 0.0958]	0.356	A
u_e	0.1	far	near	0	0.4305	0.0612	[0.0497, 0.0747]	0.823	A
u_e	0.1	far	near	1	0.4149	0.0604	[0.0487, 0.0736]	0.809	A
u_e	0.1	far	near	2	0.4136	0.0595	[0.0480, 0.0728]	0.814	A
u_e	0.1	far	near	3	0.4095	0.0574	[0.0460, 0.0704]	0.806	A
u_e	0.1	far	near	4	0.4102	0.0562	[0.0449, 0.0689]	0.802	A
u_e	0.1	near	far	0	0.5720	0.1718	[0.1535, 0.1917]	0.678	N
u_e	0.1	near	far	1	0.5722	0.1724	[0.1539, 0.1922]	0.687	N
u_e	0.1	near	far	2	0.5799	0.1759	[0.1572, 0.1956]	0.704	N
u_e	0.1	near	far	3	0.5984	0.1813	[0.1624, 0.2011]	0.739	N
u_e	0.1	near	far	4	0.5730	0.1697	[0.1513, 0.1894]	0.679	N
u_e	0.1	near	near	0	0.5720	0.0822	[0.0687, 0.0972]	0.930	A
u_e	0.1	near	near	1	0.5722	0.0824	[0.0691, 0.0977]	0.928	A
u_e	0.1	near	near	2	0.5799	0.0820	[0.0685, 0.0972]	0.933	A
u_e	0.1	near	near	3	0.5984	0.0873	[0.0733, 0.1028]	0.944	?
u_e	0.1	near	near	4	0.5730	0.0793	[0.0659, 0.0940]	0.929	A
u_s	10^{-2}	far	far	0	0.1407	0.0031	[0.0010, 0.0071]	0.011	A
u_s	10^{-2}	far	far	1	0.1415	0.0026	[0.0008, 0.0065]	0.009	A
u_s	10^{-2}	far	far	2	0.1466	0.0072	[0.0035, 0.0128]	0.016	?
u_s	10^{-2}	far	far	3	0.1422	0.0043	[0.0018, 0.0088]	0.010	A
u_s	10^{-2}	far	far	4	0.1460	0.0055	[0.0024, 0.0107]	0.013	?
u_s	10^{-2}	far	near	0	0.1407	0.0101	[0.0057, 0.0167]	0.055	?
u_s	10^{-2}	far	near	1	0.1415	0.0074	[0.0037, 0.0134]	0.062	?
u_s	10^{-2}	far	near	2	0.1466	0.0111	[0.0064, 0.0181]	0.087	?
u_s	10^{-2}	far	near	3	0.1422	0.0086	[0.0044, 0.0147]	0.065	?
u_s	10^{-2}	far	near	4	0.1460	0.0141	[0.0088, 0.0217]	0.078	?
u_s	10^{-2}	near	far	0	0.1265	0.0024	[0.0006, 0.0064]	0.005	A
u_s	10^{-2}	near	far	1	0.1238	0.0007	[0.0000, 0.0034]	0.003	A
u_s	10^{-2}	near	far	2	0.1309	0.0022	[0.0005, 0.0058]	0.004	A
u_s	10^{-2}	near	far	3	0.1305	0.0033	[0.0010, 0.0076]	0.005	A
u_s	10^{-2}	near	far	4	0.1358	0.0040	[0.0015, 0.0083]	0.006	A
u_s	10^{-2}	near	near	0	0.1265	0.0023	[0.0006, 0.0060]	0.020	A
u_s	10^{-2}	near	near	1	0.1238	0.0009	[0.0001, 0.0039]	0.016	A
u_s	10^{-2}	near	near	2	0.1309	0.0032	[0.0011, 0.0074]	0.031	A
u_s	10^{-2}	near	near	3	0.1305	0.0021	[0.0004, 0.0059]	0.028	A
u_s	10^{-2}	near	near	4	0.1358	0.0081	[0.0040, 0.0139]	0.040	?
u_s	0.05	far	far	0	0.1861	0.0434	[0.0336, 0.0546]	0.082	?
u_s	0.05	far	far	1	0.1808	0.0320	[0.0237, 0.0420]	0.063	A
u_s	0.05	far	far	2	0.1852	0.0460	[0.0362, 0.0577]	0.082	?
u_s	0.05	far	far	3	0.1801	0.0392	[0.0301, 0.0501]	0.071	?
u_s	0.05	far	far	4	0.1809	0.0346	[0.0260, 0.0451]	0.065	A
u_s	0.05	far	near	0	0.1861	0.0485	[0.0383, 0.0608]	0.345	?
u_s	0.05	far	near	1	0.1808	0.0361	[0.0272, 0.0469]	0.311	A
u_s	0.05	far	near	2	0.1852	0.0423	[0.0324, 0.0536]	0.354	?
u_s	0.05	far	near	3	0.1801	0.0350	[0.0263, 0.0459]	0.309	A
u_s	0.05	far	near	4	0.1809	0.0407	[0.0311, 0.0519]	0.309	?
u_s	0.05	near	far	0	0.1815	0.0371	[0.0282, 0.0478]	0.070	A
u_s	0.05	near	far	1	0.1760	0.0263	[0.0188, 0.0355]	0.052	A
u_s	0.05	near	far	2	0.1783	0.0402	[0.0311, 0.0514]	0.061	?
u_s	0.05	near	far	3	0.1784	0.0379	[0.0290, 0.0487]	0.066	A
u_s	0.05	near	far	4	0.1809	0.0346	[0.0260, 0.0451]	0.065	A
u_s	0.05	near	near	0	0.1815	0.0451	[0.0351, 0.0568]	0.307	?
u_s	0.05	near	near	1	0.1760	0.0307	[0.0225, 0.0408]	0.272	A
u_s	0.05	near	near	2	0.1783	0.0363	[0.0271, 0.0469]	0.298	A
u_s	0.05	near	near	3	0.1784	0.0328	[0.0243, 0.0433]	0.295	A
u_s	0.05	near	near	4	0.1809	0.0407	[0.0311, 0.0519]	0.309	?
u_s	0.1	far	far	0	0.2129	0.1004	[0.0858, 0.1164]	0.170	?
u_s	0.1	far	far	1	0.2088	0.0867	[0.0730, 0.1017]	0.151	?
u_s	0.1	far	far	2	0.2105	0.0973	[0.0829, 0.1130]	0.170	?
u_s	0.1	far	far	3	0.2081	0.0893	[0.0755, 0.1044]	0.164	?
u_s	0.1	far	far	4	0.2081	0.0813	[0.0682, 0.0962]	0.151	A
u_s	0.1	far	near	0	0.2129	0.0691	[0.0567, 0.0830]	0.547	A
u_s	0.1	far	near	1	0.2088	0.0622	[0.0504, 0.0755]	0.525	A
u_s	0.1	far	near	2	0.2105	0.0657	[0.0534, 0.0793]	0.554	A
u_s	0.1	far	near	3	0.2081	0.0597	[0.0480, 0.0729]	0.524	A
u_s	0.1	far	near	4	0.2081	0.0625	[0.0506, 0.0758]	0.518	A
u_s	0.1	near	far	0	0.2279	0.1282	[0.1118, 0.1457]	0.232	N
u_s	0.1	near	far	1	0.2297	0.1286	[0.1125, 0.1466]	0.229	N
u_s	0.1	near	far	2	0.2351	0.1411	[0.1240, 0.1591]	0.276	N
u_s	0.1	near	far	3	0.2248	0.1204	[0.1047, 0.1375]	0.236	N
u_s	0.1	near	far	4	0.2298	0.1213	[0.1053, 0.1385]	0.238	N
u_s	0.1	near	near	0	0.2279	0.0824	[0.0691, 0.0977]	0.648	A
u_s	0.1	near	near	1	0.2297	0.0850	[0.0712, 0.1001]	0.661	?

ind.	α	sel.	use	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	out.
u_s	0.1	near	near	2	0.2351	0.0875	[0.0736, 0.1031]	0.706	?
u_s	0.1	near	near	3	0.2248	0.0743	[0.0614, 0.0889]	0.644	A
u_s	0.1	near	near	4	0.2298	0.0847	[0.0709, 0.0998]	0.668	A

3.2 KITTI, distance, labels as run and with the KITTI rule (m6 part A)

Source: `results/division_by_frame/kitti_rule_dataset_change_weather/A_kitti_range_protocol.csv`. Car detections, median cut of the grid part. Protocol `as_run`: labels of the detection pipeline; `kitti_ignore`: wrong detections in DontCare regions or on a Van removed (Section 5).

Table 4: KITTI distance (m6 A): outcome per split.

protocol	ind.	α	sel.	use	seeds 0–4
as_run	u_e	10^{-2}	far	far	AA?AA
as_run	u_e	10^{-2}	far	near	NNN??
as_run	u_e	10^{-2}	near	far	AAAAA
as_run	u_e	10^{-2}	near	near	?A???
as_run	u_e	10^{-3}	far	far	---
as_run	u_e	10^{-3}	far	near	---
as_run	u_e	10^{-3}	near	far	---
as_run	u_e	10^{-3}	near	near	---
as_run	u_e	10^{-4}	far	far	---
as_run	u_e	10^{-4}	far	near	---
as_run	u_e	10^{-4}	near	far	---
as_run	u_e	10^{-4}	near	near	---
as_run	u_e	0.05	far	far	AAAAA
as_run	u_e	0.05	far	near	?????
as_run	u_e	0.05	near	far	AAAAA
as_run	u_e	0.05	near	near	A???A
as_run	u_e	0.1	far	far	A?AAA
as_run	u_e	0.1	far	near	AAAAA
as_run	u_e	0.1	near	far	NNNNN
as_run	u_e	0.1	near	near	AAA?A
as_run	u_s	10^{-2}	far	far	?AAA?
as_run	u_s	10^{-2}	far	near	?A?AA
as_run	u_s	10^{-2}	near	far	?A?AA
as_run	u_s	10^{-2}	near	near	?A?AA
as_run	u_s	10^{-3}	far	far	---
as_run	u_s	10^{-3}	far	near	---
as_run	u_s	10^{-3}	near	far	---
as_run	u_s	10^{-3}	near	near	---
as_run	u_s	10^{-4}	far	far	---
as_run	u_s	10^{-4}	far	near	---
as_run	u_s	10^{-4}	near	far	---
as_run	u_s	10^{-4}	near	near	---
as_run	u_s	0.05	far	far	?AA?A
as_run	u_s	0.05	far	near	AAAAA
as_run	u_s	0.05	near	far	???N?
as_run	u_s	0.05	near	near	AAA?A
as_run	u_s	0.1	far	far	AAA?A
as_run	u_s	0.1	far	near	AAAAA
as_run	u_s	0.1	near	far	NNNNN
as_run	u_s	0.1	near	near	AAA?A
kitti_ignore	u_e	10^{-2}	far	far	?A?AA
kitti_ignore	u_e	10^{-2}	far	near	?????
kitti_ignore	u_e	10^{-2}	near	far	AAAAA
kitti_ignore	u_e	10^{-2}	near	near	??A??
kitti_ignore	u_e	10^{-3}	far	far	---
kitti_ignore	u_e	10^{-3}	far	near	---
kitti_ignore	u_e	10^{-3}	near	far	---
kitti_ignore	u_e	10^{-3}	near	near	---
kitti_ignore	u_e	10^{-4}	far	far	---
kitti_ignore	u_e	10^{-4}	far	near	---
kitti_ignore	u_e	10^{-4}	near	far	---
kitti_ignore	u_e	10^{-4}	near	near	---
kitti_ignore	u_e	0.05	far	far	AAAAA
kitti_ignore	u_e	0.05	far	near	AAAAA
kitti_ignore	u_e	0.05	near	far	?N?N?
kitti_ignore	u_e	0.05	near	near	AAA?A
kitti_ignore	u_e	0.1	far	far	A?AAA
kitti_ignore	u_e	0.1	far	near	AAAAA
kitti_ignore	u_e	0.1	near	far	NNNNN
kitti_ignore	u_e	0.1	near	near	AAA?A
kitti_ignore	u_s	10^{-2}	far	far	?AAA?
kitti_ignore	u_s	10^{-2}	far	near	AA?AA
kitti_ignore	u_s	10^{-2}	near	far	?A?N?
kitti_ignore	u_s	10^{-2}	near	near	?A??A
kitti_ignore	u_s	10^{-3}	far	far	---
kitti_ignore	u_s	10^{-3}	far	near	---

protocol	ind.	α	sel.	use	seeds 0–4
kitti_ignore	u_s	10^{-3}	near	far	---
kitti_ignore	u_s	10^{-3}	near	near	---
kitti_ignore	u_s	10^{-4}	far	far	---
kitti_ignore	u_s	10^{-4}	far	near	---
kitti_ignore	u_s	10^{-4}	near	far	---
kitti_ignore	u_s	10^{-4}	near	near	---
kitti_ignore	u_s	0.05	far	far	AAA?A
kitti_ignore	u_s	0.05	far	near	AAAAA
kitti_ignore	u_s	0.05	near	far	NNNNN
kitti_ignore	u_s	0.05	near	near	?AA?A
kitti_ignore	u_s	0.1	far	far	AAA?A
kitti_ignore	u_s	0.1	far	near	AAAAA
kitti_ignore	u_s	0.1	near	far	NNNNN
kitti_ignore	u_s	0.1	near	near	?AA?A

Table 5: KITTI distance (m6 A): values per split (no τ column is recorded in this file).

protocol	ind.	α	sel.	use	seed	$\hat{A}$	HB interval	$\hat{R}$	out.
as_run	u_e	10^{-2}	far	far	0	0.0039	[0.0011, 0.0094]	0.082	A
as_run	u_e	10^{-2}	far	far	1	0.0039	[0.0011, 0.0095]	0.089	A
as_run	u_e	10^{-2}	far	far	2	0.0065	[0.0027, 0.0132]	0.082	?
as_run	u_e	10^{-2}	far	far	3	0.0029	[0.0006, 0.0083]	0.062	A
as_run	u_e	10^{-2}	far	far	4	0.0006	[0.0000, 0.0043]	0.048	A
as_run	u_e	10^{-2}	far	near	0	0.0190	[0.0112, 0.0289]	0.597	N
as_run	u_e	10^{-2}	far	near	1	0.0209	[0.0130, 0.0319]	0.600	N
as_run	u_e	10^{-2}	far	near	2	0.0188	[0.0112, 0.0290]	0.605	N
as_run	u_e	10^{-2}	far	near	3	0.0171	[0.0098, 0.0267]	0.571	?
as_run	u_e	10^{-2}	far	near	4	0.0162	[0.0092, 0.0258]	0.527	?
as_run	u_e	10^{-2}	near	far	0	0.0000	[0.0000, 0.0024]	0.006	A
as_run	u_e	10^{-2}	near	far	1	0.0000	[0.0000, 0.0024]	0.007	A
as_run	u_e	10^{-2}	near	far	2	0.0000	[0.0000, 0.0024]	0.006	A
as_run	u_e	10^{-2}	near	far	3	0.0000	[0.0000, 0.0024]	0.010	A
as_run	u_e	10^{-2}	near	far	4	0.0000	[0.0000, 0.0024]	0.011	A
as_run	u_e	10^{-2}	near	near	0	0.0050	[0.0016, 0.0115]	0.316	?
as_run	u_e	10^{-2}	near	near	1	0.0023	[0.0003, 0.0075]	0.274	A
as_run	u_e	10^{-2}	near	near	2	0.0054	[0.0019, 0.0121]	0.324	?
as_run	u_e	10^{-2}	near	near	3	0.0051	[0.0016, 0.0115]	0.322	?
as_run	u_e	10^{-2}	near	near	4	0.0075	[0.0031, 0.0150]	0.346	?
as_run	u_e	0.05	far	far	0	0.0260	[0.0172, 0.0375]	0.333	A
as_run	u_e	0.05	far	far	1	0.0362	[0.0259, 0.0490]	0.366	A
as_run	u_e	0.05	far	far	2	0.0254	[0.0169, 0.0365]	0.314	A
as_run	u_e	0.05	far	far	3	0.0313	[0.0215, 0.0437]	0.338	A
as_run	u_e	0.05	far	far	4	0.0279	[0.0190, 0.0395]	0.321	A
as_run	u_e	0.05	far	near	0	0.0401	[0.0290, 0.0541]	0.811	?
as_run	u_e	0.05	far	near	1	0.0440	[0.0319, 0.0581]	0.827	?
as_run	u_e	0.05	far	near	2	0.0427	[0.0314, 0.0572]	0.799	?
as_run	u_e	0.05	far	near	3	0.0422	[0.0310, 0.0568]	0.814	?
as_run	u_e	0.05	far	near	4	0.0413	[0.0297, 0.0550]	0.807	?
as_run	u_e	0.05	near	far	0	0.0206	[0.0130, 0.0307]	0.283	A
as_run	u_e	0.05	near	far	1	0.0223	[0.0141, 0.0332]	0.266	A
as_run	u_e	0.05	near	far	2	0.0254	[0.0169, 0.0365]	0.314	A
as_run	u_e	0.05	near	far	3	0.0255	[0.0167, 0.0369]	0.316	A
as_run	u_e	0.05	near	far	4	0.0182	[0.0111, 0.0279]	0.239	A
as_run	u_e	0.05	near	near	0	0.0356	[0.0251, 0.0489]	0.783	A
as_run	u_e	0.05	near	near	1	0.0373	[0.0262, 0.0506]	0.771	?
as_run	u_e	0.05	near	near	2	0.0427	[0.0314, 0.0572]	0.799	?
as_run	u_e	0.05	near	near	3	0.0396	[0.0281, 0.0530]	0.805	?
as_run	u_e	0.05	near	near	4	0.0347	[0.0241, 0.0475]	0.760	A
as_run	u_e	0.1	far	far	0	0.0709	[0.0562, 0.0881]	0.539	A
as_run	u_e	0.1	far	far	1	0.0841	[0.0682, 0.1028]	0.554	?
as_run	u_e	0.1	far	far	2	0.0642	[0.0499, 0.0807]	0.518	A
as_run	u_e	0.1	far	far	3	0.0802	[0.0645, 0.0983]	0.543	A
as_run	u_e	0.1	far	far	4	0.0615	[0.0475, 0.0777]	0.501	A
as_run	u_e	0.1	far	near	0	0.0581	[0.0444, 0.0740]	0.895	A
as_run	u_e	0.1	far	near	1	0.0627	[0.0482, 0.0790]	0.902	A
as_run	u_e	0.1	far	near	2	0.0573	[0.0439, 0.0734]	0.890	A
as_run	u_e	0.1	far	near	3	0.0618	[0.0477, 0.0781]	0.903	A
as_run	u_e	0.1	far	near	4	0.0554	[0.0422, 0.0712]	0.883	A
as_run	u_e	0.1	near	far	0	0.1277	[0.1079, 0.1491]	0.679	N
as_run	u_e	0.1	near	far	1	0.1330	[0.1129, 0.1548]	0.689	N
as_run	u_e	0.1	near	far	2	0.1206	[0.1016, 0.1421]	0.680	N
as_run	u_e	0.1	near	far	3	0.1443	[0.1234, 0.1667]	0.698	N
as_run	u_e	0.1	near	far	4	0.1289	[0.1091, 0.1507]	0.685	N
as_run	u_e	0.1	near	near	0	0.0789	[0.0632, 0.0970]	0.937	A
as_run	u_e	0.1	near	near	1	0.0797	[0.0638, 0.0981]	0.943	A
as_run	u_e	0.1	near	near	2	0.0750	[0.0597, 0.0929]	0.940	A
as_run	u_e	0.1	near	near	3	0.0838	[0.0678, 0.1027]	0.954	?
as_run	u_e	0.1	near	near	4	0.0793	[0.0633, 0.0973]	0.945	A
as_run	u_s	10^{-2}	far	far	0	0.0083	[0.0039, 0.0156]	0.027	?
as_run	u_s	10^{-2}	far	far	1	0.0019	[0.0003, 0.0066]	0.016	A
as_run	u_s	10^{-2}	far	far	2	0.0033	[0.0008, 0.0086]	0.019	A

protocol	ind.	α	sel.	use	seed	$\hat{A}$	HB interval	$\hat{R}$	out.
as_run	u_s	10^{-2}	far	far	3	0.0019	[0.0003, 0.0065]	0.007	A
as_run	u_s	10^{-2}	far	far	4	0.0045	[0.0015, 0.0104]	0.018	?
as_run	u_s	10^{-2}	far	near	0	0.0072	[0.0031, 0.0141]	0.138	?
as_run	u_s	10^{-2}	far	near	1	0.0017	[0.0002, 0.0063]	0.086	A
as_run	u_s	10^{-2}	far	near	2	0.0072	[0.0031, 0.0142]	0.090	?
as_run	u_s	10^{-2}	far	near	3	0.0020	[0.0003, 0.0067]	0.076	A
as_run	u_s	10^{-2}	far	near	4	0.0039	[0.0012, 0.0096]	0.111	A
as_run	u_s	10^{-2}	near	far	0	0.0083	[0.0039, 0.0156]	0.027	?
as_run	u_s	10^{-2}	near	far	1	0.0006	[0.0000, 0.0042]	0.004	A
as_run	u_s	10^{-2}	near	far	2	0.0052	[0.0019, 0.0114]	0.032	?
as_run	u_s	10^{-2}	near	far	3	0.0019	[0.0003, 0.0065]	0.007	A
as_run	u_s	10^{-2}	near	far	4	0.0032	[0.0008, 0.0086]	0.007	A
as_run	u_s	10^{-2}	near	near	0	0.0072	[0.0031, 0.0141]	0.138	?
as_run	u_s	10^{-2}	near	near	1	0.0007	[0.0000, 0.0044]	0.029	A
as_run	u_s	10^{-2}	near	near	2	0.0082	[0.0035, 0.0159]	0.118	?
as_run	u_s	10^{-2}	near	near	3	0.0020	[0.0003, 0.0067]	0.076	A
as_run	u_s	10^{-2}	near	near	4	0.0026	[0.0005, 0.0077]	0.043	A
as_run	u_s	0.05	far	far	0	0.0373	[0.0268, 0.0502]	0.116	?
as_run	u_s	0.05	far	far	1	0.0320	[0.0221, 0.0445]	0.132	A
as_run	u_s	0.05	far	far	2	0.0326	[0.0228, 0.0449]	0.131	A
as_run	u_s	0.05	far	far	3	0.0461	[0.0341, 0.0606]	0.142	?
as_run	u_s	0.05	far	far	4	0.0299	[0.0206, 0.0417]	0.109	A
as_run	u_s	0.05	far	near	0	0.0229	[0.0143, 0.0336]	0.368	A
as_run	u_s	0.05	far	near	1	0.0197	[0.0119, 0.0303]	0.356	A
as_run	u_s	0.05	far	near	2	0.0233	[0.0149, 0.0345]	0.339	A
as_run	u_s	0.05	far	near	3	0.0232	[0.0150, 0.0347]	0.401	A
as_run	u_s	0.05	far	near	4	0.0209	[0.0128, 0.0314]	0.340	A
as_run	u_s	0.05	near	far	0	0.0469	[0.0351, 0.0611]	0.172	?
as_run	u_s	0.05	near	far	1	0.0632	[0.0490, 0.0795]	0.215	?
as_run	u_s	0.05	near	far	2	0.0578	[0.0442, 0.0735]	0.207	?
as_run	u_s	0.05	near	far	3	0.0689	[0.0541, 0.0856]	0.223	N
as_run	u_s	0.05	near	far	4	0.0602	[0.0464, 0.0762]	0.194	?
as_run	u_s	0.05	near	near	0	0.0312	[0.0213, 0.0436]	0.450	A
as_run	u_s	0.05	near	near	1	0.0357	[0.0251, 0.0490]	0.484	A
as_run	u_s	0.05	near	near	2	0.0354	[0.0246, 0.0482]	0.467	A
as_run	u_s	0.05	near	near	3	0.0381	[0.0270, 0.0515]	0.502	?
as_run	u_s	0.05	near	near	4	0.0335	[0.0230, 0.0460]	0.470	A
as_run	u_s	0.1	far	far	0	0.0770	[0.0614, 0.0945]	0.253	A
as_run	u_s	0.1	far	far	1	0.0777	[0.0623, 0.0958]	0.261	A
as_run	u_s	0.1	far	far	2	0.0786	[0.0628, 0.0964]	0.275	A
as_run	u_s	0.1	far	far	3	0.0885	[0.0721, 0.1074]	0.270	?
as_run	u_s	0.1	far	far	4	0.0793	[0.0638, 0.0976]	0.241	A
as_run	u_s	0.1	far	near	0	0.0417	[0.0302, 0.0556]	0.556	A
as_run	u_s	0.1	far	near	1	0.0399	[0.0284, 0.0536]	0.545	A
as_run	u_s	0.1	far	near	2	0.0438	[0.0319, 0.0579]	0.549	A
as_run	u_s	0.1	far	near	3	0.0444	[0.0327, 0.0590]	0.567	A
as_run	u_s	0.1	far	near	4	0.0390	[0.0280, 0.0528]	0.536	A
as_run	u_s	0.1	near	far	0	0.1947	[0.1715, 0.2200]	0.467	N
as_run	u_s	0.1	near	far	1	0.2090	[0.1849, 0.2349]	0.512	N
as_run	u_s	0.1	near	far	2	0.1953	[0.1720, 0.2203]	0.473	N
as_run	u_s	0.1	near	far	3	0.2252	[0.2004, 0.2517]	0.526	N
as_run	u_s	0.1	near	far	4	0.1952	[0.1715, 0.2204]	0.462	N
as_run	u_s	0.1	near	near	0	0.0761	[0.0608, 0.0941]	0.769	A
as_run	u_s	0.1	near	near	1	0.0774	[0.0619, 0.0959]	0.787	A
as_run	u_s	0.1	near	near	2	0.0716	[0.0568, 0.0893]	0.759	A
as_run	u_s	0.1	near	near	3	0.0838	[0.0672, 0.1020]	0.808	?
as_run	u_s	0.1	near	near	4	0.0760	[0.0604, 0.0937]	0.773	A
kitti_ignore	u_e	10^{-2}	far	far	0	0.0058	[0.0022, 0.0122]	0.121	?
kitti_ignore	u_e	10^{-2}	far	far	1	0.0023	[0.0003, 0.0073]	0.120	A
kitti_ignore	u_e	10^{-2}	far	far	2	0.0052	[0.0019, 0.0115]	0.107	?
kitti_ignore	u_e	10^{-2}	far	far	3	0.0032	[0.0008, 0.0085]	0.111	A
kitti_ignore	u_e	10^{-2}	far	far	4	0.0023	[0.0003, 0.0073]	0.108	A
kitti_ignore	u_e	10^{-2}	far	near	0	0.0098	[0.0045, 0.0179]	0.650	?
kitti_ignore	u_e	10^{-2}	far	near	1	0.0078	[0.0032, 0.0157]	0.615	?
kitti_ignore	u_e	10^{-2}	far	near	2	0.0054	[0.0019, 0.0123]	0.617	?
kitti_ignore	u_e	10^{-2}	far	near	3	0.0077	[0.0032, 0.0155]	0.619	?
kitti_ignore	u_e	10^{-2}	far	near	4	0.0062	[0.0024, 0.0135]	0.601	?
kitti_ignore	u_e	10^{-2}	near	far	0	0.0013	[0.0001, 0.0055]	0.049	A
kitti_ignore	u_e	10^{-2}	near	far	1	0.0013	[0.0001, 0.0055]	0.069	A
kitti_ignore	u_e	10^{-2}	near	far	2	0.0013	[0.0001, 0.0056]	0.035	A
kitti_ignore	u_e	10^{-2}	near	far	3	0.0032	[0.0008, 0.0085]	0.111	A
kitti_ignore	u_e	10^{-2}	near	far	4	0.0000	[0.0000, 0.0024]	0.057	A
kitti_ignore	u_e	10^{-2}	near	near	0	0.0050	[0.0015, 0.0117]	0.520	?
kitti_ignore	u_e	10^{-2}	near	near	1	0.0057	[0.0020, 0.0128]	0.539	?
kitti_ignore	u_e	10^{-2}	near	near	2	0.0014	[0.0001, 0.0059]	0.466	A
kitti_ignore	u_e	10^{-2}	near	near	3	0.0077	[0.0032, 0.0155]	0.619	?
kitti_ignore	u_e	10^{-2}	near	near	4	0.0048	[0.0016, 0.0114]	0.502	?
kitti_ignore	u_e	0.05	far	far	0	0.0323	[0.0221, 0.0446]	0.422	A
kitti_ignore	u_e	0.05	far	far	1	0.0334	[0.0233, 0.0463]	0.425	A
kitti_ignore	u_e	0.05	far	far	2	0.0242	[0.0159, 0.0352]	0.371	A
kitti_ignore	u_e	0.05	far	far	3	0.0349	[0.0248, 0.0481]	0.419	A
kitti_ignore	u_e	0.05	far	far	4	0.0245	[0.0158, 0.0358]	0.382	A
kitti_ignore	u_e	0.05	far	near	0	0.0229	[0.0146, 0.0342]	0.843	A
kitti_ignore	u_e	0.05	far	near	1	0.0226	[0.0144, 0.0343]	0.842	A
kitti_ignore	u_e	0.05	far	near	2	0.0236	[0.0152, 0.0351]	0.823	A

protocol	ind.	α	sel.	use	seed	$\hat{A}$	HB interval	$\hat{R}$	out.
kitti_ignore	u_e	0.05	far	near	3	0.0245	[0.0159, 0.0365]	0.845	A
kitti_ignore	u_e	0.05	far	near	4	0.0201	[0.0121, 0.0308]	0.819	A
kitti_ignore	u_e	0.05	near	far	0	0.0631	[0.0491, 0.0796]	0.560	?
kitti_ignore	u_e	0.05	near	far	1	0.0664	[0.0522, 0.0834]	0.574	N
kitti_ignore	u_e	0.05	near	far	2	0.0570	[0.0439, 0.0725]	0.565	?
kitti_ignore	u_e	0.05	near	far	3	0.0852	[0.0692, 0.1039]	0.615	N
kitti_ignore	u_e	0.05	near	far	4	0.0598	[0.0459, 0.0757]	0.585	?
kitti_ignore	u_e	0.05	near	near	0	0.0341	[0.0239, 0.0475]	0.898	A
kitti_ignore	u_e	0.05	near	near	1	0.0353	[0.0246, 0.0487]	0.904	A
kitti_ignore	u_e	0.05	near	near	2	0.0351	[0.0246, 0.0484]	0.903	A
kitti_ignore	u_e	0.05	near	near	3	0.0407	[0.0295, 0.0553]	0.921	?
kitti_ignore	u_e	0.05	near	near	4	0.0332	[0.0226, 0.0459]	0.905	A
kitti_ignore	u_e	0.1	far	far	0	0.0695	[0.0549, 0.0867]	0.591	A
kitti_ignore	u_e	0.1	far	far	1	0.0840	[0.0680, 0.1026]	0.630	?
kitti_ignore	u_e	0.1	far	far	2	0.0570	[0.0439, 0.0725]	0.565	A
kitti_ignore	u_e	0.1	far	far	3	0.0759	[0.0604, 0.0934]	0.591	A
kitti_ignore	u_e	0.1	far	far	4	0.0675	[0.0528, 0.0843]	0.611	A
kitti_ignore	u_e	0.1	far	near	0	0.0376	[0.0267, 0.0513]	0.906	A
kitti_ignore	u_e	0.1	far	near	1	0.0406	[0.0292, 0.0550]	0.921	A
kitti_ignore	u_e	0.1	far	near	2	0.0351	[0.0246, 0.0484]	0.903	A
kitti_ignore	u_e	0.1	far	near	3	0.0379	[0.0266, 0.0514]	0.914	A
kitti_ignore	u_e	0.1	far	near	4	0.0348	[0.0243, 0.0483]	0.911	A
kitti_ignore	u_e	0.1	near	far	0	0.1692	[0.1470, 0.1933]	0.817	N
kitti_ignore	u_e	0.1	near	far	1	0.1566	[0.1354, 0.1803]	0.825	N
kitti_ignore	u_e	0.1	near	far	2	0.1596	[0.1377, 0.1831]	0.821	N
kitti_ignore	u_e	0.1	near	far	3	0.1857	[0.1629, 0.2108]	0.824	N
kitti_ignore	u_e	0.1	near	far	4	0.1704	[0.1479, 0.1944]	0.833	N
kitti_ignore	u_e	0.1	near	near	0	0.0790	[0.0631, 0.0973]	0.974	A
kitti_ignore	u_e	0.1	near	near	1	0.0697	[0.0544, 0.0871]	0.977	A
kitti_ignore	u_e	0.1	near	near	2	0.0736	[0.0585, 0.0918]	0.977	A
kitti_ignore	u_e	0.1	near	near	3	0.0863	[0.0698, 0.1057]	0.981	?
kitti_ignore	u_e	0.1	near	near	4	0.0811	[0.0648, 0.0997]	0.977	A
kitti_ignore	u_s	10^{-2}	far	far	0	0.0078	[0.0035, 0.0148]	0.030	?
kitti_ignore	u_s	10^{-2}	far	far	1	0.0013	[0.0001, 0.0055]	0.014	A
kitti_ignore	u_s	10^{-2}	far	far	2	0.0039	[0.0011, 0.0096]	0.030	A
kitti_ignore	u_s	10^{-2}	far	far	3	0.0013	[0.0001, 0.0055]	0.005	A
kitti_ignore	u_s	10^{-2}	far	far	4	0.0055	[0.0019, 0.0123]	0.024	?
kitti_ignore	u_s	10^{-2}	far	near	0	0.0040	[0.0012, 0.0098]	0.150	A
kitti_ignore	u_s	10^{-2}	far	near	1	0.0007	[0.0000, 0.0045]	0.055	A
kitti_ignore	u_s	10^{-2}	far	near	2	0.0060	[0.0023, 0.0126]	0.116	?
kitti_ignore	u_s	10^{-2}	far	near	3	0.0000	[0.0000, 0.0025]	0.042	A
kitti_ignore	u_s	10^{-2}	far	near	4	0.0020	[0.0003, 0.0069]	0.105	A
kitti_ignore	u_s	10^{-2}	near	far	0	0.0091	[0.0044, 0.0165]	0.047	?
kitti_ignore	u_s	10^{-2}	near	far	1	0.0019	[0.0003, 0.0066]	0.018	A
kitti_ignore	u_s	10^{-2}	near	far	2	0.0065	[0.0027, 0.0132]	0.056	?
kitti_ignore	u_s	10^{-2}	near	far	3	0.0200	[0.0126, 0.0300]	0.102	N
kitti_ignore	u_s	10^{-2}	near	far	4	0.0104	[0.0053, 0.0183]	0.040	?
kitti_ignore	u_s	10^{-2}	near	near	0	0.0054	[0.0019, 0.0123]	0.199	?
kitti_ignore	u_s	10^{-2}	near	near	1	0.0014	[0.0001, 0.0058]	0.081	A
kitti_ignore	u_s	10^{-2}	near	near	2	0.0073	[0.0028, 0.0144]	0.202	?
kitti_ignore	u_s	10^{-2}	near	near	3	0.0080	[0.0032, 0.0155]	0.312	?
kitti_ignore	u_s	10^{-2}	near	near	4	0.0024	[0.0004, 0.0076]	0.169	A
kitti_ignore	u_s	0.05	far	far	0	0.0350	[0.0248, 0.0476]	0.164	A
kitti_ignore	u_s	0.05	far	far	1	0.0354	[0.0249, 0.0485]	0.196	A
kitti_ignore	u_s	0.05	far	far	2	0.0278	[0.0186, 0.0398]	0.161	A
kitti_ignore	u_s	0.05	far	far	3	0.0393	[0.0286, 0.0526]	0.180	?
kitti_ignore	u_s	0.05	far	far	4	0.0277	[0.0185, 0.0396]	0.163	A
kitti_ignore	u_s	0.05	far	near	0	0.0142	[0.0078, 0.0238]	0.434	A
kitti_ignore	u_s	0.05	far	near	1	0.0135	[0.0070, 0.0227]	0.436	A
kitti_ignore	u_s	0.05	far	near	2	0.0130	[0.0069, 0.0222]	0.395	A
kitti_ignore	u_s	0.05	far	near	3	0.0117	[0.0060, 0.0208]	0.438	A
kitti_ignore	u_s	0.05	far	near	4	0.0100	[0.0046, 0.0182]	0.394	A
kitti_ignore	u_s	0.05	near	far	0	0.1218	[0.1028, 0.1433]	0.425	N
kitti_ignore	u_s	0.05	near	far	1	0.1172	[0.0984, 0.1383]	0.437	N
kitti_ignore	u_s	0.05	near	far	2	0.1032	[0.0852, 0.1232]	0.398	N
kitti_ignore	u_s	0.05	near	far	3	0.1456	[0.1246, 0.1681]	0.444	N
kitti_ignore	u_s	0.05	near	far	4	0.1115	[0.0931, 0.1323]	0.402	N
kitti_ignore	u_s	0.05	near	near	0	0.0372	[0.0262, 0.0505]	0.719	?
kitti_ignore	u_s	0.05	near	near	1	0.0333	[0.0228, 0.0464]	0.711	A
kitti_ignore	u_s	0.05	near	near	2	0.0329	[0.0229, 0.0461]	0.685	A
kitti_ignore	u_s	0.05	near	near	3	0.0387	[0.0272, 0.0522]	0.740	?
kitti_ignore	u_s	0.05	near	near	4	0.0274	[0.0181, 0.0396]	0.692	A
kitti_ignore	u_s	0.1	far	far	0	0.0702	[0.0554, 0.0874]	0.302	A
kitti_ignore	u_s	0.1	far	far	1	0.0732	[0.0580, 0.0906]	0.329	A
kitti_ignore	u_s	0.1	far	far	2	0.0631	[0.0491, 0.0798]	0.305	A
kitti_ignore	u_s	0.1	far	far	3	0.1000	[0.0827, 0.1200]	0.353	?
kitti_ignore	u_s	0.1	far	far	4	0.0742	[0.0587, 0.0915]	0.320	A
kitti_ignore	u_s	0.1	far	near	0	0.0233	[0.0151, 0.0350]	0.604	A
kitti_ignore	u_s	0.1	far	near	1	0.0212	[0.0128, 0.0319]	0.605	A
kitti_ignore	u_s	0.1	far	near	2	0.0242	[0.0157, 0.0359]	0.576	A
kitti_ignore	u_s	0.1	far	near	3	0.0290	[0.0193, 0.0413]	0.650	A
kitti_ignore	u_s	0.1	far	near	4	0.0212	[0.0132, 0.0324]	0.610	A
kitti_ignore	u_s	0.1	near	far	0	0.2401	[0.2143, 0.2667]	0.632	N
kitti_ignore	u_s	0.1	near	far	1	0.2215	[0.1964, 0.2476]	0.639	N
kitti_ignore	u_s	0.1	near	far	2	0.2376	[0.2118, 0.2644]	0.635	N
kitti_ignore	u_s	0.1	near	far	3	0.2672	[0.2408, 0.2950]	0.674	N
kitti_ignore	u_s	0.1	near	far	4	0.2131	[0.1887, 0.2393]	0.600	N

protocol	ind.	α	sel.	use	seed	$\hat{A}$	HB interval	$\hat{R}$	out.
kitti_ignore	u_s	0.1	near	near	0	0.0830	[0.0667, 0.1016]	0.866	?
kitti_ignore	u_s	0.1	near	near	1	0.0720	[0.0568, 0.0901]	0.859	A
kitti_ignore	u_s	0.1	near	near	2	0.0801	[0.0639, 0.0983]	0.868	A
kitti_ignore	u_s	0.1	near	near	3	0.0895	[0.0723, 0.1087]	0.883	?
kitti_ignore	u_s	0.1	near	near	4	0.0642	[0.0496, 0.0811]	0.852	A

3.3 Change of dataset and detector, KITTI and nuScenes (m4, one split; m6 part B, five splits)

Sources: `results/division_by_frame/dataset_change_one_split/shift_matrix.csv` (one split by frame, 50 % selection, 50 % test part; listed as seed 0) and `results/division_by_frame/kitti_rule_dataset_change_weather/B_shift_seeds_protocol.csv` (seeds 0 to 4, labels as run and with the KITTI rule).

Table 6: Change of dataset and detector (m4): outcome of the recorded split.

ind.	α	source	target	split
u_e	10^{-2}	kitti	kitti	?
u_e	10^{-2}	kitti	nuscenes	?
u_e	10^{-2}	nuscenes	kitti	A
u_e	10^{-2}	nuscenes	nuscenes	A
u_e	10^{-3}	kitti	kitti	-
u_e	10^{-3}	kitti	nuscenes	-
u_e	10^{-3}	nuscenes	kitti	-
u_e	10^{-3}	nuscenes	nuscenes	-
u_e	10^{-4}	kitti	kitti	-
u_e	10^{-4}	kitti	nuscenes	-
u_e	10^{-4}	nuscenes	kitti	-
u_e	10^{-4}	nuscenes	nuscenes	-
u_e	0.05	kitti	kitti	A
u_e	0.05	kitti	nuscenes	A
u_e	0.05	nuscenes	kitti	?
u_e	0.05	nuscenes	nuscenes	?
u_e	0.1	kitti	kitti	A
u_e	0.1	kitti	nuscenes	A
u_e	0.1	nuscenes	kitti	?
u_e	0.1	nuscenes	nuscenes	?
u_s	10^{-2}	kitti	kitti	?
u_s	10^{-2}	kitti	nuscenes	A
u_s	10^{-2}	nuscenes	kitti	N
u_s	10^{-2}	nuscenes	nuscenes	A
u_s	10^{-3}	kitti	kitti	-
u_s	10^{-3}	kitti	nuscenes	-
u_s	10^{-3}	nuscenes	kitti	-
u_s	10^{-3}	nuscenes	nuscenes	-
u_s	10^{-4}	kitti	kitti	-
u_s	10^{-4}	kitti	nuscenes	-
u_s	10^{-4}	nuscenes	kitti	-
u_s	10^{-4}	nuscenes	nuscenes	-
u_s	0.05	kitti	kitti	A
u_s	0.05	kitti	nuscenes	A
u_s	0.05	nuscenes	kitti	N
u_s	0.05	nuscenes	nuscenes	?
u_s	0.1	kitti	kitti	A
u_s	0.1	kitti	nuscenes	A
u_s	0.1	nuscenes	kitti	N
u_s	0.1	nuscenes	nuscenes	A

Table 7: Change of dataset and detector (m4): values (n: test frames).

ind.	α	source	target	split	τ	$\hat{A}$	HB interval	$\hat{R}$	n	out.
u_e	10^{-2}	kitti	kitti	0	0.1100	0.0079	[0.0036, 0.0144]	0.285	1857	?
u_e	10^{-2}	kitti	nuscenes	0	0.1100	0.0059	[0.0028, 0.0103]	0.097	2850	?
u_e	10^{-2}	nuscenes	kitti	0	0.1000	0.0044	[0.0015, 0.0100]	0.235	1857	A
u_e	10^{-2}	nuscenes	nuscenes	0	0.1000	0.0039	[0.0017, 0.0080]	0.079	2850	A
u_e	0.05	kitti	kitti	0	0.2200	0.0327	[0.0234, 0.0439]	0.625	1857	A
u_e	0.05	kitti	nuscenes	0	0.2200	0.0303	[0.0231, 0.0390]	0.303	2850	A
u_e	0.05	nuscenes	kitti	0	0.2600	0.0439	[0.0332, 0.0567]	0.697	1857	?
u_e	0.05	nuscenes	nuscenes	0	0.2600	0.0424	[0.0337, 0.0522]	0.379	2850	?
u_e	0.1	kitti	kitti	0	0.3700	0.0777	[0.0636, 0.0940]	0.823	1857	A
u_e	0.1	kitti	nuscenes	0	0.3700	0.0713	[0.0603, 0.0838]	0.567	2850	A
u_e	0.1	nuscenes	kitti	0	0.4400	0.1092	[0.0923, 0.1275]	0.877	1857	?
u_e	0.1	nuscenes	nuscenes	0	0.4400	0.0876	[0.0753, 0.1010]	0.675	2850	?
u_s	10^{-2}	kitti	kitti	0	0.0800	0.0075	[0.0036, 0.0138]	0.054	1857	?
u_s	10^{-2}	kitti	nuscenes	0	0.0800	0.0000	[0.0000, 0.0013]	0.000	2850	A
u_s	10^{-2}	nuscenes	kitti	0	0.1200	0.0627	[0.0499, 0.0775]	0.458	1857	N
u_s	10^{-2}	nuscenes	nuscenes	0	0.1200	0.0026	[0.0008, 0.0061]	0.008	2850	A

ind.	α	source	target	split	τ	$\hat{A}$	HB interval	$\hat{R}$	n	out.
u_s	0.05	kitti	kitti	0	0.1000	0.0284	[0.0197, 0.0390]	0.225	1857	A
u_s	0.05	kitti	nuscenes	0	0.1000	0.0004	[0.0000, 0.0023]	0.001	2850	A
u_s	0.05	nuscenes	kitti	0	0.1700	0.2019	[0.1799, 0.2248]	0.812	1857	N
u_s	0.05	nuscenes	nuscenes	0	0.1700	0.0445	[0.0356, 0.0545]	0.153	2850	?
u_s	0.1	kitti	kitti	0	0.1200	0.0627	[0.0499, 0.0775]	0.458	1857	A
u_s	0.1	kitti	nuscenes	0	0.1200	0.0026	[0.0008, 0.0061]	0.008	2850	A
u_s	0.1	nuscenes	kitti	0	0.2000	0.2877	[0.2630, 0.3136]	0.894	1857	N
u_s	0.1	nuscenes	nuscenes	0	0.2000	0.0779	[0.0662, 0.0905]	0.330	2850	A

Table 8: Change of dataset and detector (m6 B): outcome per split.

protocol	ind.	α	source	target	seeds 0–4
as_run	u_e	10^{-2}	kitti	kitti	?A???
as_run	u_e	10^{-2}	kitti	nuscenes	?AAA?
as_run	u_e	10^{-2}	nuscenes	kitti	AA???
as_run	u_e	10^{-2}	nuscenes	nuscenes	AAAA?
as_run	u_e	10^{-3}	kitti	kitti	---
as_run	u_e	10^{-3}	kitti	nuscenes	---
as_run	u_e	10^{-3}	nuscenes	kitti	---
as_run	u_e	10^{-3}	nuscenes	nuscenes	---
as_run	u_e	10^{-4}	kitti	kitti	---
as_run	u_e	10^{-4}	kitti	nuscenes	---
as_run	u_e	10^{-4}	nuscenes	kitti	---
as_run	u_e	10^{-4}	nuscenes	nuscenes	---
as_run	u_e	0.05	kitti	kitti	AAA?A
as_run	u_e	0.05	kitti	nuscenes	AAAAA
as_run	u_e	0.05	nuscenes	kitti	?????
as_run	u_e	0.05	nuscenes	nuscenes	?A?AA
as_run	u_e	0.1	kitti	kitti	AAA?A
as_run	u_e	0.1	kitti	nuscenes	AAAAA
as_run	u_e	0.1	nuscenes	kitti	?????
as_run	u_e	0.1	nuscenes	nuscenes	?A?AA
as_run	u_s	10^{-2}	kitti	kitti	?A?A?
as_run	u_s	10^{-2}	kitti	nuscenes	AAAAA
as_run	u_s	10^{-2}	nuscenes	kitti	NNNNN
as_run	u_s	10^{-2}	nuscenes	nuscenes	AAA??
as_run	u_s	10^{-3}	kitti	kitti	---
as_run	u_s	10^{-3}	kitti	nuscenes	---
as_run	u_s	10^{-3}	nuscenes	kitti	---
as_run	u_s	10^{-3}	nuscenes	nuscenes	---
as_run	u_s	10^{-4}	kitti	kitti	---
as_run	u_s	10^{-4}	kitti	nuscenes	---
as_run	u_s	10^{-4}	nuscenes	kitti	---
as_run	u_s	10^{-4}	nuscenes	nuscenes	---
as_run	u_s	0.05	kitti	kitti	AAA?A
as_run	u_s	0.05	kitti	nuscenes	AAAAA
as_run	u_s	0.05	nuscenes	kitti	NNNNN
as_run	u_s	0.05	nuscenes	nuscenes	?AAAA
as_run	u_s	0.1	kitti	kitti	AAA?A
as_run	u_s	0.1	kitti	nuscenes	AAAAA
as_run	u_s	0.1	nuscenes	kitti	NNNNN
as_run	u_s	0.1	nuscenes	nuscenes	AAAAA
kitti_ignore	u_e	10^{-2}	kitti	kitti	??A??
kitti_ignore	u_e	10^{-2}	kitti	nuscenes	?????
kitti_ignore	u_e	10^{-2}	nuscenes	kitti	AAAAA
kitti_ignore	u_e	10^{-2}	nuscenes	nuscenes	AAAA?
kitti_ignore	u_e	10^{-3}	kitti	kitti	---
kitti_ignore	u_e	10^{-3}	kitti	nuscenes	---
kitti_ignore	u_e	10^{-3}	nuscenes	kitti	---
kitti_ignore	u_e	10^{-3}	nuscenes	nuscenes	---
kitti_ignore	u_e	10^{-4}	kitti	kitti	---
kitti_ignore	u_e	10^{-4}	kitti	nuscenes	---
kitti_ignore	u_e	10^{-4}	nuscenes	kitti	---
kitti_ignore	u_e	10^{-4}	nuscenes	nuscenes	---
kitti_ignore	u_e	0.05	kitti	kitti	AAA?A
kitti_ignore	u_e	0.05	kitti	nuscenes	?????
kitti_ignore	u_e	0.05	nuscenes	kitti	AAAAA
kitti_ignore	u_e	0.05	nuscenes	nuscenes	?A?AA
kitti_ignore	u_e	0.1	kitti	kitti	?AA?A
kitti_ignore	u_e	0.1	kitti	nuscenes	?????
kitti_ignore	u_e	0.1	nuscenes	kitti	AAAAA
kitti_ignore	u_e	0.1	nuscenes	nuscenes	?A?AA
kitti_ignore	u_s	10^{-2}	kitti	kitti	?A?AA
kitti_ignore	u_s	10^{-2}	kitti	nuscenes	AAAAA
kitti_ignore	u_s	10^{-2}	nuscenes	kitti	NNNNN
kitti_ignore	u_s	10^{-2}	nuscenes	nuscenes	AAA??
kitti_ignore	u_s	10^{-3}	kitti	kitti	---
kitti_ignore	u_s	10^{-3}	kitti	nuscenes	---
kitti_ignore	u_s	10^{-3}	nuscenes	kitti	---
kitti_ignore	u_s	10^{-3}	nuscenes	nuscenes	---
kitti_ignore	u_s	10^{-4}	kitti	kitti	---

protocol	ind.	α	source	target	seeds 0–4
kitti_ignore	u_s	10^{-4}	kitti	nuscenes	---
kitti_ignore	u_s	10^{-4}	nuscenes	kitti	---
kitti_ignore	u_s	10^{-4}	nuscenes	nuscenes	---
kitti_ignore	u_s	0.05	kitti	kitti	AAA?A
kitti_ignore	u_s	0.05	kitti	nuscenes	AAAAA
kitti_ignore	u_s	0.05	nuscenes	kitti	NNNNN
kitti_ignore	u_s	0.05	nuscenes	nuscenes	?AAAA
kitti_ignore	u_s	0.1	kitti	kitti	AAA?A
kitti_ignore	u_s	0.1	kitti	nuscenes	AAAAA
kitti_ignore	u_s	0.1	nuscenes	kitti	NNNNN
kitti_ignore	u_s	0.1	nuscenes	nuscenes	AAAAA

Table 9: Change of dataset and detector (m6 B): values per split.

protocol	ind.	α	source	target	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	out.
as_run	u_e	10^{-2}	kitti	kitti	0	0.1100	0.0079	[0.0036, 0.0144]	0.285	?
as_run	u_e	10^{-2}	kitti	kitti	1	0.0900	0.0016	[0.0002, 0.0055]	0.188	A
as_run	u_e	10^{-2}	kitti	kitti	2	0.1000	0.0051	[0.0019, 0.0109]	0.232	?
as_run	u_e	10^{-2}	kitti	kitti	3	0.1000	0.0047	[0.0016, 0.0102]	0.240	?
as_run	u_e	10^{-2}	kitti	kitti	4	0.1100	0.0078	[0.0036, 0.0145]	0.285	?
as_run	u_e	10^{-2}	kitti	nuscenes	0	0.1100	0.0059	[0.0028, 0.0103]	0.097	?
as_run	u_e	10^{-2}	kitti	nuscenes	1	0.0900	0.0024	[0.0007, 0.0057]	0.066	A
as_run	u_e	10^{-2}	kitti	nuscenes	2	0.1000	0.0044	[0.0019, 0.0086]	0.087	A
as_run	u_e	10^{-2}	kitti	nuscenes	3	0.1000	0.0039	[0.0017, 0.0080]	0.082	A
as_run	u_e	10^{-2}	kitti	nuscenes	4	0.1100	0.0067	[0.0036, 0.0117]	0.104	?
as_run	u_e	10^{-2}	nuscenes	kitti	0	0.1000	0.0044	[0.0015, 0.0100]	0.235	A
as_run	u_e	10^{-2}	nuscenes	kitti	1	0.1000	0.0038	[0.0012, 0.0091]	0.236	A
as_run	u_e	10^{-2}	nuscenes	kitti	2	0.1000	0.0051	[0.0019, 0.0109]	0.232	?
as_run	u_e	10^{-2}	nuscenes	kitti	3	0.1000	0.0047	[0.0016, 0.0102]	0.240	?
as_run	u_e	10^{-2}	nuscenes	kitti	4	0.1100	0.0078	[0.0036, 0.0145]	0.285	?
as_run	u_e	10^{-2}	nuscenes	nuscenes	0	0.1000	0.0039	[0.0017, 0.0080]	0.079	A
as_run	u_e	10^{-2}	nuscenes	nuscenes	1	0.1000	0.0031	[0.0010, 0.0067]	0.085	A
as_run	u_e	10^{-2}	nuscenes	nuscenes	2	0.1000	0.0044	[0.0019, 0.0086]	0.087	A
as_run	u_e	10^{-2}	nuscenes	nuscenes	3	0.1000	0.0039	[0.0017, 0.0080]	0.082	A
as_run	u_e	10^{-2}	nuscenes	nuscenes	4	0.1100	0.0067	[0.0036, 0.0117]	0.104	?
as_run	u_e	0.05	kitti	kitti	0	0.2200	0.0327	[0.0234, 0.0439]	0.625	A
as_run	u_e	0.05	kitti	kitti	1	0.2300	0.0365	[0.0266, 0.0482]	0.639	A
as_run	u_e	0.05	kitti	kitti	2	0.2300	0.0368	[0.0272, 0.0490]	0.636	A
as_run	u_e	0.05	kitti	kitti	3	0.2400	0.0431	[0.0327, 0.0561]	0.667	?
as_run	u_e	0.05	kitti	kitti	4	0.2300	0.0366	[0.0267, 0.0483]	0.638	A
as_run	u_e	0.05	kitti	nuscenes	0	0.2200	0.0303	[0.0231, 0.0390]	0.303	A
as_run	u_e	0.05	kitti	nuscenes	1	0.2300	0.0277	[0.0207, 0.0360]	0.332	A
as_run	u_e	0.05	kitti	nuscenes	2	0.2300	0.0318	[0.0245, 0.0408]	0.330	A
as_run	u_e	0.05	kitti	nuscenes	3	0.2400	0.0343	[0.0265, 0.0433]	0.341	A
as_run	u_e	0.05	kitti	nuscenes	4	0.2300	0.0292	[0.0220, 0.0376]	0.327	A
as_run	u_e	0.05	nuscenes	kitti	0	0.2600	0.0439	[0.0332, 0.0567]	0.697	?
as_run	u_e	0.05	nuscenes	kitti	1	0.2500	0.0417	[0.0313, 0.0543]	0.674	?
as_run	u_e	0.05	nuscenes	kitti	2	0.2600	0.0441	[0.0333, 0.0569]	0.689	?
as_run	u_e	0.05	nuscenes	kitti	3	0.2500	0.0444	[0.0336, 0.0573]	0.686	?
as_run	u_e	0.05	nuscenes	kitti	4	0.2500	0.0418	[0.0314, 0.0544]	0.672	?
as_run	u_e	0.05	nuscenes	nuscenes	0	0.2600	0.0424	[0.0337, 0.0522]	0.379	?
as_run	u_e	0.05	nuscenes	nuscenes	1	0.2500	0.0330	[0.0254, 0.0419]	0.368	A
as_run	u_e	0.05	nuscenes	nuscenes	2	0.2600	0.0416	[0.0329, 0.0513]	0.387	?
as_run	u_e	0.05	nuscenes	nuscenes	3	0.2500	0.0351	[0.0274, 0.0444]	0.361	A
as_run	u_e	0.05	nuscenes	nuscenes	4	0.2500	0.0355	[0.0276, 0.0447]	0.365	A
as_run	u_e	0.1	kitti	kitti	0	0.3700	0.0777	[0.0636, 0.0940]	0.823	A
as_run	u_e	0.1	kitti	kitti	1	0.3700	0.0776	[0.0636, 0.0940]	0.822	A
as_run	u_e	0.1	kitti	kitti	2	0.3700	0.0760	[0.0618, 0.0919]	0.826	A
as_run	u_e	0.1	kitti	kitti	3	0.3900	0.0929	[0.0774, 0.1102]	0.837	?
as_run	u_e	0.1	kitti	kitti	4	0.3700	0.0761	[0.0622, 0.0924]	0.824	A
as_run	u_e	0.1	kitti	nuscenes	0	0.3700	0.0713	[0.0603, 0.0838]	0.567	A
as_run	u_e	0.1	kitti	nuscenes	1	0.3700	0.0677	[0.0569, 0.0799]	0.580	A
as_run	u_e	0.1	kitti	nuscenes	2	0.3700	0.0698	[0.0587, 0.0819]	0.576	A
as_run	u_e	0.1	kitti	nuscenes	3	0.3900	0.0745	[0.0632, 0.0871]	0.605	A
as_run	u_e	0.1	kitti	nuscenes	4	0.3700	0.0659	[0.0554, 0.0781]	0.574	A
as_run	u_e	0.1	nuscenes	kitti	0	0.4400	0.1092	[0.0923, 0.1275]	0.877	?
as_run	u_e	0.1	nuscenes	kitti	1	0.4300	0.1028	[0.0864, 0.1207]	0.870	?
as_run	u_e	0.1	nuscenes	kitti	2	0.4400	0.1032	[0.0866, 0.1210]	0.881	?
as_run	u_e	0.1	nuscenes	kitti	3	0.4400	0.1101	[0.0933, 0.1286]	0.877	?
as_run	u_e	0.1	nuscenes	kitti	4	0.4200	0.0978	[0.0820, 0.1156]	0.867	?
as_run	u_e	0.1	nuscenes	nuscenes	0	0.4400	0.0876	[0.0753, 0.1010]	0.675	?
as_run	u_e	0.1	nuscenes	nuscenes	1	0.4300	0.0817	[0.0700, 0.0950]	0.668	A
as_run	u_e	0.1	nuscenes	nuscenes	2	0.4400	0.0869	[0.0747, 0.1004]	0.681	?
as_run	u_e	0.1	nuscenes	nuscenes	3	0.4400	0.0862	[0.0740, 0.0994]	0.680	A
as_run	u_e	0.1	nuscenes	nuscenes	4	0.4200	0.0771	[0.0655, 0.0898]	0.652	A
as_run	u_s	10^{-2}	kitti	kitti	0	0.0800	0.0075	[0.0036, 0.0138]	0.054	?
as_run	u_s	10^{-2}	kitti	kitti	1	0.0700	0.0000	[0.0000, 0.0020]	0.017	A
as_run	u_s	10^{-2}	kitti	kitti	2	0.0800	0.0059	[0.0025, 0.0117]	0.052	?
as_run	u_s	10^{-2}	kitti	kitti	3	0.0700	0.0011	[0.0001, 0.0046]	0.016	A
as_run	u_s	10^{-2}	kitti	kitti	4	0.0800	0.0057	[0.0022, 0.0116]	0.057	?
as_run	u_s	10^{-2}	kitti	nuscenes	0	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
as_run	u_s	10^{-2}	kitti	nuscenes	1	0.0700	0.0000	[0.0000, 0.0013]	0.000	A

protocol	ind.	α	source	target	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	out.
as_run	u_s	10^{-2}	kitti	nuscenes	2	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
as_run	u_s	10^{-2}	kitti	nuscenes	3	0.0700	0.0000	[0.0000, 0.0013]	0.000	A
as_run	u_s	10^{-2}	kitti	nuscenes	4	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
as_run	u_s	10^{-2}	nuscenes	kitti	0	0.1200	0.0627	[0.0499, 0.0775]	0.458	N
as_run	u_s	10^{-2}	nuscenes	kitti	1	0.1200	0.0628	[0.0500, 0.0776]	0.457	N
as_run	u_s	10^{-2}	nuscenes	kitti	2	0.1200	0.0622	[0.0496, 0.0772]	0.453	N
as_run	u_s	10^{-2}	nuscenes	kitti	3	0.1300	0.0929	[0.0774, 0.1102]	0.560	N
as_run	u_s	10^{-2}	nuscenes	kitti	4	0.1300	0.0866	[0.0716, 0.1034]	0.549	N
as_run	u_s	10^{-2}	nuscenes	nuscenes	0	0.1200	0.0026	[0.0008, 0.0061]	0.008	A
as_run	u_s	10^{-2}	nuscenes	nuscenes	1	0.1200	0.0009	[0.0001, 0.0034]	0.008	A
as_run	u_s	10^{-2}	nuscenes	nuscenes	2	0.1200	0.0025	[0.0008, 0.0057]	0.008	A
as_run	u_s	10^{-2}	nuscenes	nuscenes	3	0.1300	0.0058	[0.0028, 0.0103]	0.018	?
as_run	u_s	10^{-2}	nuscenes	nuscenes	4	0.1300	0.0067	[0.0036, 0.0113]	0.017	?
as_run	u_s	0.05	kitti	kitti	0	0.1000	0.0284	[0.0197, 0.0390]	0.225	A
as_run	u_s	0.05	kitti	kitti	1	0.1000	0.0249	[0.0171, 0.0353]	0.229	A
as_run	u_s	0.05	kitti	kitti	2	0.1000	0.0262	[0.0180, 0.0366]	0.226	A
as_run	u_s	0.05	kitti	kitti	3	0.1100	0.0459	[0.0351, 0.0591]	0.347	?
as_run	u_s	0.05	kitti	kitti	4	0.1000	0.0292	[0.0207, 0.0403]	0.233	A
as_run	u_s	0.05	kitti	nuscenes	0	0.1000	0.0004	[0.0000, 0.0023]	0.001	A
as_run	u_s	0.05	kitti	nuscenes	1	0.1000	0.0000	[0.0000, 0.0013]	0.001	A
as_run	u_s	0.05	kitti	nuscenes	2	0.1000	0.0000	[0.0000, 0.0013]	0.001	A
as_run	u_s	0.05	kitti	nuscenes	3	0.1100	0.0011	[0.0001, 0.0036]	0.003	A
as_run	u_s	0.05	kitti	nuscenes	4	0.1000	0.0007	[0.0001, 0.0030]	0.001	A
as_run	u_s	0.05	nuscenes	kitti	0	0.1700	0.2019	[0.1799, 0.2248]	0.812	N
as_run	u_s	0.05	nuscenes	kitti	1	0.1600	0.1720	[0.1517, 0.1941]	0.771	N
as_run	u_s	0.05	nuscenes	kitti	2	0.1600	0.1678	[0.1475, 0.1895]	0.770	N
as_run	u_s	0.05	nuscenes	kitti	3	0.1600	0.1779	[0.1572, 0.2000]	0.778	N
as_run	u_s	0.05	nuscenes	kitti	4	0.1600	0.1686	[0.1483, 0.1903]	0.765	N
as_run	u_s	0.05	nuscenes	nuscenes	0	0.1700	0.0445	[0.0356, 0.0545]	0.153	?
as_run	u_s	0.05	nuscenes	nuscenes	1	0.1600	0.0252	[0.0186, 0.0332]	0.111	A
as_run	u_s	0.05	nuscenes	nuscenes	2	0.1600	0.0291	[0.0220, 0.0376]	0.112	A
as_run	u_s	0.05	nuscenes	nuscenes	3	0.1600	0.0247	[0.0182, 0.0327]	0.110	A
as_run	u_s	0.05	nuscenes	nuscenes	4	0.1600	0.0300	[0.0229, 0.0388]	0.106	A
as_run	u_s	0.1	kitti	kitti	0	0.1200	0.0627	[0.0499, 0.0775]	0.458	A
as_run	u_s	0.1	kitti	kitti	1	0.1200	0.0628	[0.0500, 0.0776]	0.457	A
as_run	u_s	0.1	kitti	kitti	2	0.1200	0.0622	[0.0496, 0.0772]	0.453	A
as_run	u_s	0.1	kitti	kitti	3	0.1300	0.0929	[0.0774, 0.1102]	0.560	?
as_run	u_s	0.1	kitti	kitti	4	0.1200	0.0637	[0.0510, 0.0789]	0.452	A
as_run	u_s	0.1	kitti	nuscenes	0	0.1200	0.0026	[0.0008, 0.0061]	0.008	A
as_run	u_s	0.1	kitti	nuscenes	1	0.1200	0.0009	[0.0001, 0.0034]	0.008	A
as_run	u_s	0.1	kitti	nuscenes	2	0.1200	0.0025	[0.0008, 0.0057]	0.008	A
as_run	u_s	0.1	kitti	nuscenes	3	0.1300	0.0058	[0.0028, 0.0103]	0.018	A
as_run	u_s	0.1	kitti	nuscenes	4	0.1200	0.0026	[0.0008, 0.0062]	0.008	A
as_run	u_s	0.1	nuscenes	kitti	0	0.2000	0.2877	[0.2630, 0.3136]	0.894	N
as_run	u_s	0.1	nuscenes	kitti	1	0.2000	0.2901	[0.2653, 0.3160]	0.893	N
as_run	u_s	0.1	nuscenes	kitti	2	0.2000	0.2847	[0.2597, 0.3102]	0.895	N
as_run	u_s	0.1	nuscenes	kitti	3	0.2000	0.2985	[0.2733, 0.3244]	0.897	N
as_run	u_s	0.1	nuscenes	kitti	4	0.2000	0.2846	[0.2598, 0.3102]	0.890	N
as_run	u_s	0.1	nuscenes	nuscenes	0	0.2000	0.0779	[0.0662, 0.0905]	0.330	A
as_run	u_s	0.1	nuscenes	nuscenes	1	0.2000	0.0765	[0.0651, 0.0893]	0.336	A
as_run	u_s	0.1	nuscenes	nuscenes	2	0.2000	0.0751	[0.0639, 0.0879]	0.343	A
as_run	u_s	0.1	nuscenes	nuscenes	3	0.2000	0.0724	[0.0612, 0.0848]	0.333	A
as_run	u_s	0.1	nuscenes	nuscenes	4	0.2000	0.0753	[0.0638, 0.0879]	0.334	A
kitti_ignore	u_e	10^{-2}	kitti	kitti	0	0.1400	0.0050	[0.0019, 0.0109]	0.418	?
kitti_ignore	u_e	10^{-2}	kitti	kitti	1	0.1400	0.0055	[0.0022, 0.0116]	0.412	?
kitti_ignore	u_e	10^{-2}	kitti	kitti	2	0.1200	0.0019	[0.0003, 0.0061]	0.328	A
kitti_ignore	u_e	10^{-2}	kitti	kitti	3	0.1400	0.0055	[0.0022, 0.0116]	0.424	?
kitti_ignore	u_e	10^{-2}	kitti	kitti	4	0.1500	0.0064	[0.0026, 0.0124]	0.448	?
kitti_ignore	u_e	10^{-2}	kitti	nuscenes	0	0.1400	0.0113	[0.0071, 0.0172]	0.153	?
kitti_ignore	u_e	10^{-2}	kitti	nuscenes	1	0.1400	0.0080	[0.0044, 0.0130]	0.162	?
kitti_ignore	u_e	10^{-2}	kitti	nuscenes	2	0.1200	0.0073	[0.0039, 0.0122]	0.124	?
kitti_ignore	u_e	10^{-2}	kitti	nuscenes	3	0.1400	0.0101	[0.0060, 0.0155]	0.159	?
kitti_ignore	u_e	10^{-2}	kitti	nuscenes	4	0.1500	0.0126	[0.0080, 0.0186]	0.181	?
kitti_ignore	u_e	10^{-2}	nuscenes	kitti	0	0.1000	0.0023	[0.0004, 0.0067]	0.235	A
kitti_ignore	u_e	10^{-2}	nuscenes	kitti	1	0.1000	0.0024	[0.0005, 0.0070]	0.236	A
kitti_ignore	u_e	10^{-2}	nuscenes	kitti	2	0.1000	0.0011	[0.0001, 0.0046]	0.232	A
kitti_ignore	u_e	10^{-2}	nuscenes	kitti	3	0.1000	0.0024	[0.0005, 0.0070]	0.240	A
kitti_ignore	u_e	10^{-2}	nuscenes	kitti	4	0.1100	0.0036	[0.0010, 0.0087]	0.285	A
kitti_ignore	u_e	10^{-2}	nuscenes	nuscenes	0	0.1000	0.0039	[0.0017, 0.0080]	0.079	A
kitti_ignore	u_e	10^{-2}	nuscenes	nuscenes	1	0.1000	0.0031	[0.0010, 0.0067]	0.085	A
kitti_ignore	u_e	10^{-2}	nuscenes	nuscenes	2	0.1000	0.0044	[0.0019, 0.0086]	0.087	A
kitti_ignore	u_e	10^{-2}	nuscenes	nuscenes	3	0.1000	0.0039	[0.0017, 0.0080]	0.082	A
kitti_ignore	u_e	10^{-2}	nuscenes	nuscenes	4	0.1100	0.0067	[0.0036, 0.0117]	0.104	?
kitti_ignore	u_e	0.05	kitti	kitti	0	0.3100	0.0351	[0.0253, 0.0465]	0.766	A
kitti_ignore	u_e	0.05	kitti	kitti	1	0.3100	0.0351	[0.0253, 0.0465]	0.760	A
kitti_ignore	u_e	0.05	kitti	kitti	2	0.3100	0.0343	[0.0249, 0.0460]	0.766	A
kitti_ignore	u_e	0.05	kitti	kitti	3	0.3200	0.0395	[0.0295, 0.0520]	0.779	?
kitti_ignore	u_e	0.05	kitti	kitti	4	0.3000	0.0308	[0.0217, 0.0417]	0.749	A
kitti_ignore	u_e	0.05	kitti	nuscenes	0	0.3100	0.0566	[0.0468, 0.0679]	0.468	?
kitti_ignore	u_e	0.05	kitti	nuscenes	1	0.3100	0.0538	[0.0440, 0.0647]	0.479	?
kitti_ignore	u_e	0.05	kitti	nuscenes	2	0.3100	0.0547	[0.0451, 0.0659]	0.476	?
kitti_ignore	u_e	0.05	kitti	nuscenes	3	0.3200	0.0586	[0.0486, 0.0701]	0.490	?
kitti_ignore	u_e	0.05	kitti	nuscenes	4	0.3000	0.0495	[0.0402, 0.0601]	0.456	?
kitti_ignore	u_e	0.05	nuscenes	kitti	0	0.2600	0.0262	[0.0180, 0.0366]	0.697	A
kitti_ignore	u_e	0.05	nuscenes	kitti	1	0.2500	0.0228	[0.0153, 0.0329]	0.674	A
kitti_ignore	u_e	0.05	nuscenes	kitti	2	0.2600	0.0263	[0.0180, 0.0367]	0.689	A

protocol	ind.	α	source	target	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	out.
kitti_ignore	u_e	0.05	nuscenes	kitti	3	0.2500	0.0260	[0.0180, 0.0366]	0.686	A
kitti_ignore	u_e	0.05	nuscenes	kitti	4	0.2500	0.0222	[0.0149, 0.0323]	0.672	A
kitti_ignore	u_e	0.05	nuscenes	nuscenes	0	0.2600	0.0424	[0.0337, 0.0522]	0.379	?
kitti_ignore	u_e	0.05	nuscenes	nuscenes	1	0.2500	0.0330	[0.0254, 0.0419]	0.368	A
kitti_ignore	u_e	0.05	nuscenes	nuscenes	2	0.2600	0.0416	[0.0329, 0.0513]	0.387	?
kitti_ignore	u_e	0.05	nuscenes	nuscenes	3	0.2500	0.0351	[0.0274, 0.0444]	0.361	A
kitti_ignore	u_e	0.05	nuscenes	nuscenes	4	0.2500	0.0355	[0.0276, 0.0447]	0.365	A
kitti_ignore	u_e	0.1	kitti	kitti	0	0.4700	0.0862	[0.0712, 0.1030]	0.897	?
kitti_ignore	u_e	0.1	kitti	kitti	1	0.4500	0.0736	[0.0598, 0.0895]	0.884	A
kitti_ignore	u_e	0.1	kitti	kitti	2	0.4500	0.0737	[0.0600, 0.0897]	0.887	A
kitti_ignore	u_e	0.1	kitti	kitti	3	0.4700	0.0855	[0.0707, 0.1024]	0.898	?
kitti_ignore	u_e	0.1	kitti	kitti	4	0.4600	0.0792	[0.0648, 0.0955]	0.893	A
kitti_ignore	u_e	0.1	kitti	nuscenes	0	0.4700	0.0984	[0.0855, 0.1125]	0.714	?
kitti_ignore	u_e	0.1	kitti	nuscenes	1	0.4500	0.0871	[0.0749, 0.1006]	0.696	?
kitti_ignore	u_e	0.1	kitti	nuscenes	2	0.4500	0.0901	[0.0777, 0.1037]	0.695	?
kitti_ignore	u_e	0.1	kitti	nuscenes	3	0.4700	0.0967	[0.0838, 0.1106]	0.722	?
kitti_ignore	u_e	0.1	kitti	nuscenes	4	0.4600	0.0877	[0.0756, 0.1014]	0.708	?
kitti_ignore	u_e	0.1	nuscenes	kitti	0	0.4400	0.0775	[0.0633, 0.0936]	0.877	A
kitti_ignore	u_e	0.1	nuscenes	kitti	1	0.4300	0.0690	[0.0554, 0.0842]	0.870	A
kitti_ignore	u_e	0.1	nuscenes	kitti	2	0.4400	0.0709	[0.0575, 0.0868]	0.881	A
kitti_ignore	u_e	0.1	nuscenes	kitti	3	0.4400	0.0762	[0.0623, 0.0925]	0.877	A
kitti_ignore	u_e	0.1	nuscenes	kitti	4	0.4200	0.0659	[0.0526, 0.0808]	0.867	A
kitti_ignore	u_e	0.1	nuscenes	nuscenes	0	0.4400	0.0876	[0.0753, 0.1010]	0.675	?
kitti_ignore	u_e	0.1	nuscenes	nuscenes	1	0.4300	0.0817	[0.0700, 0.0950]	0.668	A
kitti_ignore	u_e	0.1	nuscenes	nuscenes	2	0.4400	0.0869	[0.0747, 0.1004]	0.681	?
kitti_ignore	u_e	0.1	nuscenes	nuscenes	3	0.4400	0.0862	[0.0740, 0.0994]	0.680	A
kitti_ignore	u_e	0.1	nuscenes	nuscenes	4	0.4200	0.0771	[0.0655, 0.0898]	0.652	A
kitti_ignore	u_s	10^{-2}	kitti	kitti	0	0.0800	0.0054	[0.0022, 0.0109]	0.054	?
kitti_ignore	u_s	10^{-2}	kitti	kitti	1	0.0800	0.0030	[0.0007, 0.0078]	0.050	A
kitti_ignore	u_s	10^{-2}	kitti	kitti	2	0.0800	0.0046	[0.0016, 0.0102]	0.052	?
kitti_ignore	u_s	10^{-2}	kitti	kitti	3	0.0800	0.0027	[0.0007, 0.0071]	0.054	A
kitti_ignore	u_s	10^{-2}	kitti	kitti	4	0.0800	0.0041	[0.0012, 0.0095]	0.057	A
kitti_ignore	u_s	10^{-2}	kitti	nuscenes	0	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
kitti_ignore	u_s	10^{-2}	kitti	nuscenes	1	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
kitti_ignore	u_s	10^{-2}	kitti	nuscenes	2	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
kitti_ignore	u_s	10^{-2}	kitti	nuscenes	3	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
kitti_ignore	u_s	10^{-2}	kitti	nuscenes	4	0.0800	0.0000	[0.0000, 0.0013]	0.000	A
kitti_ignore	u_s	10^{-2}	nuscenes	kitti	0	0.1200	0.0379	[0.0281, 0.0502]	0.458	N
kitti_ignore	u_s	10^{-2}	nuscenes	kitti	1	0.1200	0.0387	[0.0286, 0.0508]	0.457	N
kitti_ignore	u_s	10^{-2}	nuscenes	kitti	2	0.1200	0.0381	[0.0282, 0.0503]	0.453	N
kitti_ignore	u_s	10^{-2}	nuscenes	kitti	3	0.1300	0.0641	[0.0510, 0.0789]	0.560	N
kitti_ignore	u_s	10^{-2}	nuscenes	kitti	4	0.1300	0.0538	[0.0419, 0.0678]	0.549	N
kitti_ignore	u_s	10^{-2}	nuscenes	nuscenes	0	0.1200	0.0026	[0.0008, 0.0061]	0.008	A
kitti_ignore	u_s	10^{-2}	nuscenes	nuscenes	1	0.1200	0.0009	[0.0001, 0.0034]	0.008	A
kitti_ignore	u_s	10^{-2}	nuscenes	nuscenes	2	0.1200	0.0025	[0.0008, 0.0057]	0.008	A
kitti_ignore	u_s	10^{-2}	nuscenes	nuscenes	3	0.1300	0.0058	[0.0028, 0.0103]	0.018	?
kitti_ignore	u_s	10^{-2}	nuscenes	nuscenes	4	0.1300	0.0067	[0.0036, 0.0113]	0.017	?
kitti_ignore	u_s	0.05	kitti	kitti	0	0.1100	0.0254	[0.0175, 0.0360]	0.342	A
kitti_ignore	u_s	0.05	kitti	kitti	1	0.1100	0.0233	[0.0158, 0.0335]	0.340	A
kitti_ignore	u_s	0.05	kitti	kitti	2	0.1100	0.0253	[0.0171, 0.0354]	0.339	A
kitti_ignore	u_s	0.05	kitti	kitti	3	0.1200	0.0459	[0.0352, 0.0593]	0.459	?
kitti_ignore	u_s	0.05	kitti	kitti	4	0.1100	0.0233	[0.0158, 0.0335]	0.341	A
kitti_ignore	u_s	0.05	kitti	nuscenes	0	0.1100	0.0007	[0.0000, 0.0030]	0.003	A
kitti_ignore	u_s	0.05	kitti	nuscenes	1	0.1100	0.0000	[0.0000, 0.0013]	0.002	A
kitti_ignore	u_s	0.05	kitti	nuscenes	2	0.1100	0.0004	[0.0000, 0.0023]	0.003	A
kitti_ignore	u_s	0.05	kitti	nuscenes	3	0.1200	0.0032	[0.0012, 0.0066]	0.008	A
kitti_ignore	u_s	0.05	kitti	nuscenes	4	0.1100	0.0014	[0.0003, 0.0041]	0.003	A
kitti_ignore	u_s	0.05	nuscenes	kitti	0	0.1700	0.1635	[0.1433, 0.1849]	0.812	N
kitti_ignore	u_s	0.05	nuscenes	kitti	1	0.1600	0.1305	[0.1123, 0.1502]	0.771	N
kitti_ignore	u_s	0.05	nuscenes	kitti	2	0.1600	0.1283	[0.1105, 0.1482]	0.770	N
kitti_ignore	u_s	0.05	nuscenes	kitti	3	0.1600	0.1405	[0.1219, 0.1610]	0.778	N
kitti_ignore	u_s	0.05	nuscenes	kitti	4	0.1600	0.1277	[0.1099, 0.1476]	0.765	N
kitti_ignore	u_s	0.05	nuscenes	nuscenes	0	0.1700	0.0445	[0.0356, 0.0545]	0.153	?
kitti_ignore	u_s	0.05	nuscenes	nuscenes	1	0.1600	0.0252	[0.0186, 0.0332]	0.111	A
kitti_ignore	u_s	0.05	nuscenes	nuscenes	2	0.1600	0.0291	[0.0220, 0.0376]	0.112	A
kitti_ignore	u_s	0.05	nuscenes	nuscenes	3	0.1600	0.0247	[0.0182, 0.0327]	0.110	A
kitti_ignore	u_s	0.05	nuscenes	nuscenes	4	0.1600	0.0300	[0.0229, 0.0388]	0.106	A
kitti_ignore	u_s	0.1	kitti	kitti	0	0.1400	0.0806	[0.0662, 0.0971]	0.648	A
kitti_ignore	u_s	0.1	kitti	kitti	1	0.1400	0.0739	[0.0598, 0.0895]	0.641	A
kitti_ignore	u_s	0.1	kitti	kitti	2	0.1400	0.0777	[0.0634, 0.0938]	0.634	A
kitti_ignore	u_s	0.1	kitti	kitti	3	0.1400	0.0876	[0.0726, 0.1047]	0.645	?
kitti_ignore	u_s	0.1	kitti	kitti	4	0.1400	0.0744	[0.0604, 0.0903]	0.634	A
kitti_ignore	u_s	0.1	kitti	nuscenes	0	0.1400	0.0098	[0.0060, 0.0151]	0.037	A
kitti_ignore	u_s	0.1	kitti	nuscenes	1	0.1400	0.0086	[0.0049, 0.0139]	0.038	A
kitti_ignore	u_s	0.1	kitti	nuscenes	2	0.1400	0.0091	[0.0052, 0.0143]	0.039	A
kitti_ignore	u_s	0.1	kitti	nuscenes	3	0.1400	0.0109	[0.0068, 0.0164]	0.039	A
kitti_ignore	u_s	0.1	kitti	nuscenes	4	0.1400	0.0120	[0.0077, 0.0177]	0.037	A
kitti_ignore	u_s	0.1	nuscenes	kitti	0	0.2000	0.2438	[0.2203, 0.2684]	0.894	N
kitti_ignore	u_s	0.1	nuscenes	kitti	1	0.2000	0.2454	[0.2219, 0.2701]	0.893	N
kitti_ignore	u_s	0.1	nuscenes	kitti	2	0.2000	0.2391	[0.2156, 0.2634]	0.895	N
kitti_ignore	u_s	0.1	nuscenes	kitti	3	0.2000	0.2531	[0.2292, 0.2779]	0.897	N
kitti_ignore	u_s	0.1	nuscenes	kitti	4	0.2000	0.2376	[0.2144, 0.2621]	0.890	N
kitti_ignore	u_s	0.1	nuscenes	nuscenes	0	0.2000	0.0779	[0.0662, 0.0905]	0.330	A
kitti_ignore	u_s	0.1	nuscenes	nuscenes	1	0.2000	0.0765	[0.0651, 0.0893]	0.336	A
kitti_ignore	u_s	0.1	nuscenes	nuscenes	2	0.2000	0.0751	[0.0639, 0.0879]	0.343	A
kitti_ignore	u_s	0.1	nuscenes	nuscenes	3	0.2000	0.0724	[0.0612, 0.0848]	0.333	A

kitti_ignore	u_s	0.1	nuscenes	nuscenes	4	0.2000	0.0753	[0.0638, 0.0879]	0.334	A

3.4 nuScenes, recorded rain against clear daytime (m6 part C)

Source: `results/division_by_frame/kitti_rule_dataset_change_weather/C_weather_nuscenes.csv`. A frame's condition comes from its scene description; night and rain-at-night frames are excluded. Rain frames hold airborne scatterers and a wet road together, so the condition is recorded rain, not one physical process.

Table 10: Rain (m6 C): outcome per split.

ind.	α	sel.	use	seeds 0–4
u_e	10^{-2}	clear	clear	AAA??
u_e	10^{-2}	clear	rain	?????
u_e	10^{-2}	rain	clear	AAAAA
u_e	10^{-2}	rain	rain	?AAA?
u_e	10^{-3}	clear	clear	---
u_e	10^{-3}	clear	rain	---
u_e	10^{-3}	rain	clear	---
u_e	10^{-3}	rain	rain	---
u_e	10^{-4}	clear	clear	---
u_e	10^{-4}	clear	rain	---
u_e	10^{-4}	rain	clear	---
u_e	10^{-4}	rain	rain	---
u_e	0.05	clear	clear	?A?A?
u_e	0.05	clear	rain	N????
u_e	0.05	rain	clear	AAAAA
u_e	0.05	rain	rain	AA??A
u_e	0.1	clear	clear	?A??A
u_e	0.1	clear	rain	NNNN?
u_e	0.1	rain	clear	AAAAA
u_e	0.1	rain	rain	?A??A
u_s	10^{-2}	clear	clear	A--?
u_s	10^{-2}	clear	rain	?--?
u_s	10^{-2}	rain	clear	---
u_s	10^{-2}	rain	rain	---
u_s	10^{-3}	clear	clear	---
u_s	10^{-3}	clear	rain	---
u_s	10^{-3}	rain	clear	---
u_s	10^{-3}	rain	rain	---
u_s	10^{-4}	clear	clear	---
u_s	10^{-4}	clear	rain	---
u_s	10^{-4}	rain	clear	---
u_s	10^{-4}	rain	rain	---
u_s	0.05	clear	clear	?AAA?
u_s	0.05	clear	rain	?????
u_s	0.05	rain	clear	AAAAA
u_s	0.05	rain	rain	?AAAA
u_s	0.1	clear	clear	?A??A
u_s	0.1	clear	rain	N????
u_s	0.1	rain	clear	AAAAA
u_s	0.1	rain	rain	AAAAA

Table 11: Rain (m6 C): values per split (n_{eval}: test frames of the condition of use).

ind.	α	sel.	use	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	n_{eval}	out.
u_e	10^{-2}	clear	clear	0	0.0927	0.0040	[0.0012, 0.0092]	0.063	1900	A
u_e	10^{-2}	clear	clear	1	0.0891	0.0017	[0.0002, 0.0056]	0.060	1915	A
u_e	10^{-2}	clear	clear	2	0.0939	0.0032	[0.0009, 0.0082]	0.069	1898	A
u_e	10^{-2}	clear	clear	3	0.1008	0.0049	[0.0018, 0.0106]	0.081	1902	?
u_e	10^{-2}	clear	clear	4	0.1145	0.0066	[0.0028, 0.0128]	0.110	1896	?
u_e	10^{-2}	clear	rain	0	0.0927	0.0024	[0.0000, 0.0158]	0.087	414	?
u_e	10^{-2}	clear	rain	1	0.0891	0.0025	[0.0000, 0.0162]	0.091	403	?
u_e	10^{-2}	clear	rain	2	0.0939	0.0037	[0.0001, 0.0188]	0.102	403	?
u_e	10^{-2}	clear	rain	3	0.1008	0.0050	[0.0004, 0.0209]	0.102	404	?
u_e	10^{-2}	clear	rain	4	0.1145	0.0101	[0.0020, 0.0292]	0.126	398	?
u_e	10^{-2}	rain	clear	0	0.0672	0.0016	[0.0002, 0.0053]	0.026	1900	A
u_e	10^{-2}	rain	clear	1	0.0538	0.0000	[0.0000, 0.0019]	0.012	1915	A
u_e	10^{-2}	rain	clear	2	0.0581	0.0000	[0.0000, 0.0019]	0.017	1898	A
u_e	10^{-2}	rain	clear	3	0.0708	0.0016	[0.0002, 0.0053]	0.030	1902	A
u_e	10^{-2}	rain	clear	4	0.0893	0.0038	[0.0012, 0.0090]	0.058	1896	A
u_e	10^{-2}	rain	rain	0	0.0672	0.0024	[0.0000, 0.0158]	0.049	414	?
u_e	10^{-2}	rain	rain	1	0.0538	0.0000	[0.0000, 0.0091]	0.019	403	A
u_e	10^{-2}	rain	rain	2	0.0581	0.0000	[0.0000, 0.0091]	0.035	403	A
u_e	10^{-2}	rain	rain	3	0.0708	0.0000	[0.0000, 0.0091]	0.045	404	A
u_e	10^{-2}	rain	rain	4	0.0893	0.0088	[0.0013, 0.0281]	0.089	398	?
u_e	0.05	clear	clear	0	0.2647	0.0398	[0.0297, 0.0519]	0.382	1900	?
u_e	0.05	clear	clear	1	0.2503	0.0295	[0.0209, 0.0402]	0.356	1915	A

ind.	α	sel.	use	seed	τ	$\hat{A}$	HB interval	$\hat{R}$	n_{eval}	out.
u_e	0.05	clear	clear	2	0.2581	0.0395	[0.0292, 0.0513]	0.373	1898	?
u_e	0.05	clear	clear	3	0.2615	0.0376	[0.0278, 0.0495]	0.378	1902	A
u_e	0.05	clear	clear	4	0.2653	0.0405	[0.0302, 0.0526]	0.383	1896	?
u_e	0.05	clear	rain	0	0.2647	0.0871	[0.0573, 0.1278]	0.421	414	N
u_e	0.05	clear	rain	1	0.2503	0.0608	[0.0351, 0.0963]	0.402	403	?
u_e	0.05	clear	rain	2	0.2581	0.0694	[0.0409, 0.1052]	0.433	403	?
u_e	0.05	clear	rain	3	0.2615	0.0665	[0.0389, 0.1020]	0.417	404	?
u_e	0.05	clear	rain	4	0.2653	0.0650	[0.0375, 0.1005]	0.439	398	?
u_e	0.05	rain	clear	0	0.1694	0.0157	[0.0094, 0.0239]	0.202	1900	A
u_e	0.05	rain	clear	1	0.1608	0.0118	[0.0065, 0.0193]	0.200	1915	A
u_e	0.05	rain	clear	2	0.1876	0.0168	[0.0103, 0.0252]	0.245	1898	A
u_e	0.05	rain	clear	3	0.1759	0.0155	[0.0094, 0.0239]	0.222	1902	A
u_e	0.05	rain	clear	4	0.1592	0.0141	[0.0082, 0.0220]	0.192	1896	A
u_e	0.05	rain	rain	0	0.1694	0.0240	[0.0088, 0.0484]	0.237	414	A
u_e	0.05	rain	rain	1	0.1608	0.0205	[0.0071, 0.0458]	0.224	403	A
u_e	0.05	rain	rain	2	0.1876	0.0376	[0.0185, 0.0689]	0.289	403	?
u_e	0.05	rain	rain	3	0.1759	0.0267	[0.0103, 0.0529]	0.243	404	?
u_e	0.05	rain	rain	4	0.1592	0.0226	[0.0087, 0.0470]	0.226	398	A
u_e	0.1	clear	clear	0	0.4639	0.0884	[0.0732, 0.1049]	0.703	1900	?
u_e	0.1	clear	clear	1	0.4503	0.0813	[0.0669, 0.0974]	0.687	1915	A
u_e	0.1	clear	clear	2	0.4633	0.0904	[0.0752, 0.1073]	0.706	1898	?
u_e	0.1	clear	clear	3	0.4508	0.0840	[0.0693, 0.1003]	0.687	1902	?
u_e	0.1	clear	clear	4	0.4510	0.0812	[0.0666, 0.0972]	0.686	1896	A
u_e	0.1	clear	rain	0	0.4639	0.1699	[0.1279, 0.2196]	0.722	414	N
u_e	0.1	clear	rain	1	0.4503	0.1449	[0.1052, 0.1927]	0.723	403	N
u_e	0.1	clear	rain	2	0.4633	0.1527	[0.1117, 0.2009]	0.747	403	N
u_e	0.1	clear	rain	3	0.4508	0.1571	[0.1158, 0.2059]	0.714	404	N
u_e	0.1	clear	rain	4	0.4510	0.1350	[0.0957, 0.1812]	0.729	398	?
u_e	0.1	rain	clear	0	0.2647	0.0398	[0.0297, 0.0519]	0.382	1900	A
u_e	0.1	rain	clear	1	0.2383	0.0275	[0.0191, 0.0378]	0.335	1915	A
u_e	0.1	rain	clear	2	0.2581	0.0395	[0.0292, 0.0513]	0.373	1898	A
u_e	0.1	rain	clear	3	0.2418	0.0296	[0.0211, 0.0405]	0.338	1902	A
u_e	0.1	rain	clear	4	0.2260	0.0262	[0.0180, 0.0364]	0.311	1896	A
u_e	0.1	rain	rain	0	0.2647	0.0871	[0.0573, 0.1278]	0.421	414	?
u_e	0.1	rain	rain	1	0.2383	0.0565	[0.0313, 0.0903]	0.377	403	A
u_e	0.1	rain	rain	2	0.2581	0.0694	[0.0409, 0.1052]	0.433	403	?
u_e	0.1	rain	rain	3	0.2418	0.0690	[0.0408, 0.1049]	0.373	404	?
u_e	0.1	rain	rain	4	0.2260	0.0503	[0.0279, 0.0853]	0.346	398	A
u_s	10^{-2}	clear	clear	0	0.1266	0.0039	[0.0012, 0.0092]	0.013	1900	A
u_s	10^{-2}	clear	clear	4	0.1356	0.0098	[0.0050, 0.0168]	0.025	1896	?
u_s	10^{-2}	clear	rain	0	0.1266	0.0048	[0.0003, 0.0204]	0.026	414	?
u_s	10^{-2}	clear	rain	4	0.1356	0.0197	[0.0062, 0.0436]	0.038	398	?
u_s	0.05	clear	clear	0	0.1692	0.0391	[0.0292, 0.0513]	0.148	1900	?
u_s	0.05	clear	clear	1	0.1637	0.0259	[0.0178, 0.0360]	0.125	1915	A
u_s	0.05	clear	clear	2	0.1657	0.0330	[0.0238, 0.0442]	0.133	1898	A
u_s	0.05	clear	clear	3	0.1673	0.0313	[0.0224, 0.0423]	0.141	1902	A
u_s	0.05	clear	clear	4	0.1731	0.0430	[0.0325, 0.0555]	0.166	1896	?
u_s	0.05	clear	rain	0	0.1692	0.0673	[0.0398, 0.1024]	0.162	414	?
u_s	0.05	clear	rain	1	0.1637	0.0518	[0.0275, 0.0843]	0.144	403	?
u_s	0.05	clear	rain	2	0.1657	0.0589	[0.0332, 0.0933]	0.159	403	?
u_s	0.05	clear	rain	3	0.1673	0.0512	[0.0275, 0.0841]	0.170	404	?
u_s	0.05	clear	rain	4	0.1731	0.0628	[0.0375, 0.0975]	0.202	398	?
u_s	0.05	rain	clear	0	0.1455	0.0116	[0.0066, 0.0194]	0.048	1900	A
u_s	0.05	rain	clear	1	0.1416	0.0064	[0.0028, 0.0127]	0.041	1915	A
u_s	0.05	rain	clear	2	0.1411	0.0101	[0.0054, 0.0175]	0.040	1898	A
u_s	0.05	rain	clear	3	0.1417	0.0094	[0.0046, 0.0161]	0.039	1902	A
u_s	0.05	rain	clear	4	0.1455	0.0146	[0.0086, 0.0227]	0.049	1896	A
u_s	0.05	rain	rain	0	0.1455	0.0256	[0.0099, 0.0516]	0.072	414	?
u_s	0.05	rain	rain	1	0.1416	0.0203	[0.0071, 0.0455]	0.056	403	A
u_s	0.05	rain	rain	2	0.1411	0.0062	[0.0006, 0.0235]	0.054	403	A
u_s	0.05	rain	rain	3	0.1417	0.0116	[0.0024, 0.0324]	0.062	404	A
u_s	0.05	rain	rain	4	0.1455	0.0228	[0.0087, 0.0493]	0.070	398	A
u_s	0.1	clear	clear	0	0.2071	0.0865	[0.0718, 0.1032]	0.366	1900	?
u_s	0.1	clear	clear	1	0.2075	0.0831	[0.0688, 0.0996]	0.373	1915	A
u_s	0.1	clear	clear	2	0.2082	0.0877	[0.0728, 0.1045]	0.386	1898	?
u_s	0.1	clear	clear	3	0.2103	0.0889	[0.0741, 0.1060]	0.393	1902	?
u_s	0.1	clear	clear	4	0.2065	0.0823	[0.0680, 0.0989]	0.364	1896	A
u_s	0.1	clear	rain	0	0.2071	0.1401	[0.1003, 0.1851]	0.385	414	N
u_s	0.1	clear	rain	1	0.2075	0.1372	[0.0988, 0.1845]	0.382	403	?
u_s	0.1	clear	rain	2	0.2082	0.1353	[0.0966, 0.1817]	0.399	403	?
u_s	0.1	clear	rain	3	0.2103	0.1214	[0.0857, 0.1675]	0.421	404	?
u_s	0.1	clear	rain	4	0.2065	0.1219	[0.0849, 0.1671]	0.394	398	?
u_s	0.1	rain	clear	0	0.1719	0.0443	[0.0338, 0.0572]	0.159	1900	A
u_s	0.1	rain	clear	1	0.1637	0.0259	[0.0178, 0.0360]	0.125	1915	A
u_s	0.1	rain	clear	2	0.1657	0.0330	[0.0238, 0.0442]	0.133	1898	A
u_s	0.1	rain	clear	3	0.1588	0.0196	[0.0128, 0.0289]	0.100	1902	A
u_s	0.1	rain	clear	4	0.1656	0.0321	[0.0229, 0.0430]	0.128	1896	A
u_s	0.1	rain	rain	0	0.1719	0.0643	[0.0379, 0.0995]	0.174	414	A
u_s	0.1	rain	rain	1	0.1637	0.0518	[0.0275, 0.0843]	0.144	403	A
u_s	0.1	rain	rain	2	0.1657	0.0589	[0.0332, 0.0933]	0.159	403	A
u_s	0.1	rain	rain	3	0.1588	0.0425	[0.0220, 0.0750]	0.133	404	A
u_s	0.1	rain	rain	4	0.1656	0.0570	[0.0317, 0.0914]	0.161	398	A

4 False acceptance proportion per split at the exploratory limit 0.1

Division by frame, KITTI with the KITTI rule. Each entry gives the false acceptance proportion A on the evaluation part of the condition of use and the outcome code (A applicable, N not applicable, ? not determined). Over the transfers into far distance, rain or KITTI, A ranges from 0.066 to 0.267; over the transfers in the opposite direction from 0.009 to 0.098 (minimum and maximum of the listed values).

Table 12: False acceptance proportion at the exploratory limit 0.1 per split.

setting	ind.	transfer	split 0	split 1	split 2	split 3	split 4
KITTI distance	u_e	far → near	0.038 A	0.041 A	0.035 A	0.038 A	0.035 A
KITTI distance	u_e	near → far	0.169 N	0.157 N	0.160 N	0.186 N	0.170 N
KITTI distance	u_s	far → near	0.023 A	0.021 A	0.024 A	0.029 A	0.021 A
KITTI distance	u_s	near → far	0.240 N	0.221 N	0.238 N	0.267 N	0.213 N
nuScenes distance	u_e	far → near	0.061 A	0.060 A	0.059 A	0.057 A	0.056 A
nuScenes distance	u_e	near → far	0.172 N	0.172 N	0.176 N	0.181 N	0.170 N
nuScenes distance	u_s	far → near	0.069 A	0.062 A	0.066 A	0.060 A	0.062 A
nuScenes distance	u_s	near → far	0.128 N	0.129 N	0.141 N	0.120 N	0.121 N
nuScenes weather	u_e	clear → rain	0.170 N	0.145 N	0.153 N	0.157 N	0.135 ?
nuScenes weather	u_e	rain → clear	0.040 A	0.028 A	0.039 A	0.030 A	0.026 A
nuScenes weather	u_s	clear → rain	0.140 N	0.137 ?	0.135 ?	0.121 ?	0.122 ?
nuScenes weather	u_s	rain → clear	0.044 A	0.026 A	0.033 A	0.020 A	0.032 A
dataset and detector	u_e	kitti → nuscenes	0.098 ?	0.087 ?	0.090 ?	0.097 ?	0.088 ?
dataset and detector	u_e	nuscenes → kitti	0.078 A	0.069 A	0.071 A	0.076 A	0.066 A
dataset and detector	u_s	kitti → nuscenes	0.010 A	0.009 A	0.009 A	0.011 A	0.012 A
dataset and detector	u_s	nuscenes → kitti	0.244 N	0.245 N	0.239 N	0.253 N	0.238 N

5 KITTI rule: counts

The labels of the detection pipeline count a car detection as wrong when it lies in a KITTI DontCare region or on a Van. The KITTI rule removes such wrong detections (neither correct nor wrong): a wrong car detection is removed if its 3D overlap with a Van is at least 0.5, or if its projected 2D box overlaps a DontCare box by more than half of the detection's own area. Only detections labelled wrong can be removed. Sources: `results/division_by_frame/kitti_rule_dataset_change_weather/provenance.json` (key `A`) and `results/division_by_frame/kitti_rule_dataset_change_weather/summary.txt`.

Table 13: Wrong car detections and the detections the KITTI rule removes, by distance class (cut: median car range over all frames).

class	wrong	in DontCare	on Van	share removed
near	5516	167	455	0.113
far	11768	1592	240	0.156

Detections removed in total: 2454; median car range over all frames: 35.063 m. The outcome counts with and without the rule are in Tables 4 and 8.

6 Accepted detections per distance bin

Car detections of the test part of split 0 accepted by the threshold selected at near distance (u_s, exploratory limit 0.1), per 5 m bin of detection distance, divided into correct and wrong. KITTI with the KITTI rule: boundary 34.806 m, $\tau_{near} = 0.16931$; nuScenes: boundary 21.402 m, $\tau_{near} = 0.22789$. Among the accepted detections the share of wrong ones grows with distance also below the near/far boundary, so the not applicable outcomes after transfers into far distance are not explained by the position of the boundary alone (split 0, u_s, 0.1). Shares in parentheses rest on fewer than 30 accepted detections. Source: `results/supporting_analyses/distance_bins_log.txt`, the console output of `tools/figures/make_figures.pycauses`, which also drew Fig. 1.

Table 14: KITTI: accepted car detections per 5 m bin under τ_{near}.

bin (m)	accepted correct	accepted wrong	share wrong
0-5	77	14	0.154
5-10	467	31	0.062
10-15	718	18	0.024
15-20	689	41	0.056
20-25	653	51	0.072
25-30	576	104	0.153
30-35	476	137	0.223
35-40	318	154	0.326
40-45	238	134	0.360
45-50	167	104	0.384
50-55	67	85	0.559
55-60	42	54	0.562
60-65	24	31	0.564
65-70	9	16	(0.640)
70-75	0	3	(1.000)

Table 15: nuScenes: accepted car detections per 5 m bin under τ_{near}.

bin (m)	accepted correct	accepted wrong	share wrong
0-5	325	6	0.018
5-10	1619	107	0.062
10-15	1494	169	0.102
15-20	1134	200	0.150
20-25	663	172	0.206
25-30	358	150	0.295
30-35	132	90	0.405
35-40	42	55	0.567
40-45	19	33	0.635
45-50	6	22	(0.786)
50-55	0	10	(1.000)
55-60	0	1	(1.000)
60-65	0	2	(1.000)

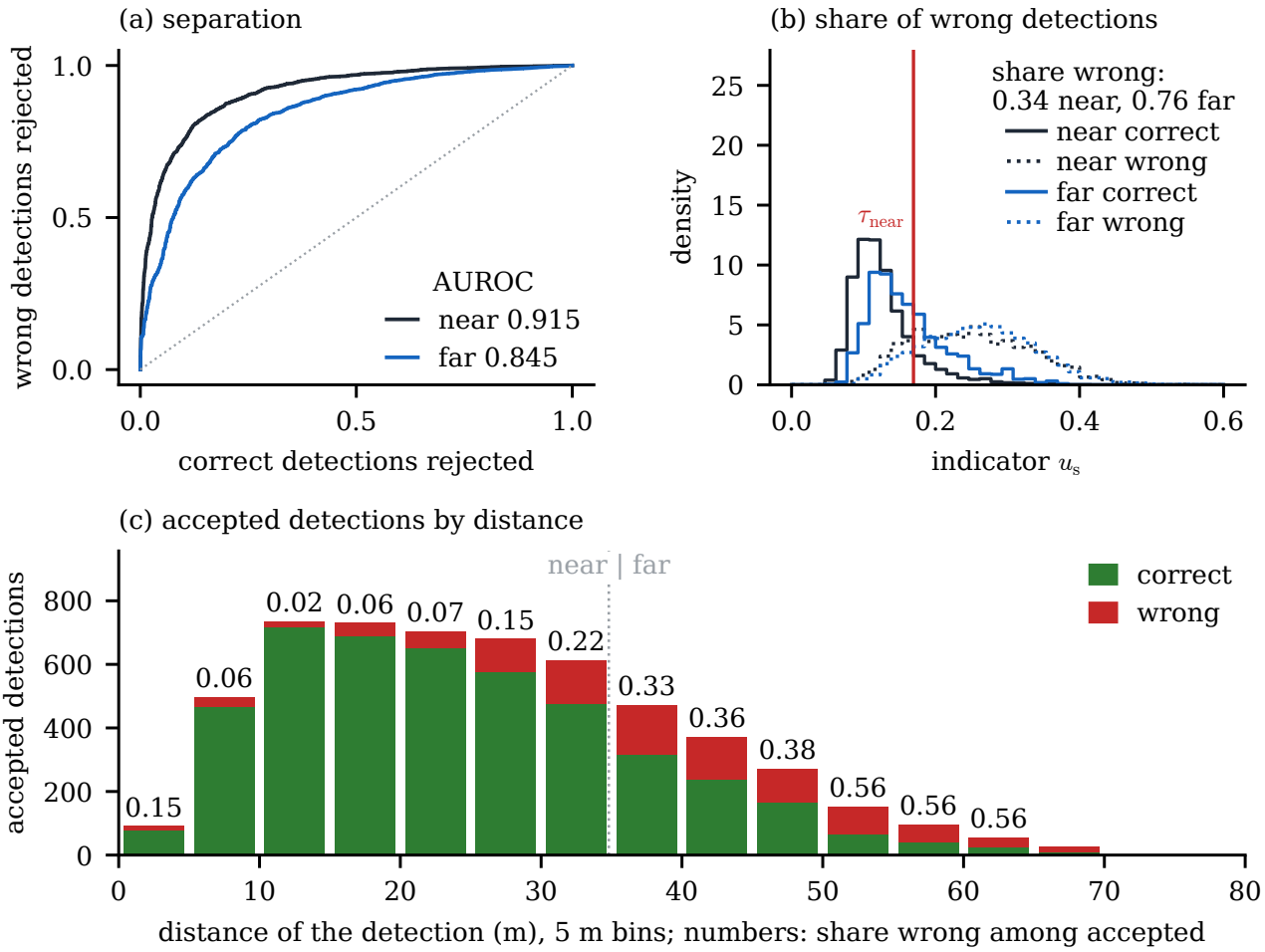


Figure 1: KITTI distance, spatial indicator, KITTI rule, split 0, threshold selected at near distance at the exploratory limit 0.1: (a) separation per distance condition, (b) indicator distributions, (c) accepted detections per 5 m bin with the share of wrong ones among them (counts in the table above).

7 Division by recorded sequence

Frames of one recorded sequence share objects, weather and road, so they are not independent draws. The analysis was therefore repeated with the grid, selection and test parts drawn by recorded sequence: nuScenes scenes (m7, 150 scenes in the validation set) and KITTI drives (m8, 45 drives in the validation frames). The sequences are permuted by the split function of the division by frame, with the same seeds 0 to 4 and the same fractions, and every frame takes the part of its sequence. Limits, δ, candidate thresholds and the boundary rule are unchanged. This analysis was added after the parameters had been fixed.

Variants. *Frame*: division by frame, as recorded. *Variant (i)*: division by recorded sequence, the frame stays the unit of the selection test and of the HB bounds. *Variant (ii)*: the same division with the sequence as the unit. The loss of sequence s in condition c is $\ell_s(\tau) = \frac{1}{|F_{s,c}|} \sum_{f \in F_{s,c}} a_f(\tau)$, where $F_{s,c}$ are the frames of s that hold a car detection of c; a sequence without accepted detections has loss 0, and each sequence has equal weight. Retention is a sum over detections and is the same for both units. The post-hoc Method A (Section 10) keeps the per-frame target exactly with the sequence as the unit.

Reproduction check. Before the division by scene, the division by frame was rerun with the scripts of the scene runs and compared with the recorded results in every row and column: "REPRODUCTION frame split: m3 1200 rows, m6 A/B/C all rows, input hashes: EQUAL" (`results/division_by_sequence/nuscenes_by_scene/REPRO_PASSED.txt`).

Table 16: nuScenes, division by recorded scene: outcome per split for the division by frame and variants (i) and (ii), and the retention of variant (ii) (–: no threshold). Cells without a selected threshold in any variant and split are not listed.

pair	ind.	α	cell	frame	(i)	(ii)	R (ii), splits 0–4
distance	u_e	10^{-2}	far→far	?A?AA	AAAA?	---	– – – – –
distance	u_e	10^{-2}	far→near	N??NN	????N	---	– – – – –
distance	u_e	10^{-2}	near→far	AAAAA	AAAAA	---	– – – – –
distance	u_e	10^{-2}	near→near	AAAA?	??AA?	---	– – – – –
distance	u_e	0.05	far→far	?A?AA	?AAA?	?????	0.01 0.02 0.02 0.03 0.02
distance	u_e	0.05	far→near	?????	???A?	?????	0.19 0.25 0.23 0.21 0.22
distance	u_e	0.05	near→far	AAAAA	AAA?N	??-??	0.00 0.00 – 0.00 0.01
distance	u_e	0.05	near→near	?A??A	??A?N	??-??	0.03 0.03 – 0.01 0.11
distance	u_e	0.1	far→far	????A	A?AA?	???A?	0.12 0.12 0.12 0.10 0.10
distance	u_e	0.1	far→near	AAAAA	AAAAA	?????	0.57 0.55 0.56 0.49 0.51
distance	u_e	0.1	near→far	NNNNN	NNNNN	A?A??	0.06 0.09 0.04 0.14 0.10
distance	u_e	0.1	near→near	AAA?A	A?AA?	??A??	0.44 0.49 0.37 0.55 0.51
distance	u_s	10^{-2}	far→far	AA?A?	A?AAA	---	– – – – –
distance	u_s	10^{-2}	far→near	?????	??AA?	---	– – – – –
distance	u_s	10^{-2}	near→far	AAAAA	AAAAA	---	– – – – –
distance	u_s	10^{-2}	near→near	AAAA?	AAAA?	---	– – – – –
distance	u_s	0.05	far→far	?A??A	A?AA?	---	– – – – –
distance	u_s	0.05	far→near	?A?A?	A?AA?	---	– – – – –
distance	u_s	0.05	near→far	AA?AA	A?A?N	--?	– – – – 0.00
distance	u_s	0.05	near→near	?AAA?	AAAAN	--?	– – – – 0.02
distance	u_s	0.1	far→far	????A	??AAN	A?AA?	0.03 0.05 0.03 0.03 0.05
distance	u_s	0.1	far→near	AAAAA	AAAAA	???A?	0.16 0.24 0.19 0.14 0.24
distance	u_s	0.1	near→far	NNNNN	NN?NN	A?A??	0.04 0.04 0.02 0.05 0.06
distance	u_s	0.1	near→near	A??AA	A?AA?	??AA?	0.17 0.20 0.11 0.20 0.27
weather	u_e	10^{-2}	clear→clear	AAA??	??AA?	---	– – – – –
weather	u_e	10^{-2}	clear→rain	?????	A??A?	---	– – – – –
weather	u_e	10^{-2}	rain→clear	AAAAA	AA-A?	---	– – – – –
weather	u_e	10^{-2}	rain→rain	?AAA?	A?-A?	---	– – – – –
weather	u_e	0.05	clear→clear	?A?A?	?AA??	---	– – – – –
weather	u_e	0.05	clear→rain	N????	?N??N	---	– – – – –
weather	u_e	0.05	rain→clear	AAAAA	AAAAA	---	– – – – –
weather	u_e	0.05	rain→rain	AA??A	A?AA?	---	– – – – –
weather	u_e	0.1	clear→clear	?A??A	??AAN	??A??	0.24 0.18 0.16 0.20 0.25
weather	u_e	0.1	clear→rain	NNNN?	?NNNN	?????	0.26 0.15 0.18 0.22 0.29
weather	u_e	0.1	rain→clear	AAAAA	AAAAA	---	– – – – –
weather	u_e	0.1	rain→rain	?A??A	A?AA?	---	– – – – –
weather	u_s	10^{-2}	clear→clear	A--?	??-A?	---	– – – – –
weather	u_s	10^{-2}	clear→rain	?--?	A?-??	---	– – – – –
weather	u_s	0.05	clear→clear	?AAA?	AAAA?	---	– – – – –
weather	u_s	0.05	clear→rain	?????	????N	---	– – – – –
weather	u_s	0.05	rain→clear	AAAAA	AAAAA	---	– – – – –
weather	u_s	0.05	rain→rain	?AAAA	AAAA?	---	– – – – –
weather	u_s	0.1	clear→clear	?A??A	??AAN	??A??	0.07 0.04 0.02 0.04 0.07
weather	u_s	0.1	clear→rain	N????	?N??N	?????	0.08 0.06 0.03 0.06 0.11
weather	u_s	0.1	rain→clear	AAAAA	AAAAA	---	– – – – –
weather	u_s	0.1	rain→rain	AAAAA	A?AA?	---	– – – – –

Table 17: nuScenes scenes per condition and part under the division by scene, per split; in brackets the frames with a car detection (`results/division_by_sequence/nuscenes_by_scene/m7_scene_split_summary.txt`, l. 94–98). A scene counts for near or far if it holds a car detection of that condition; most scenes count for both.

split	grid / sel. / test	boundary	near sel. / test	far sel. / test	clear sel. / test	rain sel. / test
0	15 / 68 / 67	19.888 m	67 / 67	68 / 67	52 (1948) / 49 (1908)	9 (337) / 10 (399)
1	15 / 68 / 67	19.975 m	67 / 67	68 / 67	48 (1768) / 50 (1952)	11 (421) / 11 (410)
2	15 / 68 / 67	21.211 m	68 / 66	68 / 67	49 (1890) / 52 (1964)	11 (394) / 9 (357)
3	15 / 68 / 67	18.154 m	67 / 67	68 / 67	52 (2028) / 47 (1746)	10 (375) / 12 (458)
4	15 / 68 / 67	19.694 m	68 / 67	68 / 67	51 (1957) / 51 (1980)	9 (343) / 10 (372)

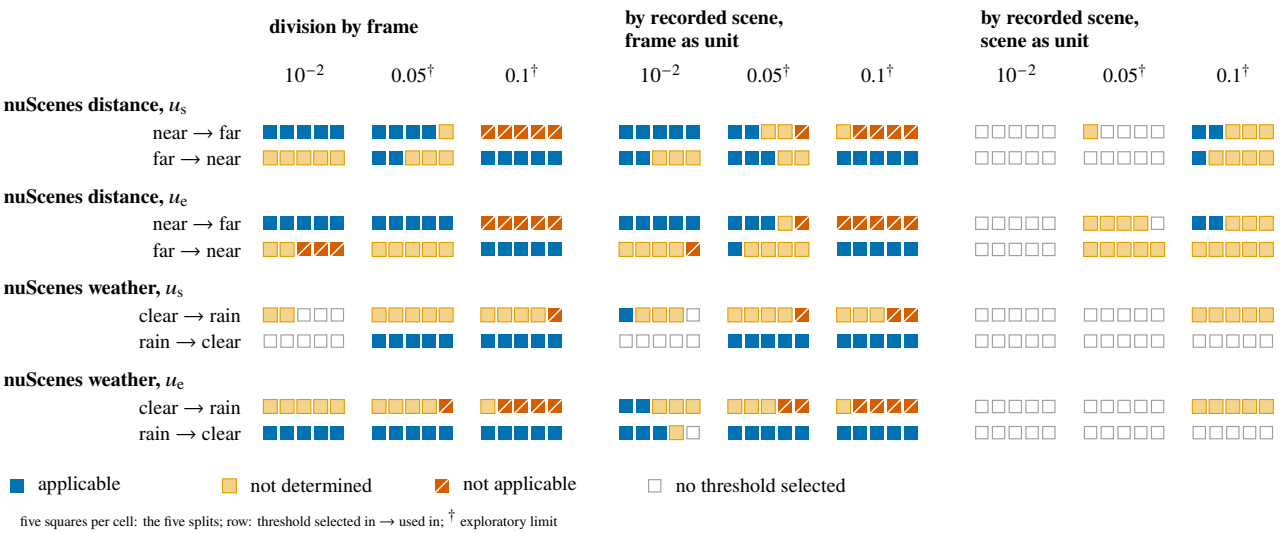


Figure 2: nuScenes: outcomes per cell on the five splits for the division by frame, by recorded scene with the frame as unit, and by recorded scene with the scene as unit (drawn by `tools/figures/supplementary/supplementary_figures.py` from `results/m7_scene_split/m7_scene_split.csv`).

8 Division of KITTI by recorded drive

The KITTI object frames were mapped to their recorded drives with the mapping files of the KITTI object development kit; the 3,769 validation frames come from 45 drives. Before the division by drive, the division by frame was run again with the same code and compared with the recorded results: "REPRODUCTION frame split: m6 A EQUAL, m6 B EQUAL, input hashes EQUAL". Variant (i) keeps the frame as the unit of the selection test and of the interval; variant (ii) uses the drive, with the mean of the per-frame false acceptance proportions over the drive's frames in the condition as its loss. In the change of dataset and detector the nuScenes side is divided by recorded scene.

Table 18: Outcome counts over the five splits, division by recorded drive (KITTI rule).

setting	unit	ind.	α	transfer	A	N	?	-
KITTI distance	drive	u_e	10^{-4}	far → near	0	0	0	5
KITTI distance	drive	u_e	10^{-4}	near → far	0	0	0	5
KITTI distance	drive	u_e	10^{-3}	far → near	0	0	0	5
KITTI distance	drive	u_e	10^{-3}	near → far	0	0	0	5
KITTI distance	drive	u_e	10^{-2}	far → near	0	0	0	5
KITTI distance	drive	u_e	10^{-2}	near → far	0	0	0	5
KITTI distance	drive	u_e	0.05	far → near	0	0	0	5
KITTI distance	drive	u_e	0.05	near → far	0	0	0	5
KITTI distance	drive	u_e	0.1	far → near	0	0	0	5
KITTI distance	drive	u_e	0.1	near → far	0	0	0	5
KITTI distance	drive	u_s	10^{-4}	far → near	0	0	0	5
KITTI distance	drive	u_s	10^{-4}	near → far	0	0	0	5
KITTI distance	drive	u_s	10^{-3}	far → near	0	0	0	5
KITTI distance	drive	u_s	10^{-3}	near → far	0	0	0	5
KITTI distance	drive	u_s	10^{-2}	far → near	0	0	0	5
KITTI distance	drive	u_s	10^{-2}	near → far	0	0	0	5
KITTI distance	drive	u_s	0.05	far → near	0	0	0	5
KITTI distance	drive	u_s	0.05	near → far	0	0	0	5
KITTI distance	drive	u_s	0.1	far → near	0	0	0	5
KITTI distance	drive	u_s	0.1	near → far	0	0	0	5
KITTI distance	frame	u_e	10^{-4}	far → near	0	0	0	5
KITTI distance	frame	u_e	10^{-4}	near → far	0	0	0	5
KITTI distance	frame	u_e	10^{-3}	far → near	0	0	0	5
KITTI distance	frame	u_e	10^{-3}	near → far	0	0	0	5
KITTI distance	frame	u_e	10^{-2}	far → near	0	0	4	1
KITTI distance	frame	u_e	10^{-2}	near → far	2	0	2	1
KITTI distance	frame	u_e	0.05	far → near	4	0	0	1
KITTI distance	frame	u_e	0.05	near → far	0	3	2	0
KITTI distance	frame	u_e	0.1	far → near	5	0	0	0
KITTI distance	frame	u_e	0.1	near → far	0	5	0	0
KITTI distance	frame	u_s	10^{-4}	far → near	0	0	0	5
KITTI distance	frame	u_s	10^{-4}	near → far	0	0	0	5
KITTI distance	frame	u_s	10^{-3}	far → near	0	0	0	5

setting	unit	ind.	α	transfer	A	N	?	-
KITTI distance	frame	u_s	10^{-3}	near → far	0	0	0	5
KITTI distance	frame	u_s	10^{-2}	far → near	3	0	1	1
KITTI distance	frame	u_s	10^{-2}	near → far	1	1	2	1
KITTI distance	frame	u_s	0.05	far → near	4	0	0	1
KITTI distance	frame	u_s	0.05	near → far	0	5	0	0
KITTI distance	frame	u_s	0.1	far → near	5	0	0	0
KITTI distance	frame	u_s	0.1	near → far	0	5	0	0
change of dataset	drive	u_e	10^{-4}	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_e	10^{-4}	nuscenes → kitti	0	0	0	5
change of dataset	drive	u_e	10^{-3}	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_e	10^{-3}	nuscenes → kitti	0	0	0	5
change of dataset	drive	u_e	10^{-2}	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_e	10^{-2}	nuscenes → kitti	0	0	0	5
change of dataset	drive	u_e	0.05	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_e	0.05	nuscenes → kitti	0	0	5	0
change of dataset	drive	u_e	0.1	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_e	0.1	nuscenes → kitti	0	0	5	0
change of dataset	drive	u_s	10^{-4}	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_s	10^{-4}	nuscenes → kitti	0	0	0	5
change of dataset	drive	u_s	10^{-3}	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_s	10^{-3}	nuscenes → kitti	0	0	0	5
change of dataset	drive	u_s	10^{-2}	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_s	10^{-2}	nuscenes → kitti	0	0	0	5
change of dataset	drive	u_s	0.05	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_s	0.05	nuscenes → kitti	0	0	5	0
change of dataset	drive	u_s	0.1	kitti → nuscenes	0	0	0	5
change of dataset	drive	u_s	0.1	nuscenes → kitti	0	0	5	0
change of dataset	frame	u_e	10^{-4}	kitti → nuscenes	0	0	0	5
change of dataset	frame	u_e	10^{-4}	nuscenes → kitti	0	0	0	5
change of dataset	frame	u_e	10^{-3}	kitti → nuscenes	0	0	0	5
change of dataset	frame	u_e	10^{-3}	nuscenes → kitti	0	0	0	5
change of dataset	frame	u_e	10^{-2}	kitti → nuscenes	2	0	3	0
change of dataset	frame	u_e	10^{-2}	nuscenes → kitti	4	0	1	0
change of dataset	frame	u_e	0.05	kitti → nuscenes	2	2	1	0
change of dataset	frame	u_e	0.05	nuscenes → kitti	5	0	0	0
change of dataset	frame	u_e	0.1	kitti → nuscenes	2	1	2	0
change of dataset	frame	u_e	0.1	nuscenes → kitti	4	0	1	0
change of dataset	frame	u_s	10^{-4}	kitti → nuscenes	0	0	0	5
change of dataset	frame	u_s	10^{-4}	nuscenes → kitti	0	0	0	5
change of dataset	frame	u_s	10^{-3}	kitti → nuscenes	0	0	0	5
change of dataset	frame	u_s	10^{-3}	nuscenes → kitti	0	0	0	5
change of dataset	frame	u_s	10^{-2}	kitti → nuscenes	5	0	0	0
change of dataset	frame	u_s	10^{-2}	nuscenes → kitti	0	5	0	0
change of dataset	frame	u_s	0.05	kitti → nuscenes	5	0	0	0
change of dataset	frame	u_s	0.05	nuscenes → kitti	0	5	0	0
change of dataset	frame	u_s	0.1	kitti → nuscenes	5	0	0	0
change of dataset	frame	u_s	0.1	nuscenes → kitti	0	5	0	0

Table 19: Drives and frames with a car detection per part and condition, division by recorded drive.

setting	split	part	condition	drives	frames
KITTI distance	0	grid	near	4	543
KITTI distance	0	grid	far	3	537
KITTI distance	0	select	near	19	714
KITTI distance	0	select	far	11	645
KITTI distance	0	eval	near	21	2065
KITTI distance	0	eval	far	12	2200
change of dataset (KITTI side)	0	select	KITTI	22	1144
change of dataset (KITTI side)	0	eval	KITTI	23	2552
KITTI distance	1	grid	near	4	295
KITTI distance	1	grid	far	3	291
KITTI distance	1	select	near	19	1057
KITTI distance	1	select	far	11	1009
KITTI distance	1	eval	near	21	1949
KITTI distance	1	eval	far	12	2097
change of dataset (KITTI side)	1	select	KITTI	22	1155
change of dataset (KITTI side)	1	eval	KITTI	23	2541
KITTI distance	2	grid	near	2	260
KITTI distance	2	grid	far	4	272
KITTI distance	2	select	near	16	1737
KITTI distance	2	select	far	20	2336
KITTI distance	2	eval	near	17	826
KITTI distance	2	eval	far	20	987
change of dataset (KITTI side)	2	select	KITTI	22	2323
change of dataset (KITTI side)	2	eval	KITTI	23	1373
KITTI distance	3	grid	near	4	297
KITTI distance	3	grid	far	2	317
KITTI distance	3	select	near	19	1091
KITTI distance	3	select	far	12	1021
KITTI distance	3	eval	near	20	1859
KITTI distance	3	eval	far	12	2086
change of dataset (KITTI side)	3	select	KITTI	22	1456

setting	split	part	condition	drives	frames
change of dataset (KITTI side)	3	eval	KITTI	23	2240
KITTI distance	4	grid	near	4	34
KITTI distance	4	grid	far	2	31
KITTI distance	4	select	near	20	1303
KITTI distance	4	select	far	11	1232
KITTI distance	4	eval	near	20	1973
KITTI distance	4	eval	far	13	2128
change of dataset (KITTI side)	4	select	KITTI	22	1412
change of dataset (KITTI side)	4	eval	KITTI	23	2284

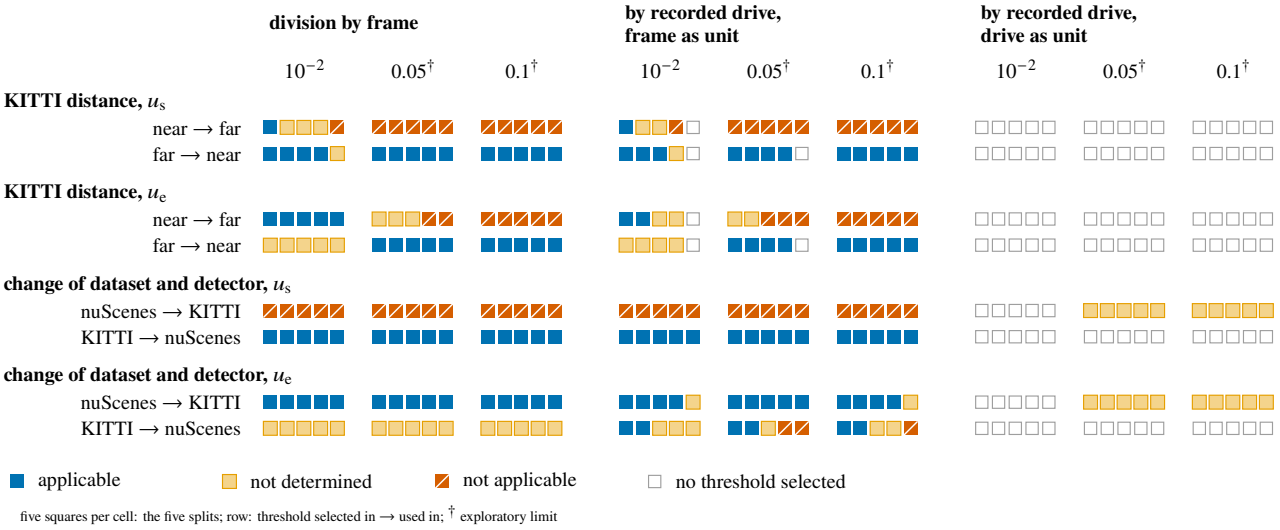


Figure 3: KITTI with the KITTI rule: outcomes per cell on the five splits for the division by frame, by recorded drive with the frame as unit, and by recorded drive with the drive as unit. In the change of dataset and detector the nuScenes side is divided by recorded scene.

9 Retention, shares of wrong detections, intervals, frames and the common threshold (division by frame)

Table 20: Every applicable transfer at 10^{-2} with its retention R (share of correct detections kept).

setting	division	ind.	transfer		split	A	$[L, U]$	R
KITTI distance	frame	u_e	near → far		0	0.0013	[0.0001, 0.0055]	0.0489
KITTI distance	frame	u_e	near → far		1	0.0013	[0.0001, 0.0055]	0.0692
KITTI distance	frame	u_e	near → far		2	0.0013	[0.0001, 0.0056]	0.0355
KITTI distance	frame	u_e	near → far		3	0.0032	[0.0008, 0.0085]	0.1113
KITTI distance	frame	u_e	near → far		4	0.0000	[0.0000, 0.0024]	0.0570
KITTI distance	frame	u_s	far → near		0	0.0040	[0.0012, 0.0098]	0.1498
KITTI distance	frame	u_s	far → near		1	0.0007	[0.0000, 0.0045]	0.0546
KITTI distance	frame	u_s	far → near		3	0.0000	[0.0000, 0.0025]	0.0417
KITTI distance	frame	u_s	far → near		4	0.0020	[0.0003, 0.0069]	0.1048
KITTI distance	frame	u_s	near → far		1	0.0019	[0.0003, 0.0066]	0.0185
change of dataset	frame	u_e	nuScenes KITTI	→	0	0.0023	[0.0004, 0.0067]	0.2347
change of dataset	frame	u_e	nuScenes KITTI	→	1	0.0024	[0.0005, 0.0070]	0.2361
change of dataset	frame	u_e	nuScenes KITTI	→	2	0.0011	[0.0001, 0.0046]	0.2316
change of dataset	frame	u_e	nuScenes KITTI	→	3	0.0024	[0.0005, 0.0070]	0.2402
change of dataset	frame	u_e	nuScenes KITTI	→	4	0.0036	[0.0010, 0.0087]	0.2851
change of dataset	frame	u_s	KITTI nuScenes	→	0	0.0000	[0.0000, 0.0013]	0.0002
change of dataset	frame	u_s	KITTI nuScenes	→	1	0.0000	[0.0000, 0.0013]	0.0000
change of dataset	frame	u_s	KITTI nuScenes	→	2	0.0000	[0.0000, 0.0013]	0.0002
change of dataset	frame	u_s	KITTI nuScenes	→	3	0.0000	[0.0000, 0.0013]	0.0000
change of dataset	frame	u_s	KITTI nuScenes	→	4	0.0000	[0.0000, 0.0013]	0.0001

setting	division	ind.	transfer	split	A	$[L, U]$	R
nuScenes distance	frame	u_e	near → far	0	0.0000	[0.0000, 0.0016]	0.0034
nuScenes distance	frame	u_e	near → far	1	0.0000	[0.0000, 0.0016]	0.0021
nuScenes distance	frame	u_e	near → far	2	0.0000	[0.0000, 0.0016]	0.0038
nuScenes distance	frame	u_e	near → far	3	0.0000	[0.0000, 0.0016]	0.0047
nuScenes distance	frame	u_e	near → far	4	0.0002	[0.0000, 0.0024]	0.0107
nuScenes distance	frame	u_s	near → far	0	0.0024	[0.0006, 0.0064]	0.0046
nuScenes distance	frame	u_s	near → far	1	0.0007	[0.0000, 0.0034]	0.0030
nuScenes distance	frame	u_s	near → far	2	0.0022	[0.0005, 0.0058]	0.0042
nuScenes distance	frame	u_s	near → far	3	0.0033	[0.0010, 0.0076]	0.0047
nuScenes distance	frame	u_s	near → far	4	0.0040	[0.0015, 0.0083]	0.0059
nuScenes distance	scene (i) frame unit	u_e	near → far	0	0.0000	[0.0000, 0.0015]	0.0044
nuScenes distance	scene (i) frame unit	u_e	near → far	1	0.0000	[0.0000, 0.0015]	0.0090
nuScenes distance	scene (i) frame unit	u_e	near → far	2	0.0000	[0.0000, 0.0016]	0.0040
nuScenes distance	scene (i) frame unit	u_e	near → far	3	0.0000	[0.0000, 0.0016]	0.0024
nuScenes distance	scene (i) frame unit	u_e	near → far	4	0.0023	[0.0006, 0.0062]	0.0150
nuScenes distance	scene (i) frame unit	u_s	far → near	2	0.0030	[0.0008, 0.0074]	0.0252
nuScenes distance	scene (i) frame unit	u_s	far → near	3	0.0031	[0.0009, 0.0075]	0.0312
nuScenes distance	scene (i) frame unit	u_s	near → far	0	0.0019	[0.0004, 0.0054]	0.0055
nuScenes distance	scene (i) frame unit	u_s	near → far	1	0.0038	[0.0014, 0.0079]	0.0037
nuScenes distance	scene (i) frame unit	u_s	near → far	2	0.0011	[0.0001, 0.0042]	0.0038
nuScenes distance	scene (i) frame unit	u_s	near → far	3	0.0042	[0.0017, 0.0086]	0.0069
nuScenes distance	scene (i) frame unit	u_s	near → far	4	0.0045	[0.0017, 0.0092]	0.0075
nuScenes rain	frame	u_e	rain → clear	0	0.0016	[0.0002, 0.0053]	0.0257
nuScenes rain	frame	u_e	rain → clear	1	0.0000	[0.0000, 0.0019]	0.0116
nuScenes rain	frame	u_e	rain → clear	2	0.0000	[0.0000, 0.0019]	0.0166
nuScenes rain	frame	u_e	rain → clear	3	0.0016	[0.0002, 0.0053]	0.0299
nuScenes rain	frame	u_e	rain → clear	4	0.0038	[0.0012, 0.0090]	0.0578
nuScenes rain	scene (i) frame unit	u_e	clear → rain	0	0.0000	[0.0000, 0.0092]	0.1009
nuScenes rain	scene (i) frame unit	u_e	clear → rain	3	0.0000	[0.0000, 0.0080]	0.0393
nuScenes rain	scene (i) frame unit	u_e	rain → clear	0	0.0005	[0.0000, 0.0034]	0.0210
nuScenes rain	scene (i) frame unit	u_e	rain → clear	1	0.0029	[0.0007, 0.0075]	0.0575
nuScenes rain	scene (i) frame unit	u_e	rain → clear	3	0.0000	[0.0000, 0.0021]	0.0048
nuScenes rain	scene (i) frame unit	u_s	clear → rain	0	0.0000	[0.0000, 0.0092]	0.0214

Table 21: Car detections, share of wrong ones and median indicator value per condition, split 0 (all parts, and the test part).

setting	condition	ind.	part	det.	wrong	med.	med. corr.	med. wrong
KITTI distance (KITTI rule)	near	u_s	all	14031	0.342	0.141	0.118	0.244
KITTI distance (KITTI rule)	near	u_e	all	14031	0.342	0.216	0.132	0.748
KITTI distance (KITTI rule)	far	u_s	all	13144	0.763	0.235	0.150	0.259
KITTI distance (KITTI rule)	far	u_e	all	13144	0.763	0.779	0.327	0.823
nuScenes distance	near	u_s	all	20227	0.187	0.214	0.206	0.266
nuScenes distance	near	u_e	all	20227	0.187	0.274	0.242	0.621
nuScenes distance	far	u_s	all	19809	0.514	0.306	0.285	0.326
nuScenes distance	far	u_e	all	19809	0.514	0.604	0.480	0.738
nuScenes rain	clear	u_s	all	30727	0.335	0.254	0.230	0.310
nuScenes rain	clear	u_e	all	30727	0.335	0.436	0.331	0.719
nuScenes rain	rain	u_s	all	6858	0.416	0.263	0.231	0.311
nuScenes rain	rain	u_e	all	6858	0.416	0.451	0.301	0.702
KITTI distance (KITTI rule)	near	u_s	eval	6310	0.336	0.139	0.117	0.246
KITTI distance (KITTI rule)	near	u_e	eval	6310	0.336	0.212	0.131	0.742
KITTI distance (KITTI rule)	far	u_s	eval	5957	0.763	0.235	0.151	0.259
KITTI distance (KITTI rule)	far	u_e	eval	5957	0.763	0.777	0.329	0.823
nuScenes distance	near	u_s	eval	9099	0.189	0.215	0.206	0.266
nuScenes distance	near	u_e	eval	9099	0.189	0.275	0.244	0.614
nuScenes distance	far	u_s	eval	8990	0.516	0.305	0.283	0.328
nuScenes distance	far	u_e	eval	8990	0.516	0.601	0.482	0.734
nuScenes rain	clear	u_s	eval	13857	0.336	0.254	0.229	0.311
nuScenes rain	clear	u_e	eval	13857	0.336	0.436	0.332	0.714
nuScenes rain	rain	u_s	eval	3122	0.425	0.264	0.232	0.310
nuScenes rain	rain	u_e	eval	3122	0.425	0.465	0.313	0.700
change of dataset	KITTI (KITTI rule)	u_s	all	27175	0.546	0.184	0.125	0.254
change of dataset	KITTI (KITTI rule)	u_e	all	27175	0.546	0.545	0.168	0.804
change of dataset	KITTI (as run)	u_s	all	29629	0.583	0.187	0.125	0.250
change of dataset	KITTI (as run)	u_e	all	29629	0.583	0.573	0.168	0.794
change of dataset	nuScenes	u_s	all	40036	0.349	0.256	0.229	0.311
change of dataset	nuScenes	u_e	all	40036	0.349	0.439	0.325	0.720
change of dataset	KITTI (KITTI rule)	u_s	eval	13707	0.544	0.184	0.124	0.255
change of dataset	KITTI (KITTI rule)	u_e	eval	13707	0.544	0.543	0.168	0.803
change of dataset	KITTI (as run)	u_s	eval	14940	0.581	0.187	0.124	0.250
change of dataset	KITTI (as run)	u_e	eval	14940	0.581	0.570	0.168	0.793
change of dataset	nuScenes	u_s	eval	19999	0.350	0.255	0.229	0.311

setting	condition	ind.	part	det.	wrong	med.	med. corr.	med. wrong
change of dataset	nuScenes	u_e	eval	19999	0.350	0.443	0.329	0.714

Table 22: Every clear → rain cell with a selected threshold (frame = division by frame; scene (i)/(ii) = division by recorded scene with the frame / scene as unit).

division	ind.	α	split	A	$[L, U]$	R	out.
frame	u_e	10^{-2}	0	0.0024	[0.0000, 0.0158]	0.087	?
frame	u_e	10^{-2}	1	0.0025	[0.0000, 0.0162]	0.091	?
frame	u_e	10^{-2}	2	0.0037	[0.0001, 0.0188]	0.102	?
frame	u_e	10^{-2}	3	0.0050	[0.0004, 0.0209]	0.102	?
frame	u_e	10^{-2}	4	0.0101	[0.0020, 0.0292]	0.126	?
frame	u_e	0.05	0	0.0871	[0.0573, 0.1278]	0.421	N
frame	u_e	0.05	1	0.0608	[0.0351, 0.0963]	0.402	?
frame	u_e	0.05	2	0.0694	[0.0409, 0.1052]	0.433	?
frame	u_e	0.05	3	0.0665	[0.0389, 0.1020]	0.417	?
frame	u_e	0.05	4	0.0650	[0.0375, 0.1005]	0.439	?
frame	u_e	0.1	0	0.1699	[0.1279, 0.2196]	0.722	N
frame	u_e	0.1	1	0.1449	[0.1052, 0.1927]	0.723	N
frame	u_e	0.1	2	0.1527	[0.1117, 0.2009]	0.747	N
frame	u_e	0.1	3	0.1571	[0.1158, 0.2059]	0.714	N
frame	u_e	0.1	4	0.1350	[0.0957, 0.1812]	0.729	?
frame	u_s	10^{-2}	0	0.0048	[0.0003, 0.0204]	0.026	?
frame	u_s	10^{-2}	4	0.0197	[0.0062, 0.0436]	0.038	?
frame	u_s	0.05	0	0.0673	[0.0398, 0.1024]	0.162	?
frame	u_s	0.05	1	0.0518	[0.0275, 0.0843]	0.144	?
frame	u_s	0.05	2	0.0589	[0.0332, 0.0933]	0.159	?
frame	u_s	0.05	3	0.0512	[0.0275, 0.0841]	0.170	?
frame	u_s	0.05	4	0.0628	[0.0375, 0.0975]	0.202	?
frame	u_s	0.1	0	0.1401	[0.1003, 0.1851]	0.385	N
frame	u_s	0.1	1	0.1372	[0.0988, 0.1845]	0.382	?
frame	u_s	0.1	2	0.1353	[0.0966, 0.1817]	0.399	?
frame	u_s	0.1	3	0.1214	[0.0857, 0.1675]	0.421	?
frame	u_s	0.1	4	0.1219	[0.0849, 0.1671]	0.394	?
scene (i) frame unit	u_e	10^{-2}	0	0.0000	[0.0000, 0.0092]	0.101	A
scene (i) frame unit	u_e	10^{-2}	1	0.0085	[0.0013, 0.0273]	0.092	?
scene (i) frame unit	u_e	10^{-2}	2	0.0042	[0.0001, 0.0212]	0.084	?
scene (i) frame unit	u_e	10^{-2}	3	0.0000	[0.0000, 0.0080]	0.039	A
scene (i) frame unit	u_e	10^{-2}	4	0.0101	[0.0016, 0.0311]	0.142	?
scene (i) frame unit	u_e	0.05	0	0.0602	[0.0354, 0.0972]	0.453	?
scene (i) frame unit	u_e	0.05	1	0.0929	[0.0619, 0.1346]	0.339	N
scene (i) frame unit	u_e	0.05	2	0.0692	[0.0397, 0.1084]	0.422	?
scene (i) frame unit	u_e	0.05	3	0.0708	[0.0446, 0.1056]	0.406	?
scene (i) frame unit	u_e	0.05	4	0.1015	[0.0661, 0.1450]	0.398	N
scene (i) frame unit	u_e	0.1	0	0.1386	[0.0998, 0.1863]	0.720	?
scene (i) frame unit	u_e	0.1	1	0.2098	[0.1642, 0.2638]	0.701	N
scene (i) frame unit	u_e	0.1	2	0.1442	[0.1021, 0.1951]	0.726	N
scene (i) frame unit	u_e	0.1	3	0.1431	[0.1057, 0.1871]	0.665	N
scene (i) frame unit	u_e	0.1	4	0.2360	[0.1840, 0.2926]	0.775	N
scene (i) frame unit	u_s	10^{-2}	0	0.0000	[0.0000, 0.0092]	0.021	A
scene (i) frame unit	u_s	10^{-2}	1	0.0061	[0.0006, 0.0231]	0.025	?
scene (i) frame unit	u_s	10^{-2}	3	0.0066	[0.0009, 0.0220]	0.014	?
scene (i) frame unit	u_s	10^{-2}	4	0.0121	[0.0024, 0.0344]	0.039	?
scene (i) frame unit	u_s	0.05	0	0.0288	[0.0119, 0.0568]	0.150	?
scene (i) frame unit	u_s	0.05	1	0.0720	[0.0441, 0.1091]	0.169	?
scene (i) frame unit	u_s	0.05	2	0.0449	[0.0229, 0.0811]	0.183	?
scene (i) frame unit	u_s	0.05	3	0.0537	[0.0308, 0.0849]	0.129	?
scene (i) frame unit	u_s	0.05	4	0.1067	[0.0706, 0.1511]	0.230	N
scene (i) frame unit	u_s	0.1	0	0.0989	[0.0657, 0.1411]	0.374	?
scene (i) frame unit	u_s	0.1	1	0.1490	[0.1098, 0.1976]	0.384	N
scene (i) frame unit	u_s	0.1	2	0.1098	[0.0736, 0.1573]	0.374	?
scene (i) frame unit	u_s	0.1	3	0.1252	[0.0905, 0.1677]	0.336	?
scene (i) frame unit	u_s	0.1	4	0.2189	[0.1693, 0.2754]	0.439	N
scene (ii) scene unit	u_e	0.1	0	0.0181	[0.0000, 0.3619]	0.260	?
scene (ii) scene unit	u_e	0.1	1	0.0242	[0.0000, 0.3525]	0.155	?
scene (ii) scene unit	u_e	0.1	2	0.0037	[0.0000, 0.3508]	0.176	?
scene (ii) scene unit	u_e	0.1	3	0.0285	[0.0000, 0.3417]	0.216	?
scene (ii) scene unit	u_e	0.1	4	0.0491	[0.0000, 0.4228]	0.286	?
scene (ii) scene unit	u_s	0.1	0	0.0106	[0.0000, 0.3434]	0.078	?
scene (ii) scene unit	u_s	0.1	1	0.0220	[0.0000, 0.3477]	0.059	?
scene (ii) scene unit	u_s	0.1	2	0.0069	[0.0000, 0.3607]	0.026	?
scene (ii) scene unit	u_s	0.1	3	0.0222	[0.0000, 0.3283]	0.058	?
scene (ii) scene unit	u_s	0.1	4	0.0563	[0.0000, 0.4350]	0.113	?

Table 23: Frames per condition in the selection and test parts (division by frame).

setting	condition	split	selection frames	test frames
KITTI distance (KITTI rule)	near	0	1484	1500
KITTI distance (KITTI rule)	far	0	1514	1545
nuScenes distance	near	0	2207	2209
nuScenes distance	far	0	2270	2276

setting	condition	split	selection frames	test frames
nuScenes rain	clear	0	1917	1900
nuScenes rain	rain	0	392	414
change of dataset	KITTI (KITTI rule)	0	1844	1852
change of dataset	nuScenes	0	2849	2850
KITTI distance (KITTI rule)	near	1	1467	1461
KITTI distance (KITTI rule)	far	1	1540	1539
nuScenes distance	near	1	2214	2198
nuScenes distance	far	1	2289	2270
nuScenes rain	clear	1	1900	1915
nuScenes rain	rain	1	417	403
change of dataset	KITTI (KITTI rule)	1	1844	1852
change of dataset	nuScenes	1	2859	2840
KITTI distance (KITTI rule)	near	2	1482	1495
KITTI distance (KITTI rule)	far	2	1523	1527
nuScenes distance	near	2	2195	2175
nuScenes distance	far	2	2311	2289
nuScenes rain	clear	2	1900	1898
nuScenes rain	rain	2	422	403
change of dataset	KITTI (KITTI rule)	2	1848	1848
change of dataset	nuScenes	2	2863	2836
KITTI distance (KITTI rule)	near	3	1466	1473
KITTI distance (KITTI rule)	far	3	1525	1551
nuScenes distance	near	3	2183	2172
nuScenes distance	far	3	2310	2304
nuScenes rain	clear	3	1917	1902
nuScenes rain	rain	3	409	404
change of dataset	KITTI (KITTI rule)	3	1844	1852
change of dataset	nuScenes	3	2847	2852
KITTI distance (KITTI rule)	near	4	1471	1474
KITTI distance (KITTI rule)	far	4	1560	1536
nuScenes distance	near	4	2199	2205
nuScenes distance	far	4	2300	2265
nuScenes rain	clear	4	1932	1896
nuScenes rain	rain	4	399	398
change of dataset	KITTI (KITTI rule)	4	1847	1849
change of dataset	nuScenes	4	2861	2838

Table 24: The common threshold τ^* on the test part of each distance condition, with the retention gap $G = R(\tau_c) - R(\tau^*)$.

setting	ind.	α	condition	split	τ^*	A	$[L, U]$	out.	$R(\tau^*)$	G
KITTI distance (KITTI rule)	u_s	10^{-2}	near	0	0.0890	0.0040	[0.0012, 0.0098]	A	0.150	0.050
KITTI distance (KITTI rule)	u_s	10^{-2}	far	0	0.0890	0.0078	[0.0035, 0.0148]	?	0.030	0.000
KITTI distance (KITTI rule)	u_s	0.05	near	0	0.1098	0.0126	[0.0064, 0.0213]	A	0.408	0.311
KITTI distance (KITTI rule)	u_s	0.05	far	0	0.1098	0.0324	[0.0227, 0.0446]	A	0.140	0.024
KITTI distance (KITTI rule)	u_s	0.1	near	0	0.1270	0.0233	[0.0151, 0.0350]	A	0.604	0.262
KITTI distance (KITTI rule)	u_s	0.1	far	0	0.1270	0.0702	[0.0554, 0.0874]	A	0.302	0.000
KITTI distance (KITTI rule)	u_s	10^{-2}	near	1	0.0782	0.0007	[0.0000, 0.0045]	A	0.055	0.027
KITTI distance (KITTI rule)	u_s	10^{-2}	far	1	0.0782	0.0013	[0.0001, 0.0055]	A	0.014	0.000
KITTI distance (KITTI rule)	u_s	0.05	near	1	0.1105	0.0112	[0.0056, 0.0201]	A	0.410	0.301
KITTI distance (KITTI rule)	u_s	0.05	far	1	0.1105	0.0315	[0.0217, 0.0440]	A	0.172	0.024
KITTI distance (KITTI rule)	u_s	0.1	near	1	0.1271	0.0212	[0.0128, 0.0319]	A	0.605	0.254
KITTI distance (KITTI rule)	u_s	0.1	far	1	0.1271	0.0732	[0.0580, 0.0906]	A	0.329	0.000
KITTI distance (KITTI rule)	u_s	10^{-2}	near	2	0.0828	0.0057	[0.0019, 0.0126]	?	0.086	0.116
KITTI distance (KITTI rule)	u_s	10^{-2}	far	2	0.0828	0.0026	[0.0005, 0.0077]	A	0.018	0.012
KITTI distance (KITTI rule)	u_s	0.05	near	2	0.1082	0.0120	[0.0059, 0.0205]	A	0.369	0.316
KITTI distance (KITTI rule)	u_s	0.05	far	2	0.1082	0.0255	[0.0170, 0.0367]	A	0.155	0.006
KITTI distance (KITTI rule)	u_s	0.1	near	2	0.1260	0.0242	[0.0157, 0.0359]	A	0.576	0.292
KITTI distance (KITTI rule)	u_s	0.1	far	2	0.1260	0.0631	[0.0491, 0.0798]	A	0.305	0.000
KITTI distance (KITTI rule)	u_s	10^{-2}	near	3	0.0750	0.0000	[0.0000, 0.0025]	A	0.042	0.270
KITTI distance (KITTI rule)	u_s	10^{-2}	far	3	0.0750	0.0013	[0.0001, 0.0055]	A	0.005	0.000
KITTI distance (KITTI rule)	u_s	0.05	near	3	0.1095	0.0115	[0.0055, 0.0199]	A	0.413	0.326
KITTI distance (KITTI rule)	u_s	0.05	far	3	0.1095	0.0348	[0.0248, 0.0474]	A	0.160	0.020

setting	ind.	α	condition	split	τ^*	A	$[L, U]$	out.	$R(\tau^*)$	G
KITTI distance (KITTI rule)	u_s	0.1	near	3	0.1291	0.0258	[0.0170, 0.0381]	A	0.634	0.248
KITTI distance (KITTI rule)	u_s	0.1	far	3	0.1291	0.0943	[0.0774, 0.1137]	?	0.341	0.012
KITTI distance (KITTI rule)	u_s	10^{-2}	near	4	0.0841	0.0020	[0.0003, 0.0069]	A	0.105	0.065
KITTI distance (KITTI rule)	u_s	10^{-2}	far	4	0.0841	0.0055	[0.0019, 0.0123]	?	0.024	0.000
KITTI distance (KITTI rule)	u_s	0.05	near	4	0.1092	0.0100	[0.0046, 0.0182]	A	0.394	0.299
KITTI distance (KITTI rule)	u_s	0.05	far	4	0.1092	0.0277	[0.0185, 0.0396]	A	0.163	0.000
KITTI distance (KITTI rule)	u_s	0.1	near	4	0.1277	0.0212	[0.0132, 0.0324]	A	0.610	0.242
KITTI distance (KITTI rule)	u_s	0.1	far	4	0.1277	0.0742	[0.0587, 0.0915]	A	0.320	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	near	0	0.1354	0.0050	[0.0015, 0.0117]	?	0.520	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	far	0	0.1354	0.0013	[0.0001, 0.0055]	A	0.049	0.072
KITTI distance (KITTI rule)	u_e	0.05	near	0	0.2747	0.0224	[0.0140, 0.0334]	A	0.827	0.071
KITTI distance (KITTI rule)	u_e	0.05	far	0	0.2747	0.0294	[0.0200, 0.0416]	A	0.390	0.033
KITTI distance (KITTI rule)	u_e	0.1	near	0	0.3784	0.0376	[0.0267, 0.0513]	A	0.906	0.068
KITTI distance (KITTI rule)	u_e	0.1	far	0	0.3784	0.0695	[0.0549, 0.0867]	A	0.591	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	near	1	0.1267	0.0041	[0.0010, 0.0100]	?	0.481	0.059
KITTI distance (KITTI rule)	u_e	10^{-2}	far	1	0.1267	0.0013	[0.0001, 0.0055]	A	0.045	0.075
KITTI distance (KITTI rule)	u_e	0.05	near	1	0.2837	0.0226	[0.0144, 0.0343]	A	0.842	0.062
KITTI distance (KITTI rule)	u_e	0.05	far	1	0.2837	0.0334	[0.0233, 0.0463]	A	0.425	0.000
KITTI distance (KITTI rule)	u_e	0.1	near	1	0.3768	0.0383	[0.0274, 0.0526]	A	0.912	0.065
KITTI distance (KITTI rule)	u_e	0.1	far	1	0.3768	0.0751	[0.0597, 0.0927]	A	0.604	0.026
KITTI distance (KITTI rule)	u_e	10^{-2}	near	2	0.1266	0.0014	[0.0001, 0.0059]	A	0.466	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	far	2	0.1266	0.0013	[0.0001, 0.0056]	A	0.035	0.072
KITTI distance (KITTI rule)	u_e	0.05	near	2	0.2595	0.0220	[0.0136, 0.0328]	A	0.804	0.099
KITTI distance (KITTI rule)	u_e	0.05	far	2	0.2595	0.0232	[0.0148, 0.0344]	A	0.335	0.036
KITTI distance (KITTI rule)	u_e	0.1	near	2	0.3645	0.0351	[0.0246, 0.0484]	A	0.903	0.074
KITTI distance (KITTI rule)	u_e	0.1	far	2	0.3645	0.0570	[0.0439, 0.0725]	A	0.565	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	near	3	0.1538	0.0066	[0.0024, 0.0137]	?	0.607	0.012
KITTI distance (KITTI rule)	u_e	10^{-2}	far	3	0.1538	0.0026	[0.0005, 0.0075]	A	0.104	0.007
KITTI distance (KITTI rule)	u_e	0.05	near	3	0.2828	0.0245	[0.0159, 0.0365]	A	0.845	0.076
KITTI distance (KITTI rule)	u_e	0.05	far	3	0.2828	0.0349	[0.0248, 0.0481]	A	0.419	0.000
KITTI distance (KITTI rule)	u_e	0.1	near	3	0.3786	0.0379	[0.0266, 0.0514]	A	0.914	0.068
KITTI distance (KITTI rule)	u_e	0.1	far	3	0.3786	0.0759	[0.0604, 0.0934]	A	0.591	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	near	4	0.1305	0.0048	[0.0016, 0.0114]	?	0.502	0.000
KITTI distance (KITTI rule)	u_e	10^{-2}	far	4	0.1305	0.0000	[0.0000, 0.0024]	A	0.057	0.051
KITTI distance (KITTI rule)	u_e	0.05	near	4	0.2571	0.0189	[0.0111, 0.0292]	A	0.809	0.096
KITTI distance (KITTI rule)	u_e	0.05	far	4	0.2571	0.0233	[0.0148, 0.0342]	A	0.360	0.022
KITTI distance (KITTI rule)	u_e	0.1	near	4	0.3752	0.0348	[0.0243, 0.0483]	A	0.911	0.066
KITTI distance (KITTI rule)	u_e	0.1	far	4	0.3752	0.0675	[0.0528, 0.0843]	A	0.611	0.000
nuScenes distance	u_s	10^{-2}	near	0	0.1265	0.0023	[0.0006, 0.0060]	A	0.020	0.000
nuScenes distance	u_s	10^{-2}	far	0	0.1265	0.0024	[0.0006, 0.0064]	A	0.005	0.006
nuScenes distance	u_s	0.05	near	0	0.1815	0.0451	[0.0351, 0.0568]	?	0.307	0.000
nuScenes distance	u_s	0.05	far	0	0.1815	0.0371	[0.0282, 0.0478]	A	0.070	0.013
nuScenes distance	u_s	0.1	near	0	0.2104	0.0657	[0.0538, 0.0796]	A	0.531	0.117
nuScenes distance	u_s	0.1	far	0	0.2104	0.0944	[0.0801, 0.1099]	?	0.158	0.012
nuScenes distance	u_s	10^{-2}	near	1	0.1238	0.0009	[0.0001, 0.0039]	A	0.016	0.000
nuScenes distance	u_s	10^{-2}	far	1	0.1238	0.0007	[0.0000, 0.0034]	A	0.003	0.006
nuScenes distance	u_s	0.05	near	1	0.1710	0.0267	[0.0190, 0.0361]	A	0.233	0.039
nuScenes distance	u_s	0.05	far	1	0.1710	0.0219	[0.0150, 0.0304]	A	0.043	0.021
nuScenes distance	u_s	0.1	near	1	0.2088	0.0622	[0.0504, 0.0755]	A	0.525	0.136

setting	ind.	α	condition	split	τ^*	A	$[L, U]$	out.	$R(\tau^*)$	G
nuScenes distance	u_s	0.1	far	1	0.2088	0.0867	[0.0730, 0.1017]	?	0.151	0.000
nuScenes distance	u_s	10^{-2}	near	2	0.1309	0.0032	[0.0011, 0.0074]	A	0.031	0.000
nuScenes distance	u_s	10^{-2}	far	2	0.1309	0.0022	[0.0005, 0.0058]	A	0.004	0.012
nuScenes distance	u_s	0.05	near	2	0.1762	0.0342	[0.0255, 0.0448]	A	0.281	0.017
nuScenes distance	u_s	0.05	far	2	0.1762	0.0346	[0.0261, 0.0451]	A	0.058	0.023
nuScenes distance	u_s	0.1	near	2	0.2082	0.0631	[0.0513, 0.0768]	A	0.536	0.170
nuScenes distance	u_s	0.1	far	2	0.2082	0.0907	[0.0768, 0.1060]	?	0.160	0.010
nuScenes distance	u_s	10^{-2}	near	3	0.1305	0.0021	[0.0004, 0.0059]	A	0.028	0.000
nuScenes distance	u_s	10^{-2}	far	3	0.1305	0.0033	[0.0010, 0.0076]	A	0.005	0.005
nuScenes distance	u_s	0.05	near	3	0.1760	0.0307	[0.0224, 0.0407]	A	0.275	0.020
nuScenes distance	u_s	0.05	far	3	0.1760	0.0336	[0.0252, 0.0438]	A	0.061	0.010
nuScenes distance	u_s	0.1	near	3	0.2081	0.0597	[0.0480, 0.0729]	A	0.524	0.120
nuScenes distance	u_s	0.1	far	3	0.2081	0.0893	[0.0755, 0.1044]	?	0.164	0.000
nuScenes distance	u_s	10^{-2}	near	4	0.1262	0.0034	[0.0010, 0.0079]	A	0.019	0.020
nuScenes distance	u_s	10^{-2}	far	4	0.1262	0.0026	[0.0008, 0.0065]	A	0.004	0.010
nuScenes distance	u_s	0.05	near	4	0.1809	0.0407	[0.0311, 0.0519]	?	0.309	0.000
nuScenes distance	u_s	0.05	far	4	0.1809	0.0346	[0.0260, 0.0451]	A	0.065	0.000
nuScenes distance	u_s	0.1	near	4	0.2057	0.0628	[0.0510, 0.0763]	A	0.499	0.169
nuScenes distance	u_s	0.1	far	4	0.2057	0.0751	[0.0621, 0.0891]	A	0.141	0.011
nuScenes distance	u_e	10^{-2}	near	0	0.0925	0.0043	[0.0016, 0.0092]	A	0.101	0.000
nuScenes distance	u_e	10^{-2}	far	0	0.0925	0.0000	[0.0000, 0.0016]	A	0.003	0.034
nuScenes distance	u_e	0.05	near	0	0.2616	0.0393	[0.0299, 0.0502]	?	0.547	0.017
nuScenes distance	u_e	0.05	far	0	0.2616	0.0250	[0.0176, 0.0339]	A	0.100	0.077
nuScenes distance	u_e	0.1	near	0	0.4210	0.0609	[0.0493, 0.0742]	A	0.813	0.116
nuScenes distance	u_e	0.1	far	0	0.4210	0.0933	[0.0793, 0.1089]	?	0.375	0.019
nuScenes distance	u_e	10^{-2}	near	1	0.0773	0.0018	[0.0004, 0.0053]	A	0.070	0.000
nuScenes distance	u_e	10^{-2}	far	1	0.0773	0.0000	[0.0000, 0.0016]	A	0.002	0.036
nuScenes distance	u_e	0.05	near	1	0.2448	0.0285	[0.0205, 0.0382]	A	0.509	0.016
nuScenes distance	u_e	0.05	far	1	0.2448	0.0158	[0.0100, 0.0232]	A	0.095	0.059
nuScenes distance	u_e	0.1	near	1	0.4149	0.0604	[0.0487, 0.0736]	A	0.809	0.120
nuScenes distance	u_e	0.1	far	1	0.4149	0.0863	[0.0726, 0.1012]	?	0.365	0.000
nuScenes distance	u_e	10^{-2}	near	2	0.0872	0.0032	[0.0009, 0.0074]	A	0.097	0.000
nuScenes distance	u_e	10^{-2}	far	2	0.0872	0.0000	[0.0000, 0.0016]	A	0.004	0.039
nuScenes distance	u_e	0.05	near	2	0.2575	0.0387	[0.0295, 0.0500]	?	0.551	0.015
nuScenes distance	u_e	0.05	far	2	0.2575	0.0221	[0.0153, 0.0307]	A	0.109	0.083
nuScenes distance	u_e	0.1	near	2	0.4136	0.0595	[0.0480, 0.0728]	A	0.814	0.119
nuScenes distance	u_e	0.1	far	2	0.4136	0.0863	[0.0728, 0.1013]	?	0.377	0.000
nuScenes distance	u_e	10^{-2}	near	3	0.1002	0.0051	[0.0022, 0.0099]	A	0.130	0.000
nuScenes distance	u_e	10^{-2}	far	3	0.1002	0.0000	[0.0000, 0.0016]	A	0.005	0.032
nuScenes distance	u_e	0.05	near	3	0.2748	0.0423	[0.0324, 0.0537]	?	0.582	0.013
nuScenes distance	u_e	0.05	far	3	0.2748	0.0293	[0.0214, 0.0389]	A	0.132	0.036
nuScenes distance	u_e	0.1	near	3	0.4095	0.0574	[0.0460, 0.0704]	A	0.806	0.139
nuScenes distance	u_e	0.1	far	3	0.4095	0.0880	[0.0743, 0.1030]	?	0.366	0.000
nuScenes distance	u_e	10^{-2}	near	4	0.1141	0.0074	[0.0037, 0.0133]	?	0.169	0.000
nuScenes distance	u_e	10^{-2}	far	4	0.1141	0.0002	[0.0000, 0.0024]	A	0.011	0.026
nuScenes distance	u_e	0.05	near	4	0.2579	0.0372	[0.0283, 0.0483]	A	0.535	0.000
nuScenes distance	u_e	0.05	far	4	0.2579	0.0191	[0.0129, 0.0274]	A	0.110	0.068
nuScenes distance	u_e	0.1	near	4	0.4102	0.0562	[0.0449, 0.0689]	A	0.802	0.128
nuScenes distance	u_e	0.1	far	4	0.4102	0.0811	[0.0678, 0.0958]	A	0.356	0.000

10 Post-hoc analyses

Post-hoc. The analyses of this section were added after the parameters of the study were fixed and after the outcomes were known. They are not part of the predictions fixed in advance. The limits 0.05 and 0.1 are exploratory and are not safety targets.

Method A tests the per-frame target $A_c(\tau) = \mathrm{E}[a_f(\tau) \mid f \in c] \le \alpha$ with the recorded sequence (KITTI drive, nuScenes scene) as the independent unit. For sequence s, N_s is the number of its frames in condition c and $S_s(\tau)$ the sum of $a_f(\tau)$ over these frames. For independent sequences $A_c = \mathrm{E}[S]/\mathrm{E}[N]$, so with $N_{\max}(c)$, the largest N_s over all validation sequences of c, the value $Y_s = (S_s - \alpha N_s + \alpha N_{\max})/N_{\max}$ lies in $[0, 1]$ and $\mathrm{E}[Y] \le \alpha$ holds exactly when $A_c \le \alpha$. Selection and test are those of the paper (Hoeffding–Bentkus, $\delta = 0.05$, bounds at $\delta/2$) with Y_s as the loss of a unit, on the division by recorded sequence (Sect. 7, Sect. 8). *Method B* repeats the test with the betting confidence interval of Waudby-Smith and Ramdas (hedged capital process, each side at $\delta/2$) and keeps the thresholds of Method A. Before any new number, the recorded results were recomputed: "GATE PASSED: 1600 recomputed rows compared with the recorded rows (frame division with N_s = 1 against m6 A/B and m7 frame rows; sequence division with frame units against m7 scene_frame_unit and m8 frame-unit rows); 0 differ".

Table 25: Post-hoc. Outcome counts over the five splits, Methods A and B (A applicable, N not applicable, ? not determined, – no threshold selected; own: threshold used in its own condition of selection). Rows in which neither method selected a threshold on any split are not listed.

pair	transfer	own	ind.	α	A: A	A: N	A: ?	A: –	B: A	B: N	B: ?	B: –
change of dataset and detector	nuScenes→KITTI		u_e	0.1	0	0	5	0	0	0	5	0

pair	transfer	own	ind.	α	A: A	A: N	A: ?	A: –	B: A	B: N	B: ?	B: –
change of dataset and detector	nuScenes→KITTI		u_s	0.1	0	0	5	0	0	0	5	0
change of dataset and detector	nuScenes→nuScenes	yes	u_e	0.1	2	0	3	0	5	0	0	0
change of dataset and detector	nuScenes→nuScenes	yes	u_s	0.1	2	0	3	0	5	0	0	0
nuScenes distance	far→near		u_e	0.1	0	0	5	0	5	0	0	0
nuScenes distance	far→near		u_s	0.1	0	0	5	0	5	0	0	0
nuScenes distance	near→far		u_e	0.1	3	0	1	1	4	0	0	1
nuScenes distance	near→far		u_s	0.1	2	0	0	3	2	0	0	3
nuScenes distance	far→far	yes	u_e	0.1	2	0	3	0	5	0	0	0
nuScenes distance	far→far	yes	u_s	0.1	1	0	4	0	5	0	0	0
nuScenes distance	near→near	yes	u_e	0.1	0	0	4	1	4	0	0	1
nuScenes distance	near→near	yes	u_s	0.1	0	0	2	3	2	0	0	3
nuScenes weather	clear→rain		u_e	0.1	0	0	5	0	0	0	5	0
nuScenes weather	clear→rain		u_s	0.1	0	0	5	0	0	0	5	0
nuScenes weather	clear→clear	yes	u_e	0.1	2	0	3	0	3	0	2	0
nuScenes weather	clear→clear	yes	u_s	0.1	0	0	5	0	5	0	0	0

Table 26: Post-hoc. AUROC of the three pairs of the separation figure on the test part of split 0, with 95 % percentile bootstrap intervals over recorded sequences (2,000 replicates). Near/far: drives resampled once for both conditions; clear/rain and the two datasets: sequences resampled within each condition.

pair	ind.	quantity	AUROC	95 % interval	resampling
KITTI distance	u_s	AUROC near	0.915	[0.884, 0.946]	joint
KITTI distance	u_s	AUROC far	0.845	[0.831, 0.860]	joint
KITTI distance	u_s	AUROC far - AUROC near	-0.069	[-0.091, -0.046]	joint
nuScenes weather	u_e	AUROC clear	0.844	[0.819, 0.869]	per condition
nuScenes weather	u_e	AUROC rain	0.856	[0.821, 0.887]	per condition
nuScenes weather	u_e	AUROC rain - AUROC clear	0.012	[-0.030, 0.051]	per condition
change of dataset and detector	u_s	AUROC nuScenes	0.749	[0.734, 0.765]	per condition
change of dataset and detector	u_s	AUROC KITTI	0.905	[0.886, 0.922]	per condition
change of dataset and detector	u_s	AUROC KITTI - AUROC nuScenes	0.157	[0.133, 0.180]	per condition

Table 27: Post-hoc. Frames per sequence in each condition (sequences with $N_s \geq 1$, all parts, split 0).

pair	condition	sequences	min N_s	median N_s	max N_s
KITTI distance	near	44	1	8.0	380
KITTI distance	far	26	1	80.0	557
nuScenes distance	near	149	3	37.0	41
nuScenes distance	far	150	4	38.0	41
nuScenes weather	clear	111	12	40.0	41
nuScenes weather	rain	24	19	40.0	41
change of dataset and detector	KITTI	45	1	9.0	571
change of dataset and detector	nuScenes	150	12	40.0	41

Training epochs of the nuScenes members (post-hoc check). The training configurations of the nuScenes ensemble members set 20 epochs, but training ended after epoch 10: each training log reaches the start of epoch 11, and the checkpoints that inference loaded (`latest_model.pth`) stop shortly before the end of epoch 10. The nuScenes members were therefore trained for about 10 epochs, not 20. The configuration of seed 5 belongs to no ensemble member; inference used the members of seeds 1 to 4. Lines of the check (`results/post_hoc/epoch_check.txt`):

```
/data/inputs/configs/cbgs_pp_multihead_nuscenes_seed1.yaml:233:    NUM_EPOCHS: 20
/data/inputs/configs/cbgs_pp_multihead_nuscenes_seed2.yaml:233:    NUM_EPOCHS: 20
/data/inputs/configs/cbgs_pp_multihead_nuscenes_seed3.yaml:233:    NUM_EPOCHS: 20
/data/inputs/configs/cbgs_pp_multihead_nuscenes_seed4.yaml:233:    NUM_EPOCHS: 20
/data/inputs/configs/cbgs_pp_multihead_nuscenes_seed5.yaml:233:    NUM_EPOCHS: 20
Train:   10/20
Train:   11/20
Train:   10/20
Train:   11/20
Train:   10/20
Train:   11/20
Train:   10/20
Train:   11/20
seed 1 latest_model.pth: epoch=9 it=308860
seed 1 checkpoint_epoch_10.pth: epoch=10 it=308950
seed 2 latest_model.pth: epoch=9 it=308840
seed 2 checkpoint_epoch_10.pth: epoch=10 it=308950
seed 3 latest_model.pth: epoch=9 it=308899
seed 3 checkpoint_epoch_10.pth: epoch=10 it=308950
seed 4 latest_model.pth: epoch=9 it=308842
seed 4 checkpoint_epoch_10.pth: epoch=10 it=308950
```

11 Predictions fixed in advance and their outcomes

The predictions P1 to P5 and the outcomes that would count as their failure were fixed on 2026-10-01, before the first evaluation run (`docs/study_protocol/predictions_frozen_2026-10-01.md`, l. 3). The outcome of each prediction, the reason and the evidence file are listed below; `docs/study_protocol/outcomes_and_deviations.md` also lists the deviations from the fixed protocol.

Table 28: Predictions and their outcomes.

prediction	outcome	reason	evidence
P1. Indicator response by mechanism. (l. 33)	not tested	classes C1 and C2 were measured only together as recorded rain, and C3 was not measured, so the comparison the prediction names cannot be made	–
P2. Discrimination is more stable than the operating point. (l. 48)	not tested	how to compare the spread of AUROC with the spread of the FAP in units of interval width was not fixed in advance; the comparison was not computed	–
P3. The maximum retention gap is not zero and not total. (l. 57)	held	the retention gap G of the common threshold lies above zero and below the retention at the threshold of the condition (KITTI, u_s, split 0)	`results/supporting_analyses/common_threshold_outcomes.csv`
P4. Applicability is not symmetric. (l. 71)	held	nuScenes→KITTI is not applicable on all splits for u_s at every limit with a threshold, KITTI→nuScenes is applicable; near→far and far→near differ in the same way at 0.1	`results/division_by_frame/kitti_rule_dataset_change_weather/B_shift_seeds_protocol.csv`
P5. Some cells return "not determined". (l. 79)	held	cells with the outcome not determined occur in every pair of conditions	`docs/results.md` (outcome table) and its source files

12 Frames shown in Fig. 1

Fig. 1 of the paper shows one KITTI frame and one nuScenes frame in which the transferred threshold accepts a wrong detection that the threshold selected in the condition of use rejects (split 0, exploratory limit 0.1). The figure shows dataset images and is not part of the repository; its facts are recorded in `results/supporting_analyses/fig1_two_panel_facts.txt` and listed here. Candidates for the nuScenes frame are the rain frames of the test part of split 0 with exactly one accepted wrong car detection whose u_e lies between the threshold selected in rain and the transferred clear-weather threshold (both at 0.1) and whose box lies inside the front camera image; the yaw of its member boxes spans at most 15 degrees; its 3D IoU with every annotation is 0 or its IoU with the nearest annotated car is below 0.3; the frame holds 3 to 8 accepted correct and at most 3 rejected car detections, all within 50 m forward. Ranking: IoU 0 with all annotations first, then the larger height of the wrong box in the image, then more accepted correct detections (`results/supporting_analyses/fig1b_candidates.csv`). Rank 1 was not used, because its wrong detection lies 1.9 m below an annotated car on a ramp at the same position in the bird's-eye view, so its overlap of 0 comes from the height and not from an empty location. The chosen wrong detection (rank 2) lies 0.26 m from an annotated car with a 3D IoU of 0.265: a mislocalised detection of an annotated car, wrong under the IoU 0.5 rule. The KITTI frame 005345 was chosen by the authors from the ranked candidates in `results/supporting_analyses/frame_candidates_ranked.csv`. The scripts are in `tools/figures/paper/frame_figure/`.

```
(a) KITTI frame 005345 (drive 2011_09_26_drive_0022, raw frame 521); indicator u_s; split 0; limit 0.1 (exploratory)
threshold selected at near distance, transferred, decides acceptance: 0.16931
threshold selected at far distance: 0.12695
car detections: accepted correct 5, accepted wrong 1, rejected 1, not scored (KITTI rule) 0
near/far boundary 34.806 m
accepted wrong detection (numbered 2): distance 55.22 m, u_s 0.1454, nearest annotated car (centre distance 30.18 m): 3D IoU 0.000; member yaw spread 3.1 deg
numbered 1: accepted correct, distance 8.16 m, u_s 0.1012
(b) nuScenes frame f0f7fec4524045f4a2f44866d410ec84 (scene-0626); indicator u_e; split 0; limit 0.1 (exploratory)
threshold selected in clear weather, transferred, decides acceptance: 0.46385
threshold selected in rain: 0.26474
car detections: accepted correct 5, accepted wrong 1, rejected 2, not scored (KITTI rule) 0
accepted wrong detection (numbered 2): distance 25.06 m, u_e 0.4281, nearest annotated car (centre distance 0.26 m): 3D IoU 0.265; largest 3D IoU with any annotation (all classes) 0.265; member yaw spread
numbered 1: accepted correct, distance 13.98 m, u_e 0.1701
```

Table 29: Candidate nuScenes frames for Fig. 1(b).

rank	token	scene	distance (m)	u_e	IoU nearest car	accepted correct	rejected
1	`8b3831ab`	scene-0909	37.4	0.4631	0.000	6	3
2	`f0f7fec4`	scene-0626	25.1	0.4281	0.265	5	2

13 Source files

Every number in this document is read by `tools/supplementary/make_supplementary.py` (supp-public-1.0 (2026-10-21)) from the files below and listed with file, line or row, and column in `docs/supplementary/supplementary_sources.csv`. CSV rows are numbered by file line (header = line 1).

file	SHA-256
docs/study_protocol/predictions_frozen_2026-10-01.md	374cd6f2b8481c26c8e5e5283da108037b83e8c8969adf0279dadd560927d387
results/division_by_frame/dataset_change_one_split/shift_matrix.csv	fbd1c2ded0f9eb58eb052095d033af7657263db050c519f0dc912b6d0e542554
results/division_by_frame/kitti_rule_dataset_change_weather/A_kitti_range_protocol.csv	56a3721febba396f78880f1ed9f0af0c288da1f5f49f64017d365119ffb464c8
results/division_by_frame/kitti_rule_dataset_change_weather/B_shift_seeds_protocol.csv	7bc64acb59d6e7b14b96f16272f4a643bb54dbff2d38885cfb294a63b9b49930
results/division_by_frame/kitti_rule_dataset_change_weather/C_weather_nuscenes.csv	e0cf1586b2b2f3fa5fa14f6d69df020fb8be00f91a7fe99e00454ebb72c03af0
results/division_by_frame/kitti_rule_dataset_change_weather/provenance.json	81d9a22cf8feeba3388c8f356c8265447bbe90f5a3ffba67c8b38498387a69dd
results/division_by_frame/kitti_rule_dataset_change_weather/summary.txt	79f649f444cadee0dd60dd31da1fac00b1cf6b76964f00d3dfae8b0640bd18cb
results/division_by_frame/nuscenes_distance/matrix_all_variants.csv	12e386948dab180cedaf02d17ea09cc756ffc189ec0afd186f643be27c34b74e
results/division_by_sequence/kitti_by_drive/REPRO_PASSED.txt	8aa86eb948c468156748f370c5cf576596451addf6bc3b2d8f92486f64cf3a41
results/division_by_sequence/nuscenes_by_scene/REPRO_PASSED.txt	cae93f90717884984c296e34a6eb5485b5d756c9c2fc4a68e94b29ec66abe478
results/division_by_sequence/nuscenes_by_scene/m7_scene_split.csv	75452edfe3893c706305f816279bf263ff483c1135331e14f05d2d1638a7a3ca
results/division_by_sequence/nuscenes_by_scene/m7_scene_split_summary.txt	c60ecabafdb0966413ba067c8088e61f1ff36eaa88f4aa762f6789f1b219dfe6
results/post_hoc/GATE_PASSED.txt	c2bbbc20f5f9473b714a71800068b59dbe16120226274fa5ab53718b6c259313
results/post_hoc/auroc_bootstrap.csv	bd9a4dd02c255fe2f447c584f6625b657c695b27e46a9b235c4399730d94eaac
results/post_hoc/epoch_check.txt	2f76488d915b9f235938bfb7d20e90434e2c7255d54fb64a93cced7c3323c5fd
results/post_hoc/ns_per_condition.csv	6fa3a284054683cae5171815864d751309d3ba0f59b11d5e0313ff1b9d32158f
results/post_hoc/summary.csv	77d964cb6ba9cac9ba9f7cde14e88bfa866dae550e6b74966c698d7e8f904309
results/supporting_analyses/common_threshold_outcomes.csv	3330fa05988c7fc7e872d86d02bbd346de8de6e3e628a8f1b86fb74d1c6071bc
results/supporting_analyses/distance_bins_log.txt	54aead337fbe573e7a37ef8a8ca4de7643661fc5773d7fcd4ac61eb9c73c87c3
results/supporting_analyses/fig1_two_panel_facts.txt	7be313c9979860a67beb405706209c19afea02b010ec597a6fba3c89f2478c3
results/supporting_analyses/fig1b_candidates.csv	f335f25d407480f78417312015e6c2dd663f8e4dca7eae4c9401fb2399f3e2f5
results/supporting_analyses/frames_per_condition.csv	ffc49f8a86cca9ae035d963485fa1fd98ce7433c62ad42ae04855506d4b7cd04
results/supporting_analyses/intervals_clear_to_rain.csv	399ece8a413fd4730380bcc5bfe725b1cd8eacf7fb10d0a0f3749a97233ea007
results/supporting_analyses/m8_drives_per_part.csv	498e5711ab3638b69413d93290ad105b19169734da303934a6575ba589e4d614
results/supporting_analyses/m8_outcome_counts.csv	ae73fdeb11d931be64d9ff059f833c7793ee00317812d8f5121e93e8962376f9
results/supporting_analyses/retention_applicable_cells.csv	d17e53465028618331bc653b92971ad403537a34ce7d97d7d08096542cb0da24
results/supporting_analyses/table_and_figure_values.json	3c44a78987eda03d12e21b25a84a9aa3bf8a39467b9aed0fb406e814b3c8bee3
results/supporting_analyses/wrong_share_and_median_split0.csv	d2ca3b8f28941e29ecdbb78714d70d5ea41820349732b906f2594fc5c1fdfe99